PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Design and Development of a Web-Based Dental Clinic Management System : Patient Records, Appointments, and Billing.

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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System Development Track

Second Semester 2025-26

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INTRODUCTION

1.1 Background of the Study

Technology has become an essential part of modern business operations, providing faster, more accurate, and more efficient ways of managing information and services. Many organizations have shifted from manual processes to computerized systems in order to improve productivity, reduce errors, and enhance the quality of service provided to clients. In the healthcare sector, information systems play an important role in managing patient records, appointments, and other administrative tasks.

Despite these technological advancements, some dental clinics still rely on paper-based records and manual appointment scheduling. These traditional methods may lead to several problems such as misplaced or lost patient files, inaccurate records, scheduling conflicts, delays in service, and difficulty in tracking patient history and treatment records. In addition, manual processes can consume more time and effort for both clinic staff and patients, which may affect the overall efficiency of clinic operations and patient satisfaction.

To address these issues, this study aims to develop a web-based Dental Records and Appointment Management System that will help dental clinics store, manage, and retrieve patient information efficiently. The proposed system will also provide an automated appointment scheduling feature that can simplify the process of booking, monitoring, and managing patient appointments. Furthermore, the system is expected to improve data organization, reduce operational errors, enhance clinic productivity, and provide better service to patients through a more reliable and accessible digital platform.

1.2 Statement of the Problem

Many dental clinics still rely on manual or semi-digital methods in handling patient information, appointment scheduling, billing, and other clinic operations. These traditional processes may result in inefficiency, inaccurate record management, scheduling conflicts, data loss, and delays in providing services to patients. In addition, the lack of an integrated system can affect the overall productivity and quality of clinic operations.

This study aims to develop a web-based dental clinic management system to address these issues. Specifically, it seeks to answer the following questions:

1. How can patient records and dental histories be securely stored, organized, and managed through a digital system?

2. How can appointment scheduling and monitoring be improved through an automated process?

3. How can the proposed system minimize errors such as double booking, misplaced records, and inaccurate patient information?

4. How can a web-based dental clinic management system improve the efficiency, accuracy, and overall performance of dental clinic operations?

1.3 Objectives of the Study

1.3.1 General Objective

General Objective

The general objective of this study is to design and develop a comprehensive web-based dental clinic management system that will enhance the efficiency and accuracy of managing patient records, appointment scheduling, billing, treatment monitoring, and other clinic operations. Specifically, the system aims to streamline administrative tasks, improve communication between patients and clinic staff, ensure secure and organized data management, and provide a more convenient and reliable service for both the dental clinic personnel and patients.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. Design and develop a secure and reliable database system capable of storing, organizing, and managing patient information, dental history, and treatment records efficiently. Create an automated system for scheduling and managing patient appointments.

2. Create an automated appointment scheduling and management system that will allow clinic staff and patients to conveniently set, update, and monitor appointments. Generate reports for patient records and scheduled appointments.

3. Implement a user authentication and access control feature for dentists, clinic staff, and administrators to ensure data security and proper system authorization.

4. Generate accurate and organized reports related to patient records, appointment schedules, and clinic transactions to support effective monitoring and decision-making.

5. Improve the overall efficiency, accuracy, and productivity of dental clinic operations through the use of digital record management and automated processes.

1.4 Scope and Delimitation

This study focuses on the **design and development of a web-based dental clinic management system** that manages patient records, dental history, appointment scheduling, and basic reporting features. The system will be accessible through web browsers and will be used by authorized clinic personnel such as dentists, staff, and administrators.

However, the system has several limitations. It will support standard dental clinic management functions, including digital treatment records and basic X-ray/image uploads for diagnostic reference. However, it will not include advanced features such as automated AI medical imaging analysis, third-party insurance processing, or native mobile application (Android/iOS) support. Additionally, the system will depend on internet availability and proper system maintenance for optimal performance.

1.5 Significance of the Study

The proposed system will benefit the following:

Dental clinics – The system will improve the management of patient records, appointment scheduling, billing processes, treatment monitoring, and clinic operations. It will help clinics organize data efficiently, reduce paperwork, and improve the quality of service provided to patients.

Dentists – Dentists will benefit from faster access to patient information, treatment history, dental charts, X-ray records, prescriptions, and appointment schedules. The system will also help dentists monitor patient progress and manage treatments more efficiently.

Patients – Patients will experience faster and more convenient appointment scheduling, organized treatment records, appointment reminders, access to schedules, downloadable records, and improved communication with the dental clinic.

Future Researchers – This study may serve as a useful reference for future researchers and developers who plan to create or improve healthcare management systems, particularly in the field of dental clinic management systems.

1.6 Definition of Terms

Dental Records – Digital files that store patient information, dental history, and treatments.

Appointment Scheduling – The process of booking and managing patient visits.

Database – A system used to store and organize data digitally.

Web-Based System – A system accessed through a web browser without installation.

User Authentication – A security process that verifies a user's identity before access.

1.7 Conceptual Framework

The development of the Web-Based Dental Clinic Management System is anchored on the Input-Process-Output (IPO) paradigm. This model provides a systematic and structured approach to understanding the requirements, developmental processes, and the final expected output of the study. The IPO model is highly effective for system development research as it clearly delineates the resources needed (Input), the methodologies and procedures applied (Process), and the ultimate solution generated (Output).

Input Block: The input phase encompasses all the fundamental requirements necessary for the successful development of the system. This includes:

1. **Hardware Requirements:** Development laptops, smartphones for responsive testing, and the client's existing computer hardware at the dental clinic.

2. **Software Requirements:** Python (Flask framework) for backend logic, HTML5/CSS3/JavaScript for frontend development, Google Firebase (Cloud Firestore)

for the NoSQL database and authentication, and Visual Studio Code as the Integrated Development Environment (IDE).

3. **Data Requirements:** Existing patient records, dental treatment histories, appointment schedules, billing structures, and clinic service pricing from the client clinic.

4. **Knowledge and Methodological Requirements:** The Agile Development Methodology, system analysis and design principles, database management concepts, and knowledge of the Philippine Data Privacy Act of 2012.

Process Block: The process phase details the systematic procedures and developmental lifecycle applied to transform the inputs into the desired output. The researchers will utilize the Agile Development Model, which allows for iterative progress and continuous improvement. The phases include:

1. **Requirements Analysis:** Conducting interviews and observations with the dentist and patients to gather functional and non-functional requirements.

2. **System Design:** Creating architectural blueprints, including Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), Use Case Diagrams, Sequence Diagrams, and database schema design.

3. **Development (Coding):** Writing the frontend interfaces, developing the backend logic using Python Flask, and integrating the Firebase database for real-time data synchronization.

4. **Testing:** Performing Unit Testing for individual modules, System Testing for end-to-end workflows, and preparing for User Acceptance Testing (UAT).

5. **Deployment and Evaluation:** Deploying the prototype and evaluating the system using ISO 25010 software quality standards.

Output Block: The output phase represents the final culmination of the study. It is the fully designed, developed, and evaluated **Web-Based Dental Clinic Management System: Patient Records, Appointments, and Billing**. The system will feature secure patient record management, automated appointment scheduling, digital treatment history tracking, automated billing computation, and comprehensive report generation, ultimately aiming to modernize and streamline the clinic's operations.

1.7.2 Narrative of the Conceptual Framework

The conceptual framework illustrates the logical flow of the system development. The transition from the Input to the Process signifies the application of technical skills and Agile methodologies to address the identified problems of the dental clinic. The researchers will continuously refer to the inputs—specifically the data gathered from the clinic and the software tools selected—to ensure that the development process remains aligned with the actual needs of the users.

The Process block acts as the bridge. By strictly following the Agile methodology, the researchers ensure that each component of the system, from the user authentication module to the billing computation engine, is developed iteratively. If a flaw is discovered during the testing phase, the process allows the researchers to loop back to the development phase to refine the code, ensuring a high-quality final product.

Finally, the Output block represents the realization of the study's general objective. The resulting web-based system will directly address the inefficiencies of the manual processes currently utilized by the clinic. By centralizing data into a Firebase cloud database and providing a responsive web interface, the output will serve as a sustainable, scalable, and secure digital solution for the dental clinic, thereby completing the IPO cycle.

1.8 Theoretical Framework

To provide a strong academic foundation for the development and evaluation of the Web-Based Dental Clinic Management System, this study is anchored on the **Technology Acceptance Model (TAM)** developed by Fred Davis in 1989. TAM is one of the most widely cited and validated theories in information systems research used to predict and explain user acceptance of technology.

The model posits that the successful adoption of a new information system is primarily determined by two critical factors: **Perceived Usefulness (PU)** and **Perceived Ease of Use (PEOU)**.

1. Perceived Usefulness (PU) In the context of this study, Perceived Usefulness refers to the degree to which the dental clinic staff, dentists, and administrators believe that using the Web-Based Dental Clinic Management System will enhance their job performance and daily operations. The system is designed to directly address the clinic's pain points by automating appointment scheduling, centralizing patient records, and streamlining billing computations. If the users perceive that the system saves them time, reduces manual paperwork, and minimizes scheduling conflicts compared to their current manual methods, their Perceived Usefulness will be high, leading to a positive attitude toward the system.

2. Perceived Ease of Use (PEOU) Perceived Ease of Use refers to the degree to which the users believe that interacting with the system will be free from physical and mental effort. Because the proposed system is web-based and utilizes responsive design principles, it can be accessed via standard web browsers without the need for complex installations. The user interface is designed with intuitive navigation, clear dashboards, and logical workflows. If the clinic staff and dentists find the system easy to learn and navigate, their Perceived Ease of Use will increase.

According to TAM, a high level of Perceived Ease of Use directly influences Perceived Usefulness, and both factors collectively determine the user's **Behavioral Intention to Use** the system. By applying TAM, the researchers ensure that the system is not only technically robust but also highly user-centric. The User Acceptance Testing (UAT) phase, evaluated using the ISO 25010 standard, will specifically measure these theoretical constructs to ensure the system is readily accepted by the end-users at the dental clinic.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

2.1.1 Electronic Dental Record Systems

A foreign study conducted in the United States examined the implementation of Electronic Dental Record (EDR) systems in dental clinics and healthcare institutions. The study aimed to determine how digital record systems improve the management, storage, retrieval, and security of patient dental information. The researchers explained that traditional paper-based dental records often result in inefficiencies such as misplaced files, incomplete patient histories, and difficulty in tracking long-term treatments. These problems can negatively affect the quality of dental care provided to patients.

The study emphasized that manual record-keeping systems are prone to human error, especially in busy dental clinics where a large number of patients are being handled daily. Paper-based systems also require physical storage space and are difficult to maintain over time. In addition, retrieving old patient records can be time-consuming and may delay diagnosis and treatment planning.

The researchers explored how electronic dental record systems improve clinical operations through centralized databases and automated data storage. These systems allow dentists and staff to encode, update, and retrieve patient records in real time. The integration of digital record systems also improves accuracy in documenting dental history, treatment procedures, and diagnostic results.

The findings of the study revealed that clinics using electronic dental records experienced improved efficiency in patient management and reduced administrative workload. Dentists were able to access patient histories more quickly, which supported better clinical decision-making and treatment planning. The study also noted that digital records improved coordination among dental staff.

Furthermore, the study found that electronic systems enhanced data security through user authentication and controlled access. Sensitive patient information was better protected compared to manual systems. However, the study also identified challenges such as high implementation costs and the need for staff training.

The study concluded that Electronic Dental Record systems significantly improve efficiency, accuracy, and security in dental clinic operations. The researchers emphasized that digital transformation in healthcare is essential for improving service delivery and patient care quality. This foreign study is relevant to the present study because it supports the development of a web-based dental clinic management system that includes secure patient record storage and real-time access to dental histories.

2.1.2 Automated Appointment Scheduling Systems

A foreign study conducted in Canada focused on the effectiveness of automated appointment scheduling systems in dental clinics. The study aimed to evaluate how digital scheduling platforms improve appointment management, reduce patient waiting time, and minimize scheduling conflicts. The researchers explained that manual appointment systems often lead to inefficiencies such as double booking, missed appointments, and difficulty in tracking available time slots.

The study emphasized that traditional scheduling methods require significant administrative effort and are highly dependent on staff availability. In busy clinics, front desk personnel often struggle to manage walk-in patients while simultaneously handling phone calls for appointment bookings. This results in errors and delays in scheduling.

The researchers explored how web-based appointment systems improve clinic operations through automation and real-time scheduling updates. Patients are able to book appointments online, while clinic staff can monitor and adjust schedules efficiently. Automated reminders sent through SMS or email help ensure that patients are informed of their appointments in advance.

The findings of the study revealed that automated scheduling systems significantly reduced missed appointments and improved time management within clinics. Dental staff reported fewer scheduling conflicts and better control of daily appointment flows. Patients also experienced improved convenience and satisfaction due to easier booking processes.

Furthermore, the study highlighted that automated systems improved communication between patients and dental clinics. Notification features ensured that patients were updated regarding appointment changes, cancellations, or confirmations. However, the study noted challenges such as internet dependency and system adoption resistance among older patients.

The study concluded that automated appointment scheduling systems improve efficiency, reduce administrative workload, and enhance patient experience in dental clinics.

This foreign study is relevant to the present study because it supports the integration of automated appointment scheduling and notification features in a web-based dental clinic management system.

2.1.3 Integrated Dental Clinic Management Systems

A foreign study conducted in the United Kingdom examined the development and implementation of integrated dental clinic management systems that combine multiple clinic functions such as patient record management, billing processes, and appointment scheduling into a single centralized platform.

The study aimed to determine how system integration improves workflow efficiency, reduces redundancy, and enhances overall operational performance within dental healthcare facilities. The researchers explained that many dental clinics continue to rely on separate systems or manual processes for different administrative and clinical tasks, which often leads to fragmented data management and inefficiencies in daily operations.

The study emphasized that the use of multiple standalone systems within a dental clinic creates significant challenges in data consistency and workflow coordination. Staff members are often required to encode the same patient information repeatedly across different systems such as billing software, appointment logs, and patient record databases. This repetitive process not only consumes time but also increases the likelihood of human error, such as mismatched records, missing information, and incorrect patient details. These issues can negatively affect the quality of healthcare services and reduce the efficiency of clinic operations.
The researchers further explained that integrated dental clinic management systems address these challenges by consolidating all clinic-related data into a single centralized database. This allows dental staff, including dentists, receptionists, and administrators, to access patient records, billing history, appointment schedules, and treatment information through one unified interface. The integration of multiple modules into a single system improves data consistency, eliminates

redundancy, and ensures that all users are working with updated and accurate information in real time.

The findings of the study revealed that integrated systems significantly improved operational efficiency and reduced administrative workload within dental clinics. Clinics that implemented centralized systems experienced faster processing of patient transactions, improved accuracy in billing and financial reporting, and more organized patient record management. The researchers noted that automation of routine tasks such as appointment scheduling and record retrieval allowed staff to focus more on patient care rather than administrative duties.

Furthermore, the study found that system integration improved communication and coordination among clinic personnel. Since all data was stored in a centralized platform, dentists and administrative staff were able to share and access patient information more effectively. This improved decision-making processes, particularly in treatment planning and patient management, as all relevant information was readily available without delays caused by manual record searching or system switching.

The study also highlighted that integrated systems contributed to better financial management within dental clinics. By combining billing and patient record systems, clinics were able to track payments, generate invoices, and monitor financial transactions more accurately. This reduced the occurrence of billing discrepancies and improved transparency in financial operations. The researchers emphasized that accurate financial tracking is essential for maintaining the sustainability and profitability of healthcare institutions.

In addition, the study discussed the role of system usability and interface design in the successful implementation of integrated systems. According to the researchers, systems that were designed with user-friendly interfaces allowed dental staff to adapt more quickly and efficiently to the new digital environment. Ease of navigation, clear data presentation, and accessible features contributed to improved user satisfaction and reduced training time for employees.

Despite the positive outcomes, the study identified several challenges associated with implementing integrated dental clinic management systems. One of the main challenges was the initial cost of system development and deployment, which required significant financial investment in software, hardware, and technical support. In addition, some clinics faced difficulties during the transition phase, particularly in migrating existing manual records into the new digital system without data loss or corruption.

The researchers also noted that staff training and adaptation were critical factors in the success of system implementation. Employees needed adequate training to fully understand how to use the integrated system effectively. Without proper training, there was a risk of underutilization of system features, which could reduce the overall effectiveness of the technology. Resistance to change among staff members was also identified as a challenge in some clinics.

Overall, the study concluded that integrated dental clinic management systems significantly improve workflow efficiency, data accuracy, financial management, and communication within dental healthcare facilities. The researchers emphasized that centralized systems are essential for modernizing dental clinic operations and ensuring high-quality patient care through efficient and reliable information management.

This foreign study is highly relevant to the present study because it supports the development of a web-based dental clinic management system that integrates patient records, appointment scheduling, billing, and clinic operations into a single unified platform to improve efficiency, accuracy, and overall service delivery.

2.1.4 Data Security and Privacy in Healthcare Information Systems

A foreign study conducted in Australia focused on data security and privacy in healthcare information systems, particularly in dental clinic environments. The study aimed to assess how secure systems protect sensitive patient data and prevent unauthorized access.

The researchers explained that dental records contain confidential information such as medical history, diagnoses, and treatment plans, which must be protected under strict security measures. Traditional systems are vulnerable to data loss, theft, and unauthorized access.

The study explored how secure web-based systems use encryption, role-based access control, and authentication mechanisms to protect patient information. These technologies ensure that only authorized users such as dentists and clinic staff can access specific data.

The findings revealed that implementing strong security measures significantly reduced data breaches and improved patient trust in digital healthcare systems. Clinics using secure systems experienced better compliance with data protection regulations.

Furthermore, the study emphasized that security systems must be regularly updated to address new cyber threats. The researchers also noted that staff training is essential to ensure proper use of security features.

The study concluded that secure healthcare information systems are essential for protecting patient confidentiality and ensuring reliable clinic operations.

This foreign study is relevant to the present study because it supports the implementation of secure login systems and data protection features in a web-based dental clinic management system.

Furthermore, the study found that centralized systems enhanced decision-making processes because administrators could easily generate reports and analyze clinic performance. However, the study identified challenges such as system complexity and initial setup costs.

The study concluded that integrated dental management systems provide significant benefits in terms of efficiency, accuracy, and organizational productivity.

This foreign study is relevant to the present study because it supports the idea of developing a unified web-based system that manages dental records, appointments, billing, and clinic operations in one platform.

2.1.5 Cloud-Based Dental Clinic Management Systems

A foreign study conducted in Singapore examined the use of cloud-based dental clinic management systems to improve accessibility and data management efficiency. The study aimed to determine how cloud technology enhances the availability and reliability of patient records.

The researchers explained that traditional local storage systems limit access to patient data, especially when clinics operate in multiple locations. Cloud-based systems solve this problem by allowing authorized users to access data anytime and anywhere through internet connectivity.

The study explored how cloud systems improve collaboration among dental professionals by enabling real-time data sharing and updates. Patient records can be accessed instantly, improving diagnosis and treatment planning.

The findings revealed that cloud-based systems improved operational flexibility, reduced dependence on physical storage, and enhanced data backup and recovery processes. Clinics also experienced improved efficiency in managing large volumes of patient data.

Furthermore, the study emphasized that cloud systems provide better protection against data loss caused by hardware failure or system crashes. However, the study also identified concerns such as internet dependency and data privacy risks.

The study concluded that cloud-based dental systems significantly improve accessibility, efficiency, and data security in modern dental clinic operations.
This foreign study is relevant to the present study because it supports the integration of web-based and cloud-enabled features for secure and accessible dental record management.

2.1.6 Responsive Web Design in Healthcare Management

A foreign study conducted in Germany examined the importance of responsive web design in healthcare management systems, including applications used in dental clinics and other medical facilities. The study aimed to determine how system accessibility across different devices affects usability, user satisfaction, and overall user experience in web-based healthcare platforms. The researchers emphasized that as healthcare services become increasingly digital, the ability of systems to function effectively across desktops, laptops, tablets, and smartphones has become a critical requirement for modern healthcare delivery.

The researchers explained that healthcare staff and patients are now increasingly dependent on mobile devices such as smartphones and tablets to access online services. Dentists may need to check patient records, update treatment information, or view schedules while moving between consultation rooms, while patients often use mobile phones to book appointments or check clinic availability. Systems that are not optimized for mobile use often result in poor usability, distorted layouts, slow navigation, and limited accessibility, which can negatively affect user experience and reduce system effectiveness.

The study explored how responsive web design addresses these issues by allowing systems to automatically adjust their layout, interface components, and functionality based on the screen size and device being used. This ensures that content remains readable, navigation remains simple, and essential features remain accessible regardless of the device. The researchers highlighted that responsive design frameworks use flexible grids, scalable images, and adaptive layouts to maintain usability across different platforms without requiring separate mobile applications.

The findings of the study revealed that responsive systems significantly improved user satisfaction, accessibility, and user engagement. Patients reported that they were able to book appointments more easily and navigate clinic services without difficulty, even when using mobile devices. Likewise, clinic staff benefited from the ability to access patient information and manage schedules remotely, improving workflow efficiency and reducing delays in service delivery. The study also noted that improved accessibility contributed to better communication between patients and healthcare providers.

Furthermore, the study emphasized that responsive design plays a crucial role in improving the overall effectiveness of healthcare information systems. By ensuring that systems are adaptable to different devices, healthcare providers can extend access to services beyond traditional desktop environments. This flexibility is particularly important in emergency situations or during peak clinic hours when quick access to information is required.

The study also discussed the importance of user interface consistency in responsive systems. According to the researchers, maintaining a consistent design across devices helps users become familiar with system navigation more quickly, reducing the learning curve and minimizing user errors. This is especially important in healthcare environments where accuracy and speed of access can directly affect service quality.

Despite the positive findings, the study identified certain challenges associated with implementing responsive web design. These include increased development complexity, the need for thorough testing across multiple devices, and the requirement for continuous updates to ensure compatibility with new technologies. Developers also need to balance performance and design flexibility to ensure that system speed is not compromised.

Overall, the study concluded that responsive web design is an essential component of modern healthcare management systems. It improves usability, accessibility, and efficiency across different user groups, making healthcare services more convenient and effective.

This foreign study is highly relevant to the present research because it supports the development of a web-based dental clinic management system that is fully responsive, allowing both patients and dental staff to access system features efficiently across various devices, thereby improving service delivery and user experience.

2.1.7 Data Analytics in Dental Clinic Management

A foreign study conducted in South Korea focused on the application of data analytics in dental clinic management systems and other healthcare information systems. The study aimed to evaluate how data-driven insights contribute to improved decision-making, operational efficiency, and overall clinic performance. The researchers emphasized that modern dental clinics generate large volumes of data on a daily basis, including patient records, appointment histories, billing transactions, treatment procedures, and service utilization patterns. However, without proper analysis and interpretation, much of this data remains underutilized and does not contribute effectively to strategic planning or service improvement.

The researchers explained that many dental clinics rely on traditional reporting methods, which often involve manual compilation of data from different sources. This approach is time-consuming and prone to errors, making it difficult for administrators to obtain accurate and timely insights. As a result, important trends such as patient flow, peak service hours, and frequently performed procedures may go unnoticed, limiting the clinic's ability to optimize its operations.

The study explored how data analytics dashboards and reporting tools can address these limitations by providing real-time visualization of clinic performance indicators. These systems allow administrators to monitor key metrics such as patient volume, appointment frequency, treatment distribution, and revenue trends through graphical representations like charts and graphs. The integration of analytics tools into dental clinic systems enables faster interpretation of data and supports evidence-based decision-making.

The findings of the study revealed that clinics using data analytics systems were able to make more informed and strategic decisions regarding resource allocation, staff scheduling, and service improvement. Data visualization tools helped identify peak hours, allowing clinics to adjust staffing levels accordingly to reduce patient waiting times. Additionally, analysis of treatment trends enabled dental professionals to better understand common patient needs and improve service planning.

Furthermore, the study highlighted that analytics integration enhances overall clinic efficiency by providing administrators with a clearer understanding of operational performance. With access to accurate and organized data, decision-makers are able to identify inefficiencies in workflow processes and implement corrective actions more effectively. This leads to improved productivity, better patient management, and enhanced service quality.

The study also emphasized the importance of integrating user-friendly analytics interfaces within healthcare systems. According to the researchers, dashboards should be designed in a way that is easy to understand, even for non-technical users such as clinic staff and administrators. Simplified visualization tools improve accessibility and encourage regular use of data insights in daily operations.

Despite the benefits, the study identified challenges in implementing analytics systems, including the need for high-quality data input, proper system configuration, and staff training. Inaccurate or incomplete data can affect the reliability of analytics results, making data validation an important requirement for system effectiveness. Additionally, some clinics may face difficulties in adapting to data-driven decision-making practices due to limited technical knowledge.

Overall, the study concluded that data analytics plays a crucial role in improving dental clinic management systems by enhancing decision-making, operational efficiency, and service planning. The researchers emphasized that integrating analytics tools into healthcare systems allows clinics to transition from traditional administrative methods to more modern, data-driven approaches.

This foreign study is highly relevant to the present research because it supports the inclusion of reporting and analytics features in a web-based dental clinic management system, enabling clinic administrators to monitor operations effectively and make informed decisions based on real-time data.

2.1.8 Chatbot Systems in Healthcare

A foreign study focusing on the integration of chatbot systems in healthcare and dental clinic platforms examined how artificial intelligence-based communication tools enhance patient interaction, service accessibility, and administrative efficiency. The study aimed to evaluate the effectiveness of chatbot technology in handling common patient inquiries such as appointment availability, clinic services, dental procedures, treatment requirements, and follow-up schedules.

The researchers emphasized that dental clinics often experience a high volume of repetitive and routine questions from patients, which can significantly overload administrative staff and delay response times for more urgent concerns.
The researchers explained that manual response systems are often inefficient, particularly in clinics with limited personnel and high patient demand.

In many cases, clinic staff are required to manage front desk operations, phone inquiries, and patient scheduling simultaneously. This multitasking environment can lead to delayed responses, missed communications, and reduced service quality. As a result, patients may experience inconvenience when seeking quick information, which can negatively affect their overall satisfaction and engagement with clinic services.

The study explored how chatbot systems function as an automated communication solution that provides instant responses to user queries through pre-programmed instructions or artificial intelligence-based processing. These systems can be integrated into web-based dental platforms, allowing patients to interact with the clinic anytime without requiring direct human assistance.

Chatbots can guide users through appointment booking processes, provide clinic information, and answer frequently asked questions in real time.
The findings of the study revealed that chatbot integration significantly improved communication efficiency and reduced administrative workload within healthcare and dental clinic environments. Clinics that implemented chatbot systems experienced faster response times and improved accessibility of information for patients. Administrative staff were able to focus more on essential clinical and operational tasks rather than repeatedly answering the same inquiries.

Furthermore, the study highlighted that chatbot systems contributed to improved patient engagement and satisfaction by providing immediate assistance and reducing waiting times for responses. Patients found it more convenient to access clinic information and schedule appointments through automated systems. However, the study also identified certain limitations, including the inability of chatbots to fully understand complex medical concerns or respond appropriately to emergency situations, which still require human intervention.

The study also emphasized the importance of properly designing chatbot systems to ensure accuracy, clarity, and user-friendly interaction. Poorly designed systems may lead to misunderstanding or incorrect information delivery, which could affect patient trust and service quality. Therefore, continuous improvement and monitoring of chatbot performance are necessary for effective implementation.

Overall, the study concluded that chatbot systems enhance patient communication, accessibility, and administrative efficiency in healthcare environments by automating routine interactions and improving service responsiveness. The researchers emphasized that while chatbots cannot replace human professionals, they serve as valuable support tools in modern healthcare systems.

This foreign study is highly relevant to the present research because it supports the inclusion of automated communication features, such as chatbots or messaging assistance, in a web-based dental clinic management system to improve patient interaction, accessibility, and overall service efficiency.

2.1.9 Automated Billing Systems

A foreign study on automated billing systems in healthcare and dental clinic environments examined how digital financial processing improves accuracy, transparency, and overall operational efficiency within clinical settings. The study aimed to analyze the effectiveness of replacing traditional manual billing procedures with computerized platforms that automatically generate invoices, calculate service charges, and track patient payment histories. The researchers emphasized that efficient financial management is a critical component of healthcare operations, as it directly affects clinic sustainability, service delivery, and patient satisfaction.

The researchers explained that traditional billing systems are highly prone to human errors such as incorrect computation of charges, missing or duplicated records, and delays in updating payment statuses. In busy dental clinics, these issues are further compounded by high patient volume and the simultaneous handling of multiple administrative tasks. As a result, inconsistencies in financial records often occur, leading to confusion, disputes, and additional workload for administrative staff.

The study explored how automated billing systems address these challenges by integrating financial modules with patient records and service management systems. Through this integration, billing information is automatically generated based on recorded treatments and services rendered, ensuring accuracy and consistency in financial transactions. Clinic staff are able to access real-time payment information, generate invoices instantly, and monitor outstanding balances without manually reviewing paper-based records or separate spreadsheets.

The findings of the study revealed that automated billing systems significantly reduced errors in financial computation and improved the accuracy of billing reports. Clinics that implemented digital billing solutions experienced faster transaction processing, reduced administrative workload, and improved transparency in financial management. The automation of billing tasks also allowed staff to allocate more time to patient-related services rather than manual accounting procedures.

Furthermore, the study highlighted that automated billing systems contribute to better financial control and monitoring within healthcare institutions. Managers were able to track revenue streams, monitor payment compliance, and identify financial discrepancies more efficiently. This improved oversight helped clinics maintain better financial stability and operational planning.

The study also emphasized the importance of proper system implementation, including staff training and adequate technical infrastructure. Without proper training, users may struggle to fully utilize system features, which could reduce the effectiveness of automation. Additionally, the initial cost of system deployment and integration was identified as a potential challenge for smaller clinics.

Overall, the study concluded that automated billing systems significantly improve financial efficiency, accuracy, and transparency in healthcare and dental clinic environments. The researchers emphasized that such systems are essential for modernizing clinic operations and reducing administrative burdens.

This foreign study is highly relevant to the present research because it supports the integration of automated billing and payment management features in a web-based dental clinic management system to enhance financial accuracy, efficiency, and operational effectiveness.

2.1.10 Authentication and Access Control

A foreign study focusing on authentication and access control in healthcare information systems examined how security frameworks play a crucial role in protecting sensitive patient data within digital healthcare and dental clinic environments. The study aimed to evaluate the effectiveness of role-based access control (RBAC), multi-user authentication mechanisms, and other security protocols in preventing unauthorized access to medical and dental records. The researchers emphasized that as healthcare systems become increasingly digitized, the importance of securing patient information becomes more critical due to the rising risks of data breaches, cyberattacks, and unauthorized system access.

The researchers explained that healthcare systems handle highly confidential information, including patient personal details, medical histories, diagnostic records, and treatment plans. Because of the sensitive nature of this data, it must only be accessible to authorized personnel such as dentists, dental assistants, administrative staff, and system administrators. Without proper access control mechanisms in place, systems become vulnerable to misuse, accidental data exposure, and intentional data manipulation, which can compromise both patient privacy and clinical integrity.

The study explored how authentication technologies such as password encryption, multi-factor authentication, and role-based permission settings enhance overall system security. Through these mechanisms, each user is assigned specific roles with corresponding access privileges, ensuring that individuals can only view or modify data relevant to their responsibilities. For example, dentists may have full access to patient treatment records, while administrative staff may only access scheduling and billing modules. This structured access system helps maintain data confidentiality and reduces the risk of unauthorized data handling.

The findings of the study revealed that secure authentication systems significantly reduced incidents of unauthorized access and improved accountability among system users. The researchers observed that when users are required to verify their identity before accessing sensitive data, system transparency and responsibility are strengthened. Each action performed within the system can be traced back to a specific user, which further enhances monitoring and auditing capabilities.

Furthermore, the study emphasized that maintaining strong system security requires continuous updates and improvements to address evolving cyber threats. Healthcare systems are frequent targets of malicious attacks, and outdated security protocols may expose clinics to potential risks. Therefore, regular system maintenance, security patching, and user training are essential to ensure long-term data protection and system reliability.

The study also highlighted that implementing strong authentication systems contributes not only to data protection but also to patient trust and confidence in digital healthcare services. Patients are more likely to engage with clinics that demonstrate a commitment to safeguarding their personal and medical information.

Overall, the study concluded that robust authentication and access control systems are essential components of modern healthcare information systems. These mechanisms ensure data integrity, confidentiality, and accountability while supporting efficient and secure clinic operations.

This foreign study is highly relevant to the present research because it supports the implementation of secure login systems, role-based access control, and data protection features in a web-based dental clinic management system to safeguard patient records and ensure secure system usage.

2.1.11 Emergency Appointment Prioritization

A foreign study conducted on emergency appointment prioritization systems in healthcare and dental clinic environments examined how scheduling flexibility improves patient care, particularly for urgent and critical dental cases. The study aimed to evaluate how emergency queue management systems enhance the ability of clinics to prioritize patients with severe or time-sensitive dental conditions while maintaining efficient scheduling for regular appointments. The researchers emphasized that effective appointment management is a key factor in ensuring timely healthcare delivery and improving patient outcomes.

The researchers explained that traditional appointment scheduling systems often treat all patient bookings equally, regardless of the severity of their conditions. In dental clinics, this approach can create significant challenges, especially when patients with urgent needs such as severe tooth infections, dental trauma, or extreme pain are required to wait for available time slots. These delays can worsen patient conditions and negatively affect overall patient satisfaction and trust in clinic services.

The study explored how emergency prioritization systems address these limitations by allowing clinics to categorize and rank appointments based on urgency levels. Through automated scheduling logic or administrative controls, the system can adjust appointment queues in real time, ensuring that emergency cases are given priority over routine consultations. This structured approach allows clinics to maintain an organized schedule while still being responsive to urgent patient needs.

The findings of the study revealed that clinics using emergency prioritization systems experienced significantly improved response times for critical dental cases. Patients requiring urgent care were attended to more quickly, reducing the risk of complications and improving treatment outcomes. The researchers also noted that staff were able to manage appointment schedules more efficiently, as the system reduced the need for manual rearrangement and decision-making during high-demand periods.

Furthermore, the study highlighted that prioritization systems contributed to improved patient satisfaction and trust in healthcare services. Patients perceived the clinic as more responsive and organized, especially when emergency cases were handled promptly. This improvement in service quality also enhanced the overall reputation of healthcare providers.

The study also emphasized the importance of clearly defined urgency criteria and system rules to ensure fairness and consistency in appointment prioritization. Without proper guidelines, there is a risk of misclassification of cases, which could affect scheduling accuracy and service fairness.

Overall, the study concluded that emergency appointment prioritization systems significantly improve healthcare responsiveness, operational efficiency, and service quality in dental and medical clinic environments. The researchers stressed that such systems are essential for modern healthcare scheduling, where both efficiency and patient care must be balanced effectively.

This foreign study is highly relevant to the present research because it supports the inclusion of emergency scheduling and appointment prioritization features in a web-based dental clinic management system to ensure timely handling of urgent cases and improve overall clinic service efficiency.

2.1.12 Appointment Calendar Systems

A foreign study on appointment calendar systems in healthcare environments examined how digital scheduling tools improve organizational efficiency and reduce booking conflicts in clinical operations. The study aimed to analyze the impact of real-time calendar synchronization on appointment management, workflow coordination, and overall service delivery within healthcare facilities, including dental clinics. The researchers emphasized that efficient scheduling is a fundamental aspect of clinic management because it directly affects patient flow, waiting time, and staff productivity.

The researchers explained that traditional manual scheduling methods, such as paper-based logs or standalone spreadsheets, often result in multiple operational issues. These include overlapping appointments, missed bookings, double entries, and inefficient time allocation. In busy healthcare environments, these problems become more frequent due to the high volume of patients and continuous changes in schedules, which can lead to confusion among staff and dissatisfaction among patients.

The study explored how digital appointment calendar systems address these challenges by providing real-time visibility of available time slots and synchronized scheduling updates. Through these systems, both clinic staff and patients can view updated appointment schedules instantly. Any changes made by a user, such as booking, rescheduling, or cancellation, are automatically reflected across the system, ensuring that all users are working with accurate and up-to-date information.

The findings of the study revealed that the implementation of digital calendar systems significantly improved scheduling accuracy and reduced appointment conflicts. Clinics experienced fewer instances of double booking and improved coordination among staff members. The researchers also noted that the automation of scheduling processes helped streamline daily operations, allowing clinic personnel to focus more on patient care rather than manual appointment management.

Furthermore, the study highlighted that real-time calendar synchronization improved overall workflow efficiency within healthcare facilities. Staff productivity increased as the time spent on manually updating schedules and resolving booking conflicts was significantly reduced. Patients also benefited from a more organized and predictable appointment system, which contributed to better service experience and reduced waiting times.

The study also emphasized the importance of system reliability and accessibility in calendar-based scheduling systems. Since appointment scheduling is a continuous process, any system downtime or synchronization delay can negatively impact clinic operations. Therefore, stable system performance and reliable data updates are essential for maintaining efficiency.

Overall, the study concluded that digital appointment calendar systems are essential tools for modern healthcare environments, as they enhance scheduling accuracy, improve workflow coordination, and support efficient clinic management.

This foreign study is highly relevant to the present research because it supports the integration of a real-time appointment calendar feature in a web-based dental clinic management system to improve scheduling efficiency and reduce booking conflicts.

2.1.13 Digital Treatment History Systems

A foreign study examining digital treatment history systems in dental clinics focused on improving long-term patient care tracking and enhancing the continuity of dental services. The study aimed to determine how electronic treatment records contribute to better clinical decision-making, more accurate diagnosis, and improved monitoring of patient progress over time.

The researchers emphasized that maintaining complete and accessible treatment histories is essential in dental practice, as it allows healthcare providers to understand previous interventions and plan appropriate future treatments.

The researchers explained that traditional paper-based treatment history systems present several limitations in clinical environments. Physical records are often incomplete, misplaced, or difficult to retrieve, especially in clinics that handle a large number of patients. Over time, paper documents may also deteriorate or become disorganized, making it challenging for dentists to access reliable patient information when needed. These issues can delay treatment processes and reduce the quality of patient care.

The study explored how digital treatment history systems address these challenges by storing patient records in structured electronic databases. Through these systems, dentists are able to encode, update, and retrieve detailed treatment information, including previous procedures, diagnoses, prescriptions, and follow-up appointments. The digital format allows for quick and efficient access to patient history, which improves workflow and reduces the time spent searching for physical files.

The findings of the study revealed that the use of digital treatment history systems significantly improved treatment accuracy and enhanced clinical decision-making. Dentists were able to review comprehensive patient histories and identify patterns in oral health conditions, which supported more informed treatment planning. This improved continuity of care ensured that patients received consistent and appropriate dental treatments based on their medical history.

Furthermore, the study highlighted that digital systems enhanced patient monitoring by allowing healthcare providers to track treatment progress over extended periods. Dentists could easily compare past and current records to assess improvements or identify recurring dental issues. This capability contributed to more proactive and preventive dental care, ultimately improving patient outcomes.

The study also emphasized the importance of data organization and system reliability in maintaining effective digital treatment records. Proper database structuring ensures that information is stored in a logical and accessible manner, reducing the risk of data loss or duplication. In addition, secure access controls are necessary to protect sensitive patient information from unauthorized access.
Overall, the study concluded that digital treatment history systems significantly improve healthcare quality, clinical efficiency, and patient monitoring in dental clinic environments. The researchers highlighted that transitioning from manual to electronic records is essential for modernizing dental healthcare services and ensuring better continuity of care.

This foreign study is highly relevant to the present research because it supports the integration of a digital treatment history module in a web-based dental clinic management system to enhance patient record accuracy, improve clinical decision-making, and ensure continuous monitoring of dental treatments.

2.1.14 Automated Report Generation Systems

A foreign study on automated report generation systems in healthcare examined how digital reporting tools improve administrative efficiency and support data-driven decision-making in clinical environments. The study aimed to evaluate the effectiveness of system-generated reports in managing clinic operations such as patient records, appointment summaries, billing transactions, and overall performance monitoring. The researchers emphasized that accurate and timely reporting is essential in healthcare management, as it supports planning, evaluation, and service improvement.

The researchers explained that manual report preparation is often time-consuming and prone to inconsistencies, especially in busy healthcare and dental clinic environments. Administrative staff are typically required to compile data from multiple sources, such as appointment logs, patient records, and financial documents. This process not only increases workload but also introduces the risk of human error, missing information, and delayed reporting, which can negatively affect clinic decision-making.

The study explored how automated reporting systems address these challenges by generating structured and real-time reports directly from centralized databases. These systems are capable of compiling information automatically based on system inputs, ensuring that reports remain accurate, consistent, and up to date. Reports can include patient statistics, appointment trends, billing summaries, and clinic performance indicators.

The findings of the study revealed that automated report generation significantly improved accuracy and reduced the time required to produce administrative documents. Clinics experienced faster access to essential information, allowing managers and administrators to make timely and informed decisions. The researchers also noted that automation reduced the workload of administrative staff, enabling them to focus on more critical operational tasks.

Furthermore, the study highlighted that automated reporting enhances transparency and accountability within healthcare organizations. Since data is directly extracted from system records, the likelihood of manipulation or inconsistency is reduced. This improves trust in the accuracy of clinic reports and supports better organizational governance.

The study concluded that automated reporting systems are essential for improving efficiency, accuracy, and decision-making in healthcare management. The researchers emphasized that integrating reporting tools into clinic systems enhances overall operational performance.

This foreign study is highly relevant to the present research because it supports the inclusion of an automated report generation feature in a web-based dental clinic management system to improve administrative efficiency and decision-making.

2.1.15 Activity Logging Systems

A foreign study on activity logging systems in healthcare examined how system monitoring tools improve security, accountability, and data integrity within digital healthcare environments. The study aimed to analyze how user activity tracking helps detect unauthorized actions, monitor system usage, and ensure proper handling of sensitive patient data. The researchers emphasized that maintaining a secure and transparent system environment is essential in healthcare information systems due to the sensitive nature of patient records.

The researchers explained that without proper logging mechanisms, it becomes difficult to track changes made to patient information or system data. In such cases, identifying the source of errors, unauthorized modifications, or system misuse can be challenging. This lack of traceability may compromise data integrity and reduce trust in the system.

The study explored how activity logging systems record detailed user actions, including login attempts, data creation, data modification, deletion activities, and overall system usage. These logs are stored in a secure database and can be reviewed by administrators to monitor system behavior and detect suspicious activities.

The findings of the study revealed that activity logging significantly improved transparency and security monitoring within healthcare systems. The researchers observed that clinics were able to quickly identify unusual or unauthorized activities, allowing them to take corrective actions promptly. Logging systems also improved accountability, as every system action could be traced back to a specific user.

Furthermore, the study highlighted that activity logs support error tracking and system maintenance by providing detailed records of system interactions. This allows technical teams to diagnose issues more efficiently and ensure system stability. The presence of logs also encourages responsible system usage among staff, as users are aware that their actions are being recorded.

The study concluded that activity logging systems are essential for maintaining system integrity, security, and accountability in healthcare environments. The researchers emphasized that logging mechanisms are a critical component of modern digital healthcare systems.

This foreign study is highly relevant to the present research because it supports the inclusion of activity logs in a web-based dental clinic management system to ensure security, transparency, and accountability.

2.1.16 Patient Review Systems

A foreign study on patient review systems in healthcare examined how feedback mechanisms contribute to improving service quality and patient satisfaction. The study aimed to evaluate the impact of patient ratings, comments, and feedback systems on the performance of healthcare and dental clinics. The researchers emphasized that patient feedback is a valuable source of information for identifying strengths and weaknesses in healthcare service delivery.

The researchers explained that without structured feedback systems, healthcare providers may struggle to understand patient experiences and identify areas that require improvement. Traditional feedback methods, such as verbal comments or paper-based forms, are often inconsistent and difficult to analyze systematically.
The study explored how digital review systems allow patients to submit feedback, rate services, and share their experiences through web-based platforms. These systems collect and organize feedback data in a structured format, making it easier for clinic administrators to evaluate service quality and patient satisfaction levels.

The findings of the study revealed that clinics using digital feedback systems experienced improved service quality and higher levels of patient satisfaction. The researchers noted that feedback helped healthcare providers identify service gaps and implement necessary improvements in clinic operations. Patients also felt more engaged and valued when their opinions were acknowledged and considered.

Furthermore, the study highlighted that patient review systems encourage transparency and continuous improvement within healthcare organizations. By regularly analyzing feedback data, clinics can adapt their services to better meet patient needs and expectations. This contributes to a more patient-centered approach in healthcare delivery.

The study concluded that feedback systems play a crucial role in improving healthcare service quality, communication, and patient satisfaction. The researchers emphasized that incorporating patient feedback mechanisms is essential for modern healthcare systems.

This foreign study is highly relevant to the present research because it supports the inclusion of a patient feedback or review feature in a web-based dental clinic management system to improve service quality and patient engagement.

2.1.17 Centralized Healthcare Databases

A foreign study on centralized healthcare databases examined how unified data storage systems improve efficiency, accuracy, and accessibility of patient information in healthcare environments. The study aimed to evaluate the benefits of integrating all patient-related data into a single centralized system. The researchers emphasized that fragmented data storage systems often lead to inefficiencies in healthcare delivery.

The researchers explained that in many healthcare institutions, patient data is stored across multiple systems or physical records, which can cause delays in retrieving information and increase the risk of data inconsistency. This fragmentation makes it difficult for healthcare providers to obtain a complete view of patient history.

The study explored how centralized databases consolidate all patient information, including medical history, appointments, billing records, and treatment details, into one unified system. This integration allows healthcare providers to access complete and updated patient data in real time.

The findings of the study revealed that centralized database systems significantly improved data retrieval speed, accuracy, and accessibility. Healthcare professionals were able to access complete patient profiles quickly, which enhanced clinical decision-making and improved service efficiency. The researchers also noted that centralized systems reduced duplication of records and minimized inconsistencies.

Furthermore, the study highlighted that centralized databases improve coordination among healthcare staff by ensuring that all users access the same updated information. This reduces communication gaps and enhances collaboration in patient care management.

The study concluded that unified healthcare databases are essential for improving operational efficiency, data accuracy, and service delivery in healthcare environments.

This foreign study is highly relevant to the present research because it supports the integration of a centralized database structure in a web-based dental clinic management system to improve data consistency and accessibility.

2.1.18 Equipment Tracking System

A foreign study on equipment tracking systems in healthcare examined how digital inventory and asset management tools improve resource utilization and operational efficiency in clinical environments. The study aimed to evaluate how tracking systems enhance the monitoring, maintenance, and usage of medical and dental equipment.

The study found that many healthcare facilities face challenges in managing equipment due to lack of proper tracking systems, leading to issues such as misplaced tools, underutilized resources, and delayed maintenance schedules. These problems can affect service delivery and operational efficiency.

The study explored how equipment tracking systems use digital logs and monitoring tools to record equipment usage, location, and maintenance history. This allows clinics to maintain better control over their resources and ensure that equipment is properly accounted for at all times.

The findings revealed that tracking systems improved equipment monitoring and reduced instances of loss or misplacement. Clinics were able to schedule maintenance more effectively, ensuring that equipment remained in good working condition and available when needed.

The study concluded that equipment tracking systems enhance operational efficiency, resource management, and service reliability in healthcare environments.

This foreign study is relevant to the present research because it supports the inclusion of equipment tracking or resource management features in a web-based dental clinic management system.

2.1.19 Cloud Backup Systems

A foreign study on cloud backup systems in healthcare examined how cloud storage technologies improve data security, reliability, and disaster recovery capabilities. The study aimed to evaluate how cloud-based backup systems protect sensitive patient information from data loss or system failure.

The researchers explained that traditional local storage systems are vulnerable to hardware failure, accidental deletion, and physical damage, which can result in permanent data loss. These risks are particularly critical in healthcare environments where patient records are essential for continuous care.

The study explored how cloud backup systems automatically store copies of healthcare data in remote servers, ensuring that information can be restored in case of system failure or data corruption. This provides an additional layer of security and reliability for healthcare institutions.

The findings revealed that cloud-based systems significantly improved data protection and recovery efficiency. Healthcare providers were able to restore lost data quickly, minimizing disruptions in clinic operations. The researchers also noted that cloud systems enhanced data accessibility and scalability.

The study concluded that cloud backup systems are essential for ensuring data safety, continuity of service, and operational resilience in healthcare environments.
This foreign study is highly relevant to the present research because it supports the integration of cloud backup or secure data storage mechanisms in a web-based dental clinic management system.

2.1.20 Electronic Treatment Notes

A foreign study on electronic treatment notes examined how digital documentation systems improve accuracy, efficiency, and accessibility of clinical records in healthcare environments. The study aimed to evaluate the effectiveness of replacing handwritten treatment notes with electronic documentation systems.

The researchers explained that manual note-taking is often prone to errors, incomplete entries, and difficulties in retrieval, especially in busy clinical settings. These limitations can affect the continuity and quality of patient care.
The study explored how electronic treatment note systems allow healthcare providers to record patient information digitally, ensuring structured, complete, and easily accessible documentation.

The findings revealed that electronic documentation reduced errors and improved the accuracy of patient records. Healthcare providers were able to retrieve treatment notes quickly, improving clinical decision-making and patient monitoring.

The study concluded that electronic treatment notes enhance efficiency and accuracy in healthcare documentation.

This foreign study is relevant to the present research because it supports the inclusion of digital treatment note features in a web-based dental clinic management system.

2.1.21 Profile Management Systems

A foreign study on profile management systems examined how personalized user accounts improve usability, accessibility, and engagement in healthcare platforms. The study aimed to evaluate how user profile systems enhance interaction between patients and healthcare providers.

The researchers explained that profile management allows users to store and update personal information, medical history, and preferences in a structured digital format. This improves personalization and system usability.

The study found that profile systems enhanced user experience by allowing patients to manage their own information and access relevant services more efficiently. Healthcare providers also benefited from having organized patient data readily available.

The study concluded that profile management systems improve system usability, personalization, and patient engagement.

This foreign study is relevant to the present research because it supports the inclusion of user profile management features in a web-based dental clinic management system.

2.1.22 Notification Systems

A foreign study on notification systems examined how automated alerts improve communication efficiency and reduce missed appointments in healthcare environments. The study aimed to evaluate the effectiveness of system-generated notifications in patient engagement and clinic scheduling.

The researchers explained that manual reminders are often inconsistent and may be overlooked, leading to missed appointments and scheduling inefficiencies.
The study explored how automated notification systems send alerts through SMS, email, or system messages to remind patients of upcoming appointments or updates.

The findings revealed that notification systems significantly reduced missed appointments and improved coordination between patients and healthcare providers. Clinics also experienced better scheduling efficiency and reduced administrative workload.

The study concluded that automated notifications enhance communication, efficiency, and patient compliance.
This foreign study is relevant to the present research because it supports the inclusion of automated reminder and notification features in a web-based dental clinic management system.

2.1.23 Web-Based Dental Clinic Management Systems

A foreign study on web-based dental clinic management systems examined the overall impact of digital transformation in healthcare operations. The study aimed to evaluate how integrated systems improve efficiency, accuracy, and patient satisfaction in dental clinic environments.

The researchers found that web-based systems significantly reduce manual workload by automating key processes such as patient record management, appointment scheduling, billing, and reporting. These systems also improve data accuracy by centralizing information into a single platform.

The study further revealed that integrated dental clinic systems enhance operational efficiency by streamlining workflows and improving coordination among staff members. Patients also benefit from faster service delivery, easier appointment booking, and improved access to clinic services.

The study concluded that web-based dental clinic management systems are essential for modern healthcare operations, as they improve service delivery, operational efficiency, and overall patient satisfaction.

This foreign study is highly relevant to the present research because it supports the development of a comprehensive web-based dental clinic management system that integrates all core clinic functions into a unified and efficient platform.

2.2 Summary and Comparison of Local Studies

The reviewed local studies show that web-based dental clinic management systems are effective in improving healthcare operations in the Philippine setting. These systems enhance patient record management, appointment scheduling, data accessibility, and clinic workflow efficiency. They also reduce reliance on manual processes such as paper-based records and handwritten appointment logs, which often lead to errors, delays, and inefficiencies in dental clinic operations.

Most of the reviewed local studies emphasize the importance of digitalization in improving dental clinic services. Features such as centralized patient records, SMS notifications, electronic medical records, and automated scheduling systems help improve communication between patients and dental staff. These systems also enhance data accuracy, reduce administrative workload, and improve patient care efficiency.

However, the reviewed local studies generally focus on individual or partial functionalities such as SMS-based scheduling, standalone record systems, or limited clinic management tools. They often lack full system integration where all clinic processes—such as patient records, appointment scheduling, treatment history, billing, dental charts, and reporting—are combined into a single unified platform.

In contrast, the present study focuses on the development of a fully integrated web-based dental clinic management system that combines multiple clinic operations into one centralized system. The proposed system includes patient record management, automated appointment scheduling, treatment history tracking, dental chart integration, notifications, reporting features, and secure user authentication.

Furthermore, the proposed system improves existing local studies by providing real-time updates, better accessibility, and a responsive web-based interface accessible to both patients and clinic staff. Unlike previous systems that focus only on specific functions, the present study aims to deliver a complete digital solution that improves both administrative and clinical operations.

Overall, the present study builds upon the strengths of existing local systems while addressing their limitations by offering a more comprehensive, secure, and efficient dental clinic management system designed to improve healthcare delivery, patient experience, and clinic productivity.

Web-Based Dental Clinic Management Systems

Web-based dental clinic management systems are computerized solutions designed to manage patient records, appointment scheduling, treatment history, billing, and other clinic operations through a centralized digital platform. These systems help dental clinics improve data accuracy, reduce manual paperwork, and enhance overall operational efficiency. They also allow both patients and dental staff to access clinic information in real time through web-based interfaces.

According to Ho et al. (2024), web-based dental clinic management systems improve clinic productivity by integrating patient records, appointment scheduling, and administrative functions into a single platform. The study emphasizes that digitizing clinic operations reduces manual workload and improves service delivery efficiency. This supports the present study because the proposed system aims to integrate multiple dental clinic processes into a unified web-based system that enhances workflow efficiency and patient service management.

Technologies Used

The proposed system is a web-based application that utilizes modern technologies for front-end interface development, server-side processing, and database management. Web-based systems are commonly used in healthcare environments, including dental clinics, to support efficient management of patient records, appointment scheduling, billing, and other administrative and clinical processes.

These technologies allow dental clinic systems to operate in a centralized digital environment where data can be accessed and updated in real time. Through automation, tasks such as patient registration, appointment handling, and record management become more efficient, reducing the need for manual processes and paper-based documentation.

The findings of Sidek and Martins (2017) revealed that the implementation of electronic health record systems in dental clinics significantly improves operational efficiency and data management processes. The system reduces manual workload in handling patient information, resulting in faster and more reliable clinic operations. Through digital automation, patient data becomes more accurate and accessible, allowing dental staff to monitor records effectively and minimize documentation errors. The researchers also noted that integrated systems improve communication and coordination among clinic personnel, contributing to smoother workflow and better service delivery.

In addition, web-based dental clinic systems allow centralized handling of core processes such as patient record management, appointment scheduling, treatment history tracking, and billing management. By integrating these functions into a single platform, clinics can optimize operational efficiency and improve data consistency across all modules.

Furthermore, these systems enhance patient service delivery by enabling features such as automated appointment reminders, real-time schedule updates, and secure access to dental records. These improvements help reduce missed appointments, improve clinic organization, and support better patient care management.

The researchers concluded that web-based healthcare information systems play a vital role in improving clinic performance and operational efficiency. The study emphasizes that automated systems are more reliable than manual methods because they reduce human error, improve data accessibility, and support faster decision-making. Moreover, these systems highlight the importance of developing secure, scalable, and user-friendly platforms that address the operational needs of modern dental clinics.

This discussion is relevant to the present study because it demonstrates how web-based systems improve dental clinic operations through automation, centralized data management, and improved workflow efficiency. It supports the idea that a computerized dental clinic management system can simplify administrative processes, reduce manual errors, and enhance overall healthcare service delivery.

2.3 Related Literature

A web-based dental clinic management system is a software application that operates through a web browser and serves as a centralized platform for managing various clinic operations. These systems are designed to improve the efficiency of dental practices by automating administrative tasks and facilitating better management of patient information.

One of the primary functions of a dental clinic management system is patient record management. Patient records contain essential information such as personal details, medical history, treatment records, prescriptions, and dental charts. Electronic storage of these records enables faster retrieval and allows dentists and staff to view patient records instantly, improving consultation efficiency and supporting accurate treatment decisions.

Appointment management is another important feature discussed in the literature. Online appointment scheduling allows patients to select available dates and times based on the dentist's schedule. The system automatically records appointments and provides notifications to both patients and administrators, reducing scheduling conflicts and improving clinic organization.

Billing and payment management functionalities are also essential components of dental clinic systems. These features help clinics generate invoices, monitor payments, track outstanding balances, and maintain financial records accurately. Automated billing processes reduce manual errors and improve financial transparency.

Modern dental clinic systems often incorporate advanced features such as appointment calendars, emergency appointment prioritization, treatment history tracking, X-ray image uploads, equipment monitoring, analytics reporting dashboards, chatbot integration, and automated reminders. Analytics functions allow administrators to view performance metrics such as patient volume, revenue trends, appointment frequency, and service utilization.

The literature further emphasizes the importance of responsive web design in healthcare applications. Responsive systems automatically adapt to different screen sizes, ensuring accessibility across smartphones, tablets, laptops, and desktop computers.Security is another major topic in related literature.

Healthcare systems must implement robust security measures to protect confidential patient information. Common security practices include password encryption, secure authentication, role-based access control, data backups, and audit logs.

Additionally, user management features allow administrators to assign roles and permissions to different users, such as dentists, staff members, and patients. Activity logs provide records of user actions, promoting accountability and system transparency, especially when accessing or modifying patient records.

Overall, the literature demonstrates that web-based dental clinic management systems provide comprehensive solutions for managing clinical operations while improving efficiency, accessibility, secure patient record viewing, communication, and data-driven decision-making through analytics.

Dental chart management systems provide specialized tools for documenting oral health conditions. Digital charts allow dentists to identify affected teeth, record treatment plans, and update information electronically. Compared to paper-based charts, digital systems improve accuracy and accessibility.

Analytics and reporting systems have become essential components of modern healthcare management. Reports generated by these systems provide valuable insights into operational performance. Common reports include patient registration summaries, appointment statistics, billing reports, treatment frequencies, and revenue analyses. Administrators use these reports to evaluate clinic efficiency and support decision-making.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign studies, local studies, and related literature collectively highlight the significant role of web-based management systems in improving healthcare and dental clinic operations. These studies consistently emphasize the benefits of digital solutions in managing patient records, including secure storage and real-time viewing of patient records, appointment scheduling, billing processes, treatment documentation, and communication between healthcare providers and patients.

Foreign studies demonstrated that web-based healthcare systems improve operational efficiency through automation, secure information management, and advanced analytical capabilities. They also emphasized the importance of electronic medical records, appointment scheduling systems, notification services, data analytics dashboards, and responsive web technologies in delivering quality healthcare services.

Similarly, local studies confirmed the relevance of these technologies in the Philippine healthcare setting. The findings emphasized the need for efficient patient management, secure record storage, appointment monitoring, responsive interfaces, dental chart integration, patient record viewing functionality, analytics-based dashboards, and patient feedback systems. Local researchers also stressed the importance of preventing duplicate patient records and maintaining accurate appointment histories.

Based on the reviewed studies and literature, the proposed Dental Clinic Management System aims to incorporate various advanced functionalities, including appointment scheduling and monitoring, patient and dental record integration with secure viewing access, automated notifications, dashboard analytics, dental chart management, billing management, emergency appointment handling, service management, user management, patient review systems, and responsive web design.

Furthermore, the proposed system prioritizes data security, confidentiality, and data-driven decision-making through secure authentication, protected storage of patient information, and integrated analytics reporting. By integrating these features, the system seeks to improve operational efficiency, enhance patient satisfaction, reduce manual processes, and provide a reliable platform for managing and viewing dental clinic records.

The synthesis of the reviewed literature demonstrates that the proposed system aligns with current technological trends and healthcare management practices. Therefore, the implementation of the system is expected to contribute significantly to the modernization of dental clinic operations and the improvement of healthcare service delivery through improved patient record accessibility and analytics-based management.

2.5 Technological Framework

To ensure the proposed system is built on robust, scalable, and industry-standard foundations, this study is supported by a comprehensive technological framework. This framework details the specific software, platforms, and methodologies that will be utilized to address the identified problems of the dental clinic.

2.5.1 Python and the Flask Micro-Framework

The backend of the proposed system will be developed using Python, paired with the Flask web framework. Python is renowned for its readability, extensive library support, and strong community backing, making it an ideal choice for rapid and secure web development. Flask is a lightweight, modular micro-framework that provides the essential tools for web development (such as routing, request handling, and templating) without imposing unnecessary bloat. This modularity is crucial for a dental clinic system, as it allows the researchers to build specific, isolated modules (e.g., billing, scheduling) that can be developed, tested, and maintained independently. Furthermore, Flask's compatibility with various database ORMs (Object-Relational Mappers) and its strong security features (such as built-in protection against Cross-Site Request Forgery) make it highly suitable for handling sensitive healthcare data.

2.5.2 Google Firebase and Cloud Firestore

For database management and real-time synchronization, the system will utilize Google Firebase, specifically the Cloud Firestore NoSQL database. Unlike traditional relational databases that require complex schema migrations, Firestore's document-oriented structure allows for flexible and scalable data storage. This is particularly beneficial for a dental clinic, where patient records may vary in structure (e.g., some patients may have extensive treatment histories and X-ray attachments, while others may only have basic demographic data). Furthermore, Firebase provides built-in, enterprise-grade security through Firebase Authentication, ensuring that user passwords are securely hashed and that session management is handled safely. The real-time capabilities of Firestore mean that when a receptionist books an appointment, the dentist's dashboard updates instantaneously without requiring a page refresh, thereby eliminating the risk of double-booking and improving overall clinic workflow.

2.5.3 Agile Software Development Methodology

The development of the system will strictly follow the Agile methodology, specifically utilizing Scrum frameworks. Agile is an iterative approach to software development that emphasizes flexibility, continuous improvement, and frequent delivery of functional software increments. Given the dynamic nature of healthcare environments, Agile allows the researchers to gather continuous feedback from the clinic staff and the dentist during the development process. If a specific feature (e.g., the billing interface) is found to be confusing during early testing, the Agile framework allows the team to pivot and refine the user interface in the next development sprint, ensuring the final product is highly aligned with the actual needs of the end-users.

2.6 Research Gap Analysis

While numerous studies have explored the digitalization of dental clinics, a careful analysis of the reviewed literature reveals specific gaps that the present study aims to address. The table below outlines the limitations of existing systems and how the proposed Web-Based Dental Clinic Management System fills these gaps.

Reviewed Study / System	Identified Limitations / Gaps	How the Proposed System Addresses the Gap
Balderas et al. (2014) - SMS Scheduling System	Focused primarily on notifications and basic record storage. Lacked integrated billing and comprehensive treatment history tracking.	The proposed system integrates scheduling, billing, and detailed treatment history (including X-ray uploads) into a single, unified web-based platform.
Abregana et al. (2013) - LAN-Based System	Restricted to a Local Area Network (LAN), meaning records could only be accessed within the physical clinic premises, limiting remote access for the dentist.	The proposed system is cloud-based (Firebase), allowing authorized personnel to securely access real-time data from any device with an internet connection.
Prandana et al. (2025) - Waterfall Model System	Utilized the rigid Waterfall model, which does not easily accommodate changes in user requirements once the development phase has begun.	The proposed system utilizes the Agile Development Model, allowing for iterative feedback, continuous testing, and flexible adaptation to the clinic's evolving needs.

Reviewed Study / System	Identified Limitations / Gaps	How the Proposed System Addresses the Gap
General Local Clinic Practices	Reliance on fragmented tools (e.g., Excel for billing, physical logbooks for appointments, paper folders for records), leading to data redundancy and high risk of human error.	The proposed system provides a centralized NoSQL database that eliminates data redundancy, ensures a single source of truth, and automates computations to minimize human error.

Based on this gap analysis, the present study is justified as it moves beyond partial or localized solutions to offer a comprehensive, cloud-integrated, and user-centric management system tailored for modern dental practices.

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CHAPTER 3

METHODOLOGY

3.1 Research Design

This study will use a Developmental Research Design, a research approach that focuses on the systematic design, development, implementation, and evaluation of a software application intended to address a specific problem or improve an existing process. This research design is appropriate for the study because it involves the creation of a Web-Based Dental Clinic Management System that aims to improve the management of patient records, appointment scheduling, treatment documentation, and payment processing in dental clinics.

The developmental research design enables the researchers to analyze the current workflow of the clinic, identify existing problems, and develop an effective technological solution based on the needs of the users. Through this approach, the researchers can gather relevant information from clinic personnel and stakeholders to determine the functional and non-functional requirements of the system. These requirements will serve as the basis for designing and developing the proposed application.

The development process will follow a series of phases that include planning, requirements gathering, system analysis, system design, development, testing, deployment, and evaluation. During the planning and analysis stages, the researchers will conduct interviews and observations to understand the current procedures used in managing patient records, appointments, treatment documentation, and payment transactions. The information gathered will be used to identify the limitations of the existing manual system and determine the necessary features of the proposed system.

During the design and development stages, the researchers created the database structure, user interfaces, and system functionalities to support patient record management, appointment scheduling, dental chart documentation, treatment history recording, GCash payment processing, and email notifications. The system was developed using modern web technologies and cloud-based data storage to ensure accessibility, security, and real-time data management.

The evaluation phase will assess the functionality, usability, reliability, efficiency, and overall performance of the developed system. Feedback from users, including the clinic owner, clinic staff, and IT faculty advisers, will be collected to determine whether the system meets its intended objectives and user requirements. The results of the evaluation will provide valuable insights for future improvements and enhancements.

By utilizing a developmental research design, the study aims to produce a functional, reliable, and user-friendly Web-Based Dental Clinic Management System that can enhance clinic operations, improve patient service, reduce manual processes, and support efficient management of patient records, appointments, treatment documentation, and online payment processing.

3.2 System Development Methodology

This study adopted the Agile Software Development Methodology in designing and developing the Dental Clinic Management System with Appointment Scheduling, Patient Records Management, and Dental Chart Documentation. Agile is an iterative and incremental software development methodology that emphasizes continuous collaboration with stakeholders, rapid delivery of functional software, adaptability to changing requirements, and continuous improvement throughout the development process. Unlike traditional software development methodologies that complete each phase sequentially, Agile encourages repeated development cycles known as iterations or sprints, allowing developers to continuously evaluate, improve, and refine the system based on stakeholder feedback (Pressman & Maxim, 2020).

The Agile methodology was selected because it provides flexibility throughout the software development process. Since the operational requirements of the dental clinic may evolve during the development of the system, the researchers required a methodology capable of accommodating changes without disrupting the overall development process. Through Agile, system features were developed incrementally, allowing the researchers to consult with the client regularly, validate completed modules, and implement revisions whenever necessary. This iterative approach ensured that the developed system remained aligned with the operational needs of the dental clinic while minimizing development risks and improving software quality.

The Agile Software Development Methodology followed in this study consisted of five major phases: Requirements Analysis, System Design, Development, Testing, and Deployment (Prototype). Each phase played a significant role in ensuring the successful completion of the proposed Dental Clinic Management System.

3.2.1 Planning

The Planning phase served as the foundation of the entire software development process. During this phase, the researchers identified the overall objectives of the project, determined the scope of the proposed Dental Clinic Management System, established the project timeline, and identified the necessary hardware, software, and human resources required for system development.

The researchers initially coordinated with the dental clinic owner and staff to discuss the existing clinic operations and determine the feasibility of developing a computerized dental clinic management system. Preliminary meetings were conducted to understand the clinic's daily workflow, including patient registration, appointment scheduling, consultation and treatment recording, dental charting, and payment processing. The researchers also discussed the expected functionalities, system limitations, and priorities of the proposed system to ensure that it would address the clinic's operational needs.

Based on these discussions, the researchers prepared the project development plan, identified the major system modules, and established development milestones for each phase of the Agile Software Development Methodology. The planning phase also included identifying potential risks that could affect project implementation, such as changes in user requirements, data migration concerns, scheduling conflicts, technical limitations, and system integration challenges. Appropriate mitigation strategies were established to minimize the impact of these risks and ensure the successful completion of the project.

The Planning phase provided a clear direction for the development of the proposed Dental Clinic Management System, ensuring that project objectives, resources, schedules, and stakeholder expectations were properly defined before proceeding to the succeeding phases of development.

3.2.2 Requirements Analysis

The Requirements Analysis phase involved gathering detailed information regarding the operational needs of the dental clinic. The primary objective of this phase was to identify the challenges encountered in the existing clinic management process and determine the functional and non-functional requirements that the proposed Dental Clinic Management System should satisfy.

The researchers utilized interviews, direct observations, and document analysis during this phase. Interviews were conducted with the clinic owner and authorized personnel to obtain comprehensive information regarding existing workflows, operational challenges, and expected system functionalities. Direct observations enabled the researchers to understand the actual procedures involved in patient registration, appointment scheduling, consultation, treatment recording, dental charting, and payment processing. Existing documents, including patient information forms, appointment logs, treatment records, billing receipts, and other clinic records, were also reviewed to understand the clinic's current documentation procedures.

Information gathered during this phase was analyzed and organized into system requirements. Functional requirements included patient registration, appointment scheduling, patient records management, dental chart documentation, treatment history recording, GCash payment processing, user authentication, and email notifications. Non-functional requirements included usability, security, reliability, maintainability, performance efficiency, responsiveness, and accessibility.

The requirements gathered during this phase served as the primary reference for designing the system architecture, database structure, and user interface of the proposed Dental Clinic Management System.

3.2.3 System Design

The System Design phase focused on translating the identified system requirements into a detailed design for the proposed Dental Clinic Management System. During this phase, the researchers developed the overall system architecture, database structure, user interface layouts, and process flow diagrams to ensure that the system would effectively meet the operational needs of the dental clinic.

The researchers designed the system using a client-server architecture integrated with Google Firebase Cloud Firestore as the cloud-based database. The database structure was created to securely store and manage patient information, appointment schedules, treatment histories, dental chart data, payment records, and user accounts while ensuring data consistency and real-time synchronization.

User interface prototypes were also designed to provide a simple, responsive, and user-friendly experience for patients and administrators. Navigation menus, input forms, dashboards, and the dental chart interface were carefully planned to improve usability and efficiency. Security measures, including user authentication and role-based access control, were incorporated into the system design to ensure that only authorized users could access confidential patient information.

The completed system design served as the blueprint for the development phase and guided the implementation of the system's features and functionalities.

3.2.4 Development

During the Development phase, the researchers translated the completed system design into a functional web-based Dental Clinic Management System. Development was carried out incrementally following the Agile Software Development Methodology, wherein individual modules were completed, tested, and evaluated before proceeding to subsequent components.

The system was developed using HTML5, CSS3, and JavaScript for the front-end interface, while Python Flask served as the backend framework responsible for processing client requests, implementing business logic, and facilitating communication between the user interface and the database. Google Firebase Cloud Firestore was utilized as the cloud-based backend service, serving as the centralized database for storing patient records, appointment schedules, treatment histories, dental chart data, and payment records. Firebase Authentication was implemented to provide secure user authentication and role-based access for patients and administrators. Visual Studio Code was used as the integrated development environment throughout the development process.

The researchers developed the system module by module. The patient management module was designed to register and maintain patient information securely. The appointment scheduling module enabled patients to book appointments online while allowing the administrator to review, accept, or decline requests and assign a consulting dentist. The dental chart module provided an interactive odontogram for documenting tooth status and recording treatment notes with automatic balance computation. The GCash payment module integrated PayMongo to process online payments securely. The email notification module automatically sent appointment status updates to patients. Python Flask handled server-side processing for these modules, ensuring efficient communication with Firebase and providing reliable management of system transactions and user requests.

Throughout the development process, completed modules were demonstrated to the client for validation. Feedback provided by the client was incorporated into succeeding development iterations, allowing continuous refinement and improvement of the system while ensuring that it remained aligned with the operational requirements of the dental clinic.

3.2.5 Testing

System testing was conducted continuously throughout the Agile Software Development process. Rather than postponing testing until system completion, each module was tested immediately after development to identify errors early, verify system functionality, and minimize software defects.

Functional testing was performed to ensure that every module operated according to the specified system requirements. The researchers tested patient registration, appointment scheduling, dental chart recording, GCash payment processing, user authentication, and email notifications individually before integrating them into the complete system.

Integration testing was subsequently conducted to verify proper communication among the different system modules. Particular attention was given to ensuring that appointment records were correctly synchronized with patient information, treatment records were accurately linked to individual patients, payment statuses were properly updated, and all data stored in Firebase Cloud Firestore were updated and synchronized in real time.

User Acceptance Testing (UAT) was conducted with the dental clinic owner and authorized personnel after the completion of the prototype system. The users evaluated the system by performing actual clinic operations, including patient registration, appointment booking, dental chart recording, and GCash payment processing. Feedback gathered during this phase was used to correct identified issues and implement necessary improvements before the deployment of the prototype.

3.2.6 Deployment and Maintenance

Following successful system testing and evaluation, the prototype Dental Clinic Management System was deployed for demonstration and limited operational use. The deployment phase involved configuring the web-based application, preparing the Firebase database, creating user accounts for patients and administrators, and verifying that the system was accessible through supported web browsers and devices.

The researchers also conducted a system orientation for the intended users, demonstrating the major system functions, including patient registration, appointment scheduling, dental chart documentation, and GCash payment processing. This orientation ensured that users understood how to navigate the system and perform their assigned tasks efficiently.

After deployment, maintenance activities were carried out to ensure the stability, security, and reliability of the system. Minor software issues identified during prototype evaluation and user feedback were corrected through system updates. Additional improvements and feature enhancements were also incorporated based on user recommendations, following the Agile Development Model, which supports continuous refinement and iterative development without significantly affecting existing system functionalities.

The maintenance phase ensured that the proposed Dental Clinic Management System remained functional, secure, and responsive to the operational needs of the dental clinic while providing a foundation for future system enhancements.

Figure 3.1 Agile Software Development Methodology

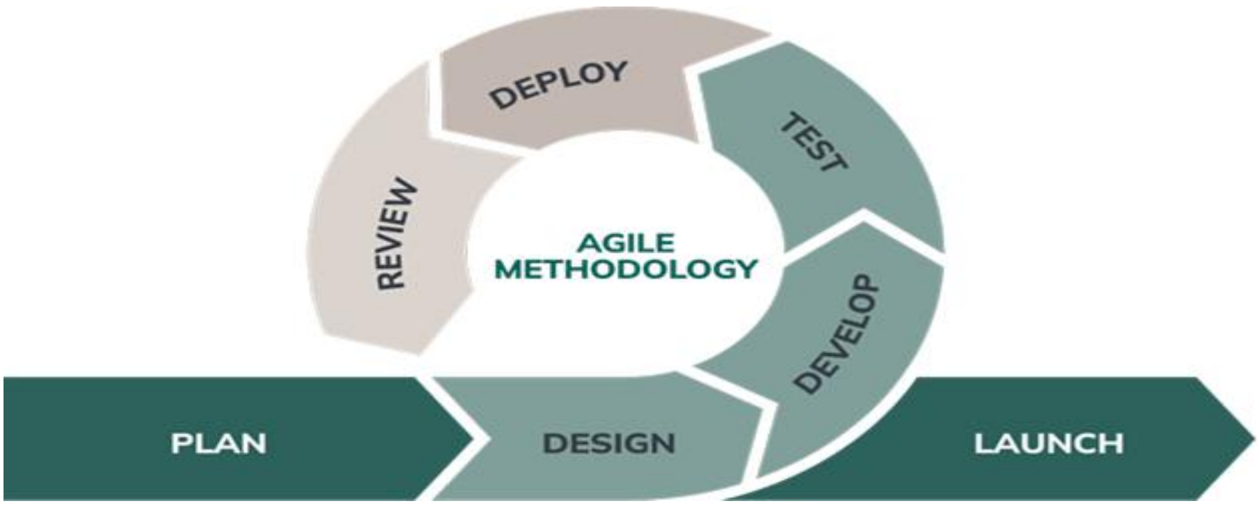


Figure 3.1 illustrates the Agile Software Development Methodology adopted in this study. Unlike traditional linear development models, Agile promotes continuous collaboration, feedback, and iterative improvement throughout the software development process. Each completed module of the Dental Clinic Management System was evaluated by the client and intended users, allowing the researchers to identify necessary revisions, address system issues, and implement enhancements during each development iteration. This iterative approach ensured that the developed system remained aligned with the operational requirements of the dental clinic while improving system quality, usability, and overall performance throughout the software development life cycle.

3.3 System Architecture

The system will use a client–server architecture with cloud integration. Users will access the system through a web-based interface, allowing dentists to manage patient records, schedule appointments, and handle billing. The server will process requests, manage system functions, and ensure secure access. All data will be stored and synchronized using Firebase, a cloud-based database that provides real-time updates, secure authentication, and efficient data management.

The client side refers to the devices used by the users, such as smartphones, tablets, laptops, or desktop computers. Through the client interface, patients can book appointments, view dental services, receive notifications, and access clinic information. Clinic staff and administrators can also manage patient records, schedules, billing information, and treatment histories using the same system interface.

The server side will handle the main processing tasks of the system. It is responsible for storing and managing data securely within the database, processing appointment requests, handling user authentication, and maintaining the overall functionality of the system. The server ensures that all information entered by users is updated in real time and can be accessed whenever needed.

The database management component will serve as the central storage of patient records, appointment schedules, billing details, dental services, and other important clinic information. This centralized approach reduces data redundancy, improves data accuracy, and makes information retrieval faster and more organized.

In addition, the system architecture supports responsive design and multi-device accessibility, allowing users to access the system conveniently regardless of the device they are using. Overall, the client-server architecture provides better efficiency, scalability, security, and reliability for the proposed dental clinic management system.

3.3.1 Functional Requirements

The functional requirements define the specific behaviors, functions, and features that the Web-Based Dental Clinic Management System must perform to satisfy the needs of the users. These are categorized by system modules.

Module 1: User Authentication and Security

- **FR-01:** The system shall allow users (Admin, Dentist, Staff, Patient) to log in using a valid email and password.
- **FR-02:** The system shall enforce Role-Based Access Control (RBAC), ensuring users can only access modules relevant to their specific role.
- **FR-03:** The system shall provide a "Forgot Password" feature that sends a secure, time-limited reset link to the user's registered email.
- **FR-04:** The system shall automatically log out users after 15 minutes of inactivity to protect sensitive patient data.

Module 2: Patient Record Management

- **FR-05:** The system shall allow authorized staff to create, read, update, and archive patient demographic profiles.
- **FR-06:** The system shall securely store patient medical histories, including allergies, current medications, and pre-existing conditions.
- **FR-07:** The system shall allow the dentist to upload and attach digital X-ray images and intraoral photos to the patient’s specific treatment record.

- **FR-08:** The system shall generate a unique, auto-incrementing Patient ID for every newly registered individual.

Module 3: Appointment Scheduling and Calendar

- **FR-09:** The system shall display a real-time, interactive visual calendar showing available and booked time slots.
- **FR-10:** The system shall prevent double-booking by automatically blocking out time slots that are already confirmed for a specific dentist.
- **FR-11:** The system shall allow patients and staff to request, confirm, reschedule, or cancel appointments.
- **FR-12:** The system shall automatically trigger automated email reminders to patients 24 hours before their scheduled appointment.

Module 4: Treatment and Clinical Records

- **FR-13:** The system shall allow the dentist to record the diagnosis, procedures performed, and prescribed medications for a specific patient visit.
- **FR-14:** The system shall link every treatment record to a specific appointment and patient profile.
- **FR-15:** The system shall lock clinical treatment notes from being edited after 24 hours to maintain medical record integrity, requiring an addendum for corrections.

Module 5: Billing and Financial Management

- **FR-16:** The system shall automatically pull the standard prices of rendered procedures from the services_catalog database.
- **FR-17:** The system shall compute the subtotal, apply discounts (if any), and calculate the final total amount due.
- **FR-18:** The system shall record partial payments and automatically calculate the remaining balance.
- **FR-19:** The system shall generate a downloadable PDF invoice/receipt upon successful payment processing.

Module 6: Reports and Analytics

- **FR-20:** The system shall generate daily, monthly, and annual reports on patient volume, revenue, and most availed services.
- **FR-21:** The system shall display visual analytics (bar charts, pie charts) on the Administrator dashboard.
- **FR-22:** The system shall allow the export of financial and patient reports to PDF and Excel formats.

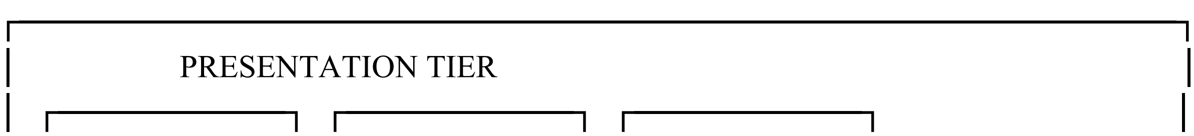
3.3.2 Non-Functional Requirements

The non-functional requirements define the quality attributes, performance constraints, and security standards of the system.

- **NFR-01 (Security):** All user passwords must be hashed using the bcrypt algorithm before being stored in the Firebase Authentication database.
- **NFR-02 (Performance):** The system shall load web pages and render dashboard data in under 3 seconds on a standard broadband connection.
- **NFR-03 (Availability):** The cloud-hosted database (Firebase) shall guarantee an uptime of 99.9% to ensure the clinic can access records during operating hours.
- **NFR-04 (Compatibility):** The frontend interface shall be fully responsive and compatible with the latest versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
- **NFR-05 (Scalability):** The NoSQL database architecture shall allow the clinic to store up to 100,000 patient records without degradation in query performance.
- **NFR-06 (Data Privacy):** The system shall comply with the Philippine Data Privacy Act of 2012 (RA 10173) by ensuring that sensitive medical data is encrypted in transit and at rest.
- **NFR-07 (Maintainability):** The backend code (Python Flask) shall be modular and well-documented, allowing future developers to easily add new features or fix bugs.
- **NFR-08 (Usability):** The system interface shall require a maximum of 3 clicks for a user to reach any core function (e.g., booking an appointment or viewing a patient record).

3.3.3 System Architecture Diagram

Figure 3.1: Three-Tier Architecture Diagram



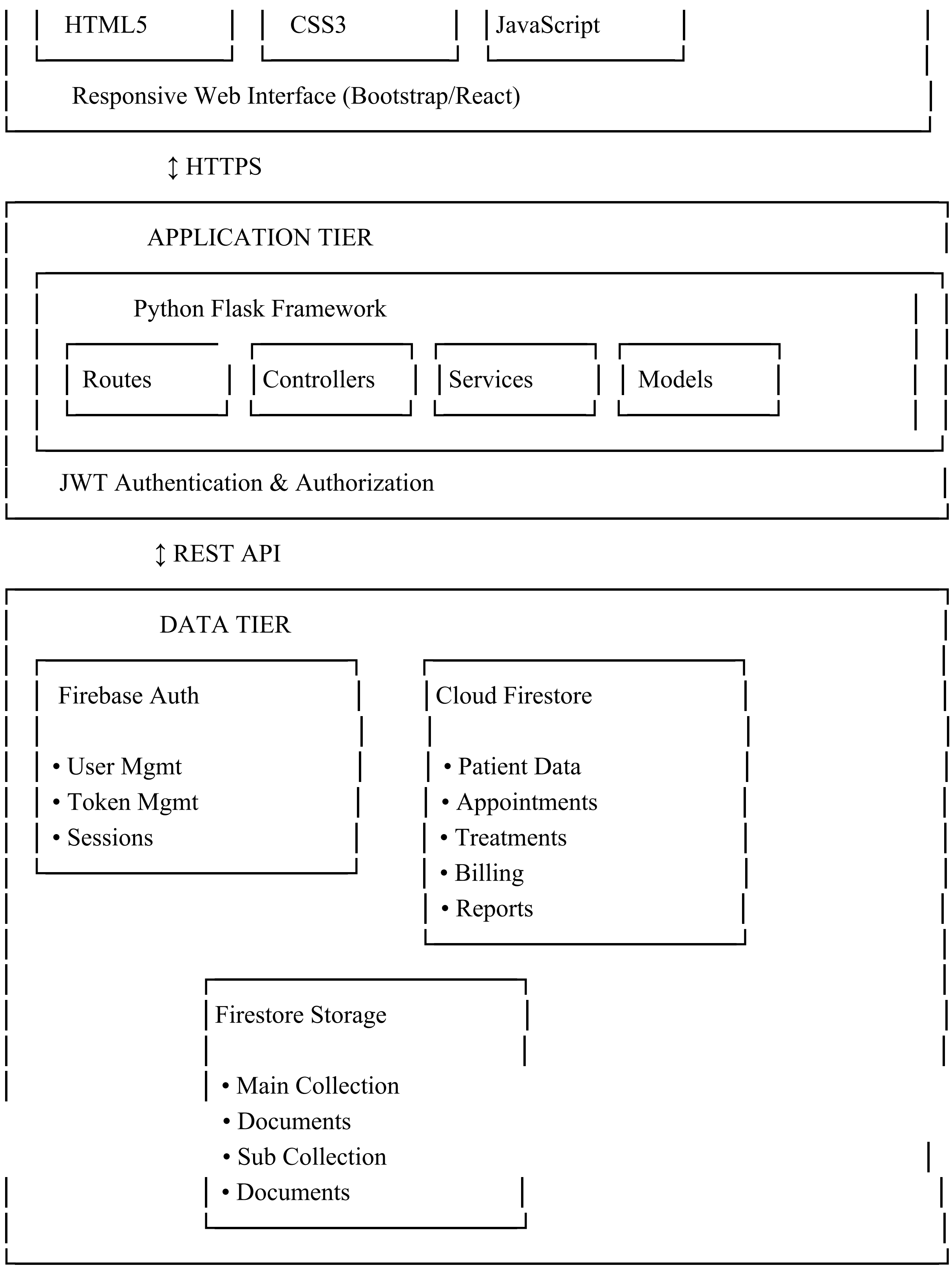
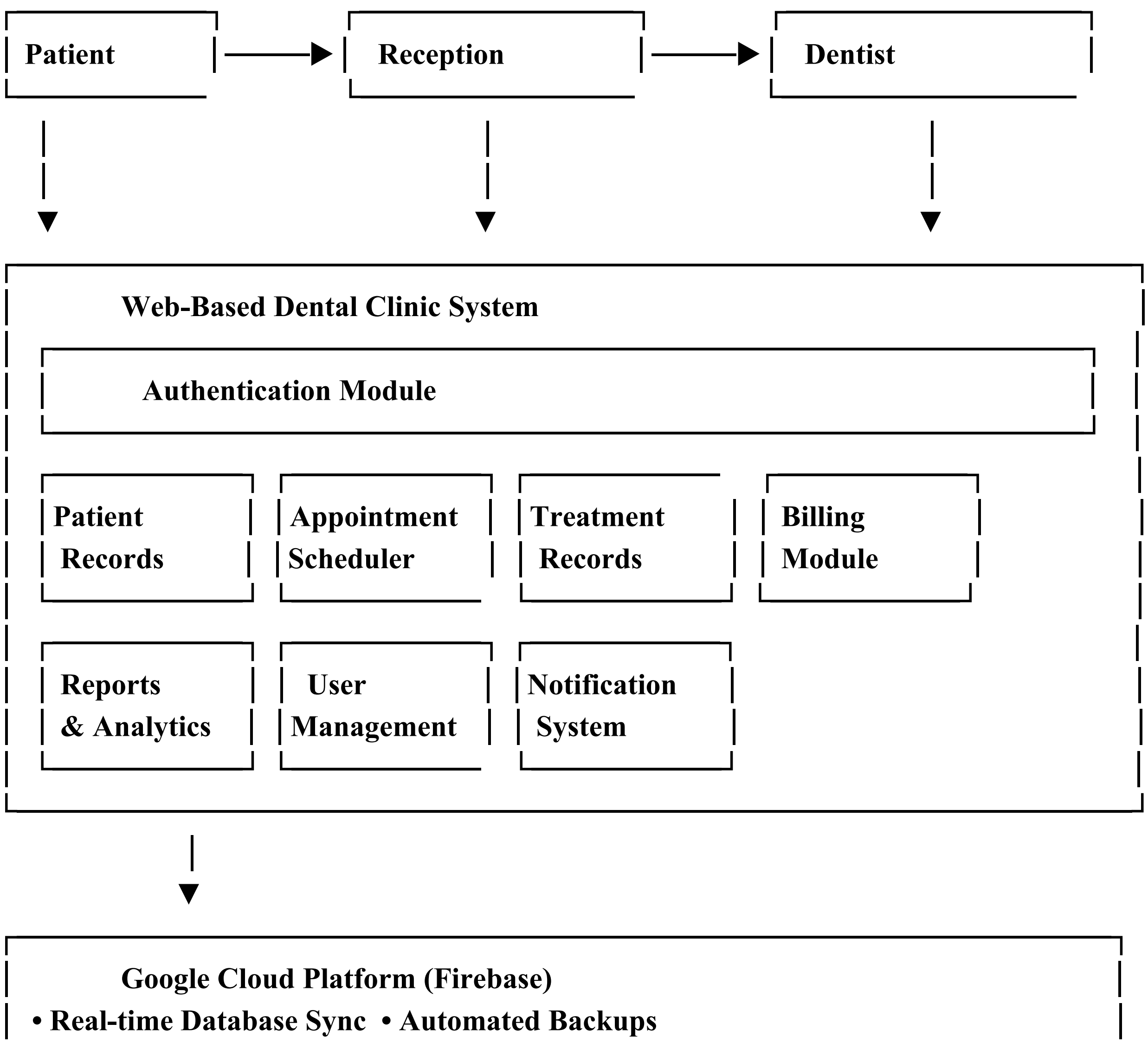

HTML5
CSS3
JavaScript
Responsive Web Interface (Bootstrap/React)
↕ HTTPS
APPLICATION TIER
Python Flask Framework
Routes
Controllers
Services
Models
JWT Authentication & Authorization
↕ REST API
DATA TIER
Firebase Auth
• User Mgmt
• Token Mgmt
• Sessions
Cloud Firestore
• Patient Data
• Appointments
• Treatments
• Billing
• Reports
Firestore Storage
• Main Collection
• Documents
• Sub Collection
• Documents

Figure 3.2: Data Flow Architecture



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3.4 Tools and Technologies Used

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python Flask
- **Database:** Google Cloud/ Cloud Firestore / Firebase
- **Development Tools:** Visual Studio Code

3.4.1 Detailed Use Case Specifications

The following use case specifications provide a granular, step-by-step description of the interactions between the system actors (Administrator, Clinic Staff, Dentist, and Patient) and the Web-Based Dental Clinic Management System. These narratives serve as the foundational blueprint for the system's backend logic and frontend interface development.

Use Case 1: System Login and Authentication

- **Use Case ID:** UC-01
- **Use Case Name:** System Login and Authentication
- **Primary Actor:** All Users (Administrator, Clinic Staff, Dentist, Patient)
- **Description:** This use case describes the process by which a user securely accesses the system by providing valid credentials. The system verifies the credentials against the Firebase Authentication database and grants role-based access.
- **Preconditions:**
    1. The user must have a registered account in the system.
    2. The user must have an active internet connection.
- **Postconditions:**
    1. The user is successfully logged in and redirected to their role-specific dashboard.
    2. A system activity log is generated recording the login timestamp and IP address.

- **Main Flow:**
    1. The user navigates to the system's login webpage.
    2. The system displays the login form requesting an Email/Username and Password.
    3. The user enters their credentials and clicks the "Login" button.
    4. The system validates the input format (e.g., checks for empty fields).
    5. The system sends the credentials to the Firebase Authentication backend.
    6. Firebase verifies the credentials and returns a success token along with the user's role (Admin, Staff, Dentist, or Patient).
    7. The system redirects the user to the appropriate dashboard based on their assigned role.
- **Alternative Flows:**
    - *A1: Forgot Password:* At Step 3, if the user clicks "Forgot Password," the system prompts for the registered email address, sends a password reset link via email, and returns to the login screen.
- **Exceptions:**
    - *E1: Invalid Credentials:* If the email or password is incorrect, the system displays an error message: "Invalid email or password. Please try again." and remains on the login page.
    - *E2: Account Locked:* If the user fails to log in after 5 consecutive attempts, the system temporarily locks the account for 15 minutes to prevent brute-force attacks.
- **Business Rules:**
    - Passwords must be at least 8 characters long and contain a mix of letters, numbers, and special characters.
    - All passwords are hashed using bcrypt before storage.

Use Case 2: Manage User Accounts

- **Use Case ID:** UC-02
- **Use Case Name:** Manage User Accounts
- **Primary Actor:** Administrator
- **Description:** This use case allows the System Administrator to create, update, deactivate, or delete user accounts for clinic staff and dentists to ensure proper system access control.
- **Preconditions:** The Administrator must be logged into the system.
- **Postconditions:** The user database is updated, and the affected user's access privileges are immediately modified.
- **Main Flow:**
    1. The Administrator navigates to the "User Management" module.
    2. The system displays a list of all current system users with their roles and status (Active/Inactive).
    3. The Administrator selects an action: "Add New User," "Edit User," or "Deactivate User."
    4. *If Add New User:* The Admin fills out a form with the user's full name, email, role, and temporary password.
    5. The Administrator clicks "Save."
    6. The system validates the data and creates a new record in the Firebase users collection.
    7. The system sends an automated welcome email to the new user with their login credentials.
- **Alternative Flows:**
    - *A1: Edit User:* The Admin selects an existing user, modifies their role or contact details, and saves. The system updates the Firestore document.
- **Exceptions:**
    - *E1: Duplicate Email:* If the Admin tries to register an email that already exists, the system displays: "Email address is already in use."

- **Business Rules:**
    - Only the Administrator can create accounts for Dentists and Clinic Staff.
    - A user cannot be deleted if they have associated historical records (e.g., past appointments or billings); they can only be set to "Inactive."

Use Case 3: Register New Patient

- **Use Case ID:** UC-03
- **Use Case Name:** Register New Patient
- **Primary Actor:** Clinic Staff (Receptionist)
- **Description:** This use case details the process of encoding a new patient's demographic and basic medical information into the centralized database.
- **Preconditions:** The Clinic Staff must be logged into the system.
- **Postconditions:** A new patient profile is created, and a unique Patient ID is generated.
- **Main Flow:**
    1. The Clinic Staff navigates to the "Patient Records" module and clicks "Add New Patient."
    2. The system displays the Patient Registration Form.
    3. The Staff enters the patient's details: Full Name, Date of Birth, Gender, Contact Number, Address, and Emergency Contact.
    4. The Staff enters basic medical history (e.g., allergies, current medications, existing medical conditions).
    5. The Staff clicks "Submit."
    6. The system validates that all required fields are filled.
    7. The system generates a unique PatientID, saves the data to the Firebase patients collection, and displays a success message with the new Patient ID.
- **Alternative Flows:**

- *A1: Existing Patient:* If the Staff searches for the patient first and finds them, the system bypasses registration and opens the existing profile for updating.

- **Exceptions:**
  - *E1: Missing Required Fields:* If mandatory fields (e.g., Name, Contact Number) are left blank, the system highlights the fields in red and prevents submission.
- **Business Rules:**
  - Every patient must have a unique contact number or email to facilitate automated appointment reminders.

Use Case 4: Schedule and Manage Appointments

- **Use Case ID:** UC-04
- **Use Case Name:** Schedule and Manage Appointments
- **Primary Actor:** Clinic Staff, Patient (via Patient Portal)
- **Description:** This use case allows authorized users to book, view, reschedule, or cancel patient appointments, ensuring no double-booking occurs.
- **Preconditions:** The user is logged in, and the patient is registered in the system.
- **Postconditions:** The appointment is saved in the database, and the clinic's daily schedule is updated in real-time.
- **Main Flow:**
  1. The user navigates to the "Appointment Calendar" module.
  2. The system displays a visual calendar showing available and booked time slots for the selected dentist.
  3. The user selects an available date and time slot.

4. The user selects the registered Patient and the type of service (e.g., Cleaning, Extraction, Consultation).

5. The user adds optional notes (e.g., "Patient requested morning slot").

6. The user clicks "Confirm Booking."

7. The system checks for scheduling conflicts. If none, it saves the appointment to the appointments collection with a status of "Pending" or "Confirmed."

8. The system triggers an automated Email reminder to the patient.

- **Alternative Flows:**
  - *A1: Reschedule:* The user selects an existing "Confirmed" appointment, chooses a new date/time, and clicks "Update." The system updates the record and sends a rescheduling notification to the patient.
  - *A2: Cancel:* The user selects an appointment and clicks "Cancel." The system changes the status to "Cancelled" and frees up the time slot on the calendar.
- **Exceptions:**
  - *E1: Double Booking:* If another user attempts to book the exact same time slot simultaneously, the system displays: "This time slot is no longer available. Please choose another."
- **Business Rules:**
  - Appointments must be scheduled at least 24 hours in advance for standard procedures.
  - Emergency slots are reserved and can be manually overridden by the Administrator or Head Dentist.

Use Case 5: Update Dental Treatment History

- **Use Case ID:** UC-05
- **Use Case Name:** Update Dental Treatment History
- **Primary Actor:** Dentist

- **Description:** This use case enables the dentist to record the diagnosis, procedures performed, prescribed medications, and clinical notes during or after a patient's visit.

- **Preconditions:** The patient has an active or completed appointment, and the Dentist is logged in.

- **Postconditions:** The patient's medical record is updated, and the treatment is linked to the billing module.

- **Main Flow:**

    1. The Dentist navigates to the "Patient Records" module and searches for the patient.
    2. The system displays the patient's profile and medical history.
    3. The Dentist clicks "Add New Treatment Record."
    4. The Dentist enters the Date of Service, Diagnosis, Procedures Performed (selectable from a dropdown or custom text), and Prescribed Medications.
    5. The Dentist optionally uploads digital X-ray images or intraoral photos.
    6. The Dentist clicks "Save Record."
    7. The system saves the data to the treatments collection, linking it to the PatientID.

- **Alternative Flows:**

    - *A1: Edit Record:* The Dentist can edit a treatment record within 24 hours of creation to correct typographical errors. After 24 hours, the record is locked for audit purposes, and an addendum must be created.

- **Exceptions:**

    - *E1: File Upload Failure:* If the X-ray image exceeds the 5MB size limit or is in an unsupported format, the system rejects the upload and prompts the user to compress or convert the file.

- **Business Rules:**

- Only the attending Dentist or the System Administrator can modify clinical treatment notes to maintain data integrity and compliance with medical record standards.

Use Case 6: Process Billing and Generate Invoice

- **Use Case ID:** UC-06
- **Use Case Name:** Process Billing and Generate Invoice
- **Primary Actor:** Clinic Dentist
- **Description:** This use case automates the computation of patient bills based on the rendered treatments and generates a printable or downloadable invoice.
- **Preconditions:** The patient has completed a treatment, and the treatment record has been saved by the Dentist.
- **Postconditions:** A billing record is created, payment status is updated, and a receipt is generated.
- **Main Flow:**
    1. The Clinic Dentist navigates to the "Billing" module and searches for the patient.
    2. The system retrieves the list of pending treatments for that patient.
    3. The Dentist selects the treatments to be billed. The system automatically pulls the corresponding prices from the services catalog.
    4. The Dentist applies any discounts (if applicable) and the system calculates the Total Amount Due.
    5. The Dentist selects the Payment Method (Cash, GCash, Credit Card).
    6. The Dentist clicks "Process Payment."
    7. The system updates the billings collection, marks the associated treatments as "Billed," and generates a PDF invoice.
- **Alternative Flows:**
    - *A1: Partial Payment:* The Staff can enter a partial payment amount. The system updates the AmountPaid and calculates the remaining Balance, keeping the billing status as "Partially Paid."

- **Exceptions:**
    - *E1: Missing Service Price: If a custom treatment was added without a price, the system prompts the Staff to manually input the cost before proceeding. Additionally, all billing actions are logged to the system_logs collection.*
    - All financial transactions must be logged. Deleted or voided receipts require Administrator approval and a stated reason.

Use Case 7: Generate Clinic Reports

- **Use Case ID:** UC-07
- Use Case Name: Generate Clinic Reports:

  Description: This use case allows the Administrator to generate analytical reports to monitor clinic performance, financial health, and patient demographics.
- **Preconditions:** The Administrator is logged in, and the system contains historical data.
- **Postconditions:** A formatted report (PDF or Excel) is generated and made available for download.
- **Main Flow:**
    1. The Administrator navigates to the "Reports and Analytics" module.
    2. The system displays available report types: Daily Patient Count, Monthly Revenue, Most Availed Services, and Appointment No-Show Rates.
    3. The Administrator selects a report type and specifies a date range (e.g., January 1, 2026, to January 31, 2026).
    4. The Administrator clicks "Generate Report."
    5. The system queries the Firebase database, aggregates the data, and renders visual charts (bar graphs, pie charts) on the screen.
    6. The Administrator clicks "Export to PDF/Excel."
    7. The system compiles the data and downloads the file to the Administrator's device.

- **Alternative Flows:**
    - *A1: Custom Report:* The Administrator can filter the report by a specific dentist or service type before generating.
- **Exceptions:**
    - *E1: No Data Found:* If the selected date range contains no records, the system displays: "No data available for the selected period."
- **Business Rules:**
    - Financial reports can only be accessed by the Administrator and the Clinic Owner.

Use Case 8: Patient Portal Access and Profile Management

- **Use Case ID:** UC-08
- **Use Case Name:** Patient Portal Access and Profile Management
- **Primary Actor:** Patient
- **Description:** This use case allows patients to log in to a dedicated portal to view their upcoming appointments, basic treatment history, and update their personal contact information.
- **Preconditions:** The patient has been registered by the Clinic Staff and provided with portal login credentials.
- **Postconditions:** The patient's profile is updated, or the patient successfully views their dashboard.
- **Main Flow:**
    1. The Patient logs into the system using their email and password.
    2. The system redirects the Patient to the "Patient Dashboard."
    3. The dashboard displays: Upcoming Appointments, Recent Treatments, and a "Update Profile" button.
    4. *If Update Profile:* The Patient clicks the button, modifies their Contact Number or Address, and clicks "Save."

5. The system validates the input and updates the patients collection in Firebase.

6. The system displays a success message: "Profile updated successfully."

- **Alternative Flows:**
    - *A1: Request Appointment:* The Patient clicks "Book Appointment," selects an available slot, and submits a request. The status is set to "Pending" until approved by the Clinic Staff.
- **Exceptions:**
    - *E1: Session Timeout:* If the Patient is inactive for 15 minutes, the system automatically logs them out for security purposes.
- **Business Rules:**
    - Patients can only *view* their clinical notes; they cannot edit or delete any treatment records entered by the dentist.

3.4.2 Comprehensive NoSQL Data Dictionary (Firebase Cloud Firestore)

A Data Dictionary is a centralized repository of information about data such as meaning, relationships to other data, origin, usage, and format. Since this system utilizes a NoSQL document-oriented database (Firebase Cloud Firestore), the schema is organized into **Collections** (equivalent to tables) and **Documents** (equivalent to rows/records).

Below is the detailed data dictionary defining the structure, data types, constraints, and descriptions of every field within the system's database.

Collection 1: users

Description: Stores authentication and profile details for all system users (Administrators and Dentists).

- **userID** (String, Required, Primary Key): Unique identifier generated by Firebase Auth (e.g., "usr_8x9a2b").
- **fullName** (String, Required): The complete legal name of the user.
- **email** (String, Required, Unique): The user's official email address used for login.
- **role** (String, Required, Enum): Defines access level. Values: ['Admin', 'Dentist', 'Staff'].

- **contactNumber** (String, Optional): Mobile number for emergency contact or two-factor authentication.

- **status** (String, Required, Default: 'Active'): Indicates if the account is Active or Inactive.

- **createdAt** (Timestamp, Required): The exact date and time the account was created.

- **lastLogin** (Timestamp, Optional): The timestamp of the user's most recent successful login.

Collection 2: patients

Description: Stores demographic and basic medical information of all registered patients.

- **patientID** (String, Required, Primary Key): Unique auto-generated identifier (e.g., "PT-2026-001").

- **firstName** (String, Required): Patient's first name.

- **lastName** (String, Required): Patient's last name.

- **dateOfBirth** (Timestamp, Required): Used to calculate the patient's age automatically.

- **gender** (String, Required, Enum): Values: ['Male', 'Female', 'Prefer not to say'].

- **contactNumber** (String, Required): Primary phone number for SMS reminders.

- **email** (String, Optional): Used for email receipts and portal access.

- **address** (String, Required): Complete residential address.

- **emergencyContactName** (String, Required): Name of the person to contact in case of medical emergency.

- **emergencyContactNumber** (String, Required): Phone number of the emergency contact.

- **allergies** (String, Optional): Known drug or material allergies (e.g., "Penicillin, Latex").

- **medicalConditions** (String, Optional): Pre-existing conditions (e.g., "Hypertension, Diabetes").

- **registrationDate** (Timestamp, Required): Date the patient was first registered in the system.

Collection 3: appointments

Description: Manages the scheduling, status, and details of patient visits.

- **appointmentID** (String, Required, Primary Key): Unique identifier (e.g., "APT-9921").
- **patientID** (String, Required, Foreign Key): References the patientID in the patients collection.
- **dentistID** (String, Required, Foreign Key): References the userID of the assigned dentist.
- **appointmentDate** (Timestamp, Required): The scheduled date of the visit.
- **startTime** (String, Required): Scheduled start time (e.g., "09:00 AM").
- **endTime** (String, Required): Scheduled end time (e.g., "10:00 AM").
- **serviceType** (String, Required): The general category of the visit (e.g., "Consultation", "Extraction", "Cleaning").
- **status** (String, Required, Enum): Current state of the appointment. Values: ['Pending', 'Confirmed', 'Completed', 'Cancelled', 'No-Show'].
- **notes** (String, Optional): Special instructions or patient requests.
- **createdAt** (Timestamp, Required): When the appointment was booked.

Collection 4: treatments (Medical Records)

Description: Stores the clinical notes, diagnoses, and procedures performed during a patient's visit.

- **treatmentID** (String, Required, Primary Key): Unique identifier (e.g., "TRT-4451").
- **patientID** (String, Required, Foreign Key): Links the record to the specific patient.
- **appointmentID** (String, Optional, Foreign Key): Links the treatment to a specific scheduled appointment.
- **dentistID** (String, Required, Foreign Key): The dentist who performed the procedure.
- **dateOfService** (Timestamp, Required): The actual date the treatment was rendered.
- **diagnosis** (Text, Required): The dentist's clinical assessment of the patient's condition.

- **proceduresPerformed** (Array of Strings, Required): List of specific actions taken (e.g., ["Tooth 18 Extraction", "Local Anesthesia Administered"]).
- **prescribedMedication** (Text, Optional): Medications prescribed, including dosage and frequency.
- **clinicalNotes** (Text, Optional): Additional observations or future treatment recommendations.
- **attachments** (Array of Strings, Optional): URLs pointing to uploaded X-ray images or intraoral photos stored in Firebase Cloud Storage.

Collection 5: services_catalog

Description: A master list of all dental services offered by the clinic and their standard pricing.

- **serviceID** (String, Required, Primary Key): Unique identifier (e.g., "SRV-001").
- **serviceName** (String, Required, Unique): Name of the procedure (e.g., "Dental Prophylaxis", "Root Canal Treatment").
- **description** (Text, Required): Brief explanation of what the service entails.
- **standardPrice** (Number/Float, Required): The default cost of the service in PHP.
- **isActive** (Boolean, Required, Default: true): Allows the Admin to hide outdated services without deleting historical billing data.

Collection 6: billings

Description: Tracks all financial transactions, invoices, and payment statuses.

- **billingID** (String, Required, Primary Key): Unique invoice number (e.g., "INV-2026-088").
- **patientID** (String, Required, Foreign Key): The patient being billed.
- **appointmentID** (String, Optional, Foreign Key): Links the bill to a specific visit.
- **treatmentIDs** (Array of Strings, Required): List of treatmentIDs included in this invoice.
- **itemizedCharges** (Array of Objects, Required): Breakdown of costs (e.g., [{service: "Cleaning", price: 1500}]).
- **subtotal** (Number/Float, Required): Sum of all itemized charges.

- **discount** (Number/Float, Default: 0): Any applied discount amount.
- **totalAmountDue** (Number/Float, Required): Final amount to be paid (subtotal - discount).
- **amountPaid** (Number/Float, Required): Amount settled by the patient.
- **balance** (Number/Float, Required): Remaining amount (totalAmountDue - amountPaid).
- **paymentMethod** (String, Required, Enum): Values: ['Cash', 'GCash', 'Credit Card', 'Bank Transfer'].
- **paymentStatus** (String, Required, Enum): Values: ['Paid in Full', 'Partially Paid', 'Unpaid'].
- **transactionDate** (Timestamp, Required): Date and time the payment was processed.
- **processedBy** (String, Required, Foreign Key): The userID of the staff member who processed the payment.

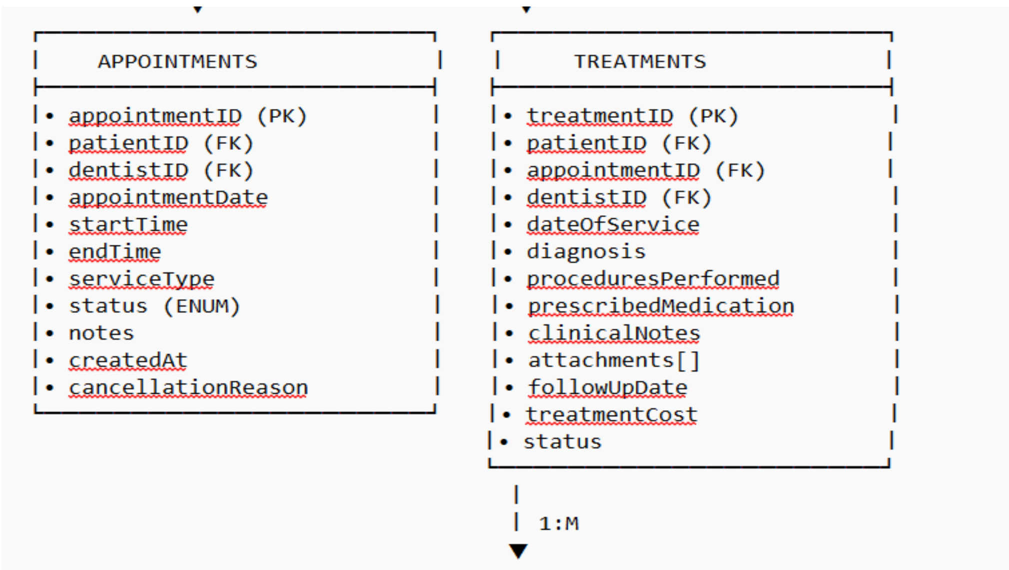
Collection 7: system_logs

Description: An audit trail for security and accountability, recording critical system actions.

- **logID** (String, Required, Primary Key): Unique identifier.
- **userID** (String, Required, Foreign Key): The user who performed the action.
- **action** (String, Required): Description of the action (e.g., "User Login", "Deleted Patient Record", "Updated Billing").
- **targetDocument** (String, Optional): The specific record affected (e.g., "patientID: PT-2026-001").
- **ipAddress** (String, Required): The IP address from which the action was performed.
- **timestamp** (Timestamp, Required): Exact date and time of the action.

3.4.2.1 Database Relationship Diagram (Detailed)

```
┌──────────────────────────────────────────────────────────┐
│                          USERS                           │
├──────────────────────────────────────────────────────────┤
│ • userID (PK)              • contactNumber               │
│ • fullName                 • status                      │
│ • email (UNIQUE)           • createdAt                   │
│ • role (ENUM)              • lastLogin                   │
│ • passwordHash             • profileImage                │
└──────────────────────────────────────────────────────────┘
                        │
                        │ 1:M
                        ▼
┌──────────────────────────────────────────────────────────┐
│                        PATIENTS                          │
├──────────────────────────────────────────────────────────┤
│ • patientID (PK)           • emergencyContactName        │
│ • firstName                • emergencyContactNumber      │
│ • lastName                 • allergies                   │
│ • dateOfBirth              • medicalConditions           │
│ • gender                   • registrationDate            │
│ • contactNumber            • assignedDentistID (FK)      │
│ • email                    • insuranceInfo               │
│ • address                  • lastVisitDate               │
└──────────────────────────────────────────────────────────┘
                        │
            ┌───────────┴──────────────┐
            │                          │
            │ 1:M                      │ 1:M
            ▼                          ▼
```



3.4.3 Advanced Behavioral and Structural Modeling

To provide a comprehensive technical blueprint for the development team and to satisfy the rigorous standards of systems analysis and design, the researchers have developed advanced Unified Modeling Language (UML) diagrams. These diagrams illustrate the dynamic behavior, state transitions, and sequential interactions within the Web-Based Dental Clinic Management System.

3.4.3.1 Sequence Diagram: Process Billing and Generate Invoice

Narrative Description:

A Sequence Diagram illustrates how objects interact in a particular scenario of a use case, arranged in time sequence. Figure X (to be inserted) depicts the sequence of interactions when a Clinic Staff processes a patient's billing.

1. The **Clinic Dentist** actor initiates the process by requesting to view pending treatments for a specific **Patient** via the **Billing Interface**.
2. The **Billing Interface** sends a query to the **Flask Backend Controller**.

3. The **Flask Backend** queries the **Firebase Firestore** (treatments and services_catalog collections) to retrieve the itemized list of procedures and their corresponding standard prices.

4. **Firebase** returns the data to the backend, which formats it and sends it back to the **Billing Interface** for the staff to review.

5. The **Clinic Dentist** inputs the payment method and the amount paid, then submits the "Process Payment" command.

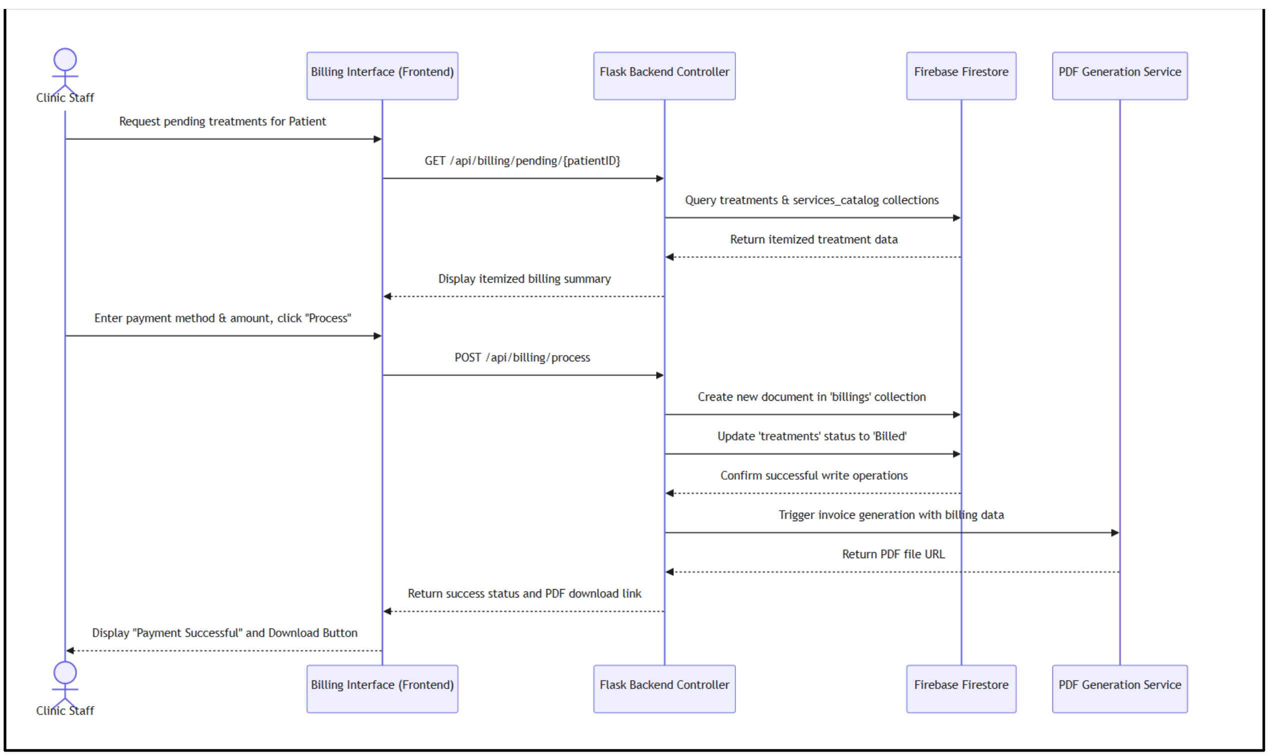
6. The **Flask Backend** validates the transaction, calculates the final balance, and creates a new document in the billings collection in **Firebase**.

7. Simultaneously, the backend updates the status of the associated treatments to "Billed".

8. Upon successful database commitment, the backend triggers the **PDF Generation Service** to create a digital receipt.

9. The **Billing Interface** displays a "Payment Successful" message and provides a download link for the generated invoice to the **Clinic Staff**.

Mermaid.js Code for Sequence Diagram:



```mermaid
sequenceDiagram
    participant Dentist as Clinic Dentis0074
    participant UI as Billing Interface
    participant API as Flask Backend
    participant DB as Firebase Firestore
    participant PDF as PDF Service

    Staff->>UI: Request pending treatments
    UI->>API: GET /api/billing/pending/{patientId}
    API->>DB: Query treatments & services_catalog
    DB-->>API: Return itemized data
    API-->>UI: Display billing summary
    Staff->>UI: Enter payment details & submit
    UI->>API: POST /api/billing/pay
    API->>DB: Create billing record & update treatment status
    API->>PDF: Trigger receipt generation
    PDF-->>API: Return PDF URL
    API-->>UI: Return success & download link
    UI-->>Staff: Display "Payment Successful"
```

3.4.3.2 Activity Diagram: Patient Appointment Lifecycle

Narrative Description:

An Activity Diagram models the workflow of a specific process, highlighting decision points and parallel activities. Figure Y illustrates the end-to-end lifecycle of a patient appointment. The process begins when an appointment request is initiated (either by the Patient via the portal or by the Clinic Staff). The system first validates if the requested time slot is available. If the slot is occupied, the system rejects the request and prompts the user to select a different time (Alternative Flow). If the slot is available, the system creates a record with a "Pending" status. A decision node follows: Does the appointment require manual approval? If yes, it waits for the Clinic Staff to confirm it, changing the status to "Confirmed". If no (e.g., auto-confirmed slots), it proceeds directly. On the day of the appointment, another decision node checks: Did the patient arrive? If the patient arrives, the status changes to "In-Progress", and the Dentist proceeds with the consultation and treatment. Upon completion, the status becomes "Completed", and the workflow transitions to the Billing module. If the patient does not arrive, the status is updated to "No-Show", and the system logs the event for future reference, freeing the slot for future bookings.

Mermaid.js Code for Activity Diagram:

```
activityDiagram
   start
   :Appointment Request Initiated\n(Patient or Staff);
   :System checks calendar for\ntime slot availability;
   if (Is time slot available?) then (No)
      :Display "Slot Unavailable" error;
      stop
   else (Yes)
      :Create Appointment Record\n(Status: Pending);
      if (Requires Manual Approval?) then (Yes)
         :Staff reviews and\napproves request;
         :Update Status to "Confirmed";
      else (No - Auto-confirm)
         :Update Status to "Confirmed";
      endif
      :System sends automated\nEmail reminder to Patient;
      :Wait for Appointment Date;
      if (Patient Arrives?) then (Yes)
         :Update Status to "In-Progress";
         :Dentist conducts consultation\nand records treatment;
         :Update Status to "Completed";
         :Trigger Billing Module;
      else (No)
         :Update Status to "No-Show";
         :Log incident in system;
      endif
   endif
   stop
```

3.4.3.3 State Machine Diagram: Appointment Entity

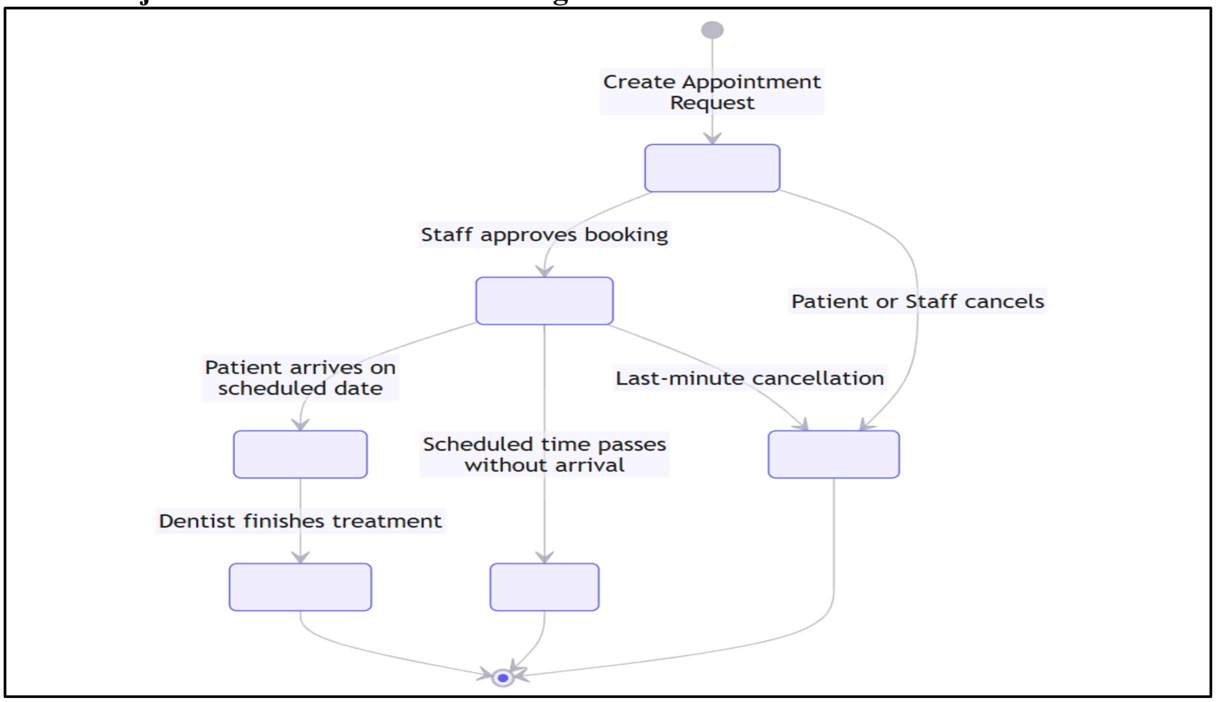
Narrative Description:

A State Machine Diagram describes the different states an object can be in during its lifecycle and the events that cause transitions between these states. Figure Z focuses specifically on the Appointment entity. The initial state is **Scheduled** (or Pending). From this state, two primary events can occur:

1. The event approve transitions the state to **Confirmed**.

2. The event cancel_request transitions the state to **Cancelled**. Once in the **Confirmed** state, the appointment awaits the scheduled date. On the day of the visit, the event patient_arrives transitions the state to **In-Progress**. Alternatively, if the time passes without the patient, the event time_elapsed transitions the state to **No-Show**. From the **In-Progress** state, the event treatment_finished transitions the appointment to its final state: **Completed**. The **Cancelled**, **No-Show**, and **Completed** states are final states (indicated by the bullseye symbol), meaning no further state transitions can occur for that specific appointment record, ensuring historical data integrity.
3.

Mermaid.js Code for State Machine Diagram:




```
stateDiagram-v2
    [*] --> Pending : Appointment Booked
    Pending --> Confirmed : Dentist Approves
    Pending --> Cancelled : Patient/Dentist Cancels
    Confirmed --> InProgress : Patient Arrives
    Confirmed --> NoShow : Time Elapsed (No Arrival)
    InProgress --> Completed : Treatment Finished & Billed
    Cancelled --> [*]
    NoShow --> [*]
    Completed --> [*]
```

3.4.4 System Module and Interface Design

The user interface (UI) and user experience (UX) of the Web-Based Dental Clinic Management System are designed to be intuitive, clean, and highly responsive. The system is divided into distinct modules, each tailored to the specific workflow of the user.

3.4.4.1 Login and Authentication Module

The entry point of the system is a secure, minimalist login page. It features the clinic's logo, a clean input field for the email address, a password field with a "show/hide" toggle, and a "Login" button. A "Forgot Password" link is provided for credential recovery. The design uses a calming color palette (typically blues and whites) to reflect a healthcare environment. Upon successful authentication, the system reads the user's role and redirects them to their specific dashboard.

3.4.4.2 Administrator Dashboard Module

The Administrator Dashboard serves as the central command center for the clinic owner or head admin. Upon logging in, the user is greeted with a summary widget displaying key performance indicators (KPIs): Total Registered Patients, Today's Appointments, Pending Appointment Requests, and Total Monthly Revenue. Below the widgets, a real-time calendar view shows the day's schedule, color-coded by status (e.g., Green for Confirmed, Yellow for Pending, Red for Cancelled). A sidebar navigation menu provides quick access to User Management, Reports, and System Settings.

3.4.4.3 Patient Records and Profile Module

This module provides a comprehensive view of a patient's demographic and clinical data. The interface is split into two main tabs: "Demographics" and "Medical History". The Demographics tab displays contact details, emergency contacts, and address. The Medical History tab highlights critical alerts in red (e.g., "Allergy: Penicillin") to ensure the dentist is immediately aware of risks. A dedicated "Files" section allows the dentist to upload and view X-ray images and digital dental charts. A search bar at the top allows staff to quickly find patients by name, contact number, or Patient ID.

3.4.4.4 Appointment Scheduling and Calendar Module

The scheduling module features an interactive, drag-and-drop calendar interface. The default view is a weekly schedule, but users can toggle to Daily or Monthly views. Available time slots are displayed in white, while booked slots are shaded. When a staff member clicks on an empty slot, a modal window pops up, prompting them to select the Patient, Service Type, and Dentist. The system provides real-time validation; if a slot is taken, it immediately turns red and displays a "Slot Unavailable" tooltip.

3.4.4.5 Treatment and Clinical Notes Module

Designed specifically for the Dentist, this module focuses on fast, accurate clinical documentation. The interface features a digital dental chart (a visual representation of the 32 adult teeth) where the dentist can click on specific teeth to mark conditions (e.g., Decay, Filling, Extraction). Below the chart is a text area for Clinical Notes, Diagnosis, and Prescribed Medications. Dropdown menus are provided for common procedures to speed up data entry. A "Save and Send to Billing" button at the bottom seamlessly transfers the selected procedures to the billing module.

3.4.4.6 Billing and Invoicing Module

The billing module is designed for the Clinic Staff to process transactions quickly. When a patient's treatment is selected, the system automatically generates an itemized list of charges. The interface displays the Subtotal, a dropdown for applying Discounts (e.g., Senior Citizen, PWD), and the Final Total Due. The staff can select the Payment Method (Cash, GCash, Card) and input the Amount Tendered. The system automatically calculates the change or remaining balance. Once processed, a digital receipt is generated, and a "Print/Download PDF" button appears.

3.4.4.7 Reports and Analytics Module

This module transforms raw database entries into actionable insights. The interface features a date-range picker at the top. Once a range is selected, the system generates visual charts. A bar chart displays daily patient volume, a pie chart shows the distribution of availed services (e.g., 40% Cleaning, 30% Extraction), and a line graph tracks monthly revenue trends. An "Export" button allows the administrator to download these reports in PDF or Excel format for accounting and auditing purposes.

3.4.4.8 UI/UX Design System and Visual Hierarchy

To ensure a consistent, professional, and intuitive user experience across all modules, the frontend development adheres to a strict Design System. The visual hierarchy is designed to reduce cognitive load for clinic staff who must navigate the system quickly during busy clinic hours. The design system utilizes the following standardized components:

1. **Color Palette:** The system utilizes a calming, healthcare-appropriate color palette. The primary brand color is **Medical Teal (Hex: #008080)**, used for primary action buttons and active navigation states. Secondary accents use **Soft Cyan (Hex: #E0F7FA)** for background highlights. Alert states are strictly standardized: **Critical Red (Hex: #D32F2F)** for allergies, double-bookings, and system errors; **Warning Amber (Hex:**

#F57C00) for pending appointments; and **Success Green (Hex: #388E3C)** for confirmed appointments and successful payments.

2. **Typography:** The system utilizes the **Inter** font family for all UI elements due to its high legibility on digital screens. Headings are rendered in bold (Font Weight: 700), while body text uses a regular weight (Font Weight: 400) with a base size of 16px to ensure readability for older users or practitioners viewing the system on smaller tablet screens.

3. **Component Library:** Reusable UI components include standardized data tables with built-in pagination and search filters, modal dialogs for confirming critical actions (e.g., "Void Receipt" or "Delete Patient"), and toast notifications for non-intrusive system feedback (e.g., "Appointment Saved Successfully").

3.4.4.9 Web Content Accessibility Guidelines (WCAG) 2.1 Compliance

Recognizing that the system will be used by a diverse range of healthcare professionals, the frontend interface is designed to comply with WCAG 2.1 Level AA standards. Key accessibility features include:

- **Keyboard Navigation:** All interactive elements, including the appointment calendar and billing forms, can be fully navigated using the Tab, Enter, and Arrow keys without requiring a mouse.

- **Screen Reader Compatibility:** All form inputs, buttons, and data tables include proper ARIA (Accessible Rich Internet Applications) labels and roles, ensuring compatibility with screen reading software for visually impaired staff.

- **Color Contrast Ratios:** All text and interactive elements maintain a minimum contrast ratio of 4.5:1 against their background colors, ensuring visibility for users with color vision deficiencies. For example, critical allergy alerts are not only highlighted in red but also include a bold text label and an icon to ensure the information is conveyed through multiple visual channels.

3.4.5 User Stories and Acceptance Criteria

To ensure the system is developed with a strong focus on user experience and functional requirements, the researchers have formulated User Stories. A User Story is an informal, natural language description of one or more features of a software system, written from the perspective of the end-user. Each user story is accompanied by Acceptance Criteria, which define the specific conditions that must be met for the story to be considered complete.

Module 1: User Authentication and Security

- **User Story 1.1:** As a registered user, I want to log in using my email and password so that I can securely access my role-specific dashboard.
    - *Acceptance Criteria:* The system must validate the email format. The system must hash the password before verification. Upon successful login, the user must be redirected to the correct dashboard (Admin, Dentist, or Staff) within 3 seconds.
- **User Story 1.2:** As a user, I want to reset my password via a secure email link so that I can regain access if I forget my credentials.
    - *Acceptance Criteria:* The system must send a unique, time-limited (15-minute) reset token to the registered email address. The token must expire after one use or after the time limit.

Module 2: Patient Record Management

- **User Story 2.1:** As a clinic staff member, I want to register a new patient with their demographic and medical history so that the dentist has accurate information before the consultation.
    - *Acceptance Criteria:* The system must generate a unique, auto-incrementing Patient ID. The system must prevent the submission of the form if mandatory fields (Name, Contact Number) are left blank.
- **User Story 2.2:** As a dentist, I want to view a patient's allergy and medical history highlighted in red so that I can avoid prescribing contraindicated medications.
    - *Acceptance Criteria:* The system must display the "Medical History" tab prominently. Any text entered into the "Allergies" field must be rendered in a bold, red font on the patient's profile dashboard.

Module 3: Appointment Scheduling and Calendar

- **User Story 3.1:** As a receptionist, I want to view a visual, color-coded calendar of the dentist's schedule so that I can quickly identify available time slots.
    - *Acceptance Criteria:* The calendar must display available slots in green, pending slots in yellow, and confirmed/booked slots in red. The calendar must update in real-time without requiring a page refresh. *. **User Story 3.2:** As a patient, I want to receive an automated email reminder 24 hours before my appointment so that I do not forget my schedule.
    - *Acceptance Criteria:* The system must automatically trigger a notification exactly 24 hours prior to the startTime of any appointment with a "Confirmed" status.

Module 4: Treatment and Clinical Records

- **User Story 4.1:** As a dentist, I want to record the procedures performed and attach digital X-ray images to a specific patient's file so that I can track their long-term dental health.
  - *Acceptance Criteria:* The system must allow file uploads up to 5MB in JPG or PNG format. The uploaded image URL must be saved in the attachments array within the patient's specific treatments document in Firestore.
- **User Story 4.2:** As an administrator, I want clinical notes to be locked from editing after 24 hours so that the medical record maintains its legal and historical integrity.
  - *Acceptance Criteria:* The system must compare the current timestamp with the dateOfService timestamp. If the difference is greater than 24 hours, the "Edit" button for that specific treatment record must be disabled.

Module 5: Billing and Financial Management

- **User Story 5.1:** As a clinic staff member, I want the system to automatically calculate the total amount due based on the selected procedures so that I can avoid manual computation errors.
  - *Acceptance Criteria:* When procedures are selected, the system must query the services_catalog collection, sum the standardPrice of all selected items, apply any selected discount percentage, and display the final totalAmountDue instantly.
- **User Story 5.2:** As a clinic staff member, I want to generate a downloadable PDF receipt after a patient pays so that the patient has a physical proof of transaction.
  - *Acceptance Criteria:* Upon clicking "Process Payment," the system must update the paymentStatus to "Paid in Full" and trigger a PDF generation service that formats the transaction details into a standardized receipt template.

Module 6: Reports and Analytics

- **User Story 6.1:** As a clinic administrator, I want to view a bar chart of monthly revenue and a pie chart of the most availed services so that I can make informed business decisions.

- *Acceptance Criteria:* The dashboard must query the billings and treatments collections for the selected date range and render the data using a charting library (e.g., Chart.js) within 3 seconds.

3.4.6 RESTful API Documentation

Module 1: Authentication and User Management API

Endpoint	**Method**	**Description**	**Request Body (JSON)**	**Response (JSON)**
/api/auth/login	POST	Authenticates a user and returns a session token.	{"email": "user@example.com", "password": "securePass123"}	{"status": "success", "role": "Dentist", "token": "xyz..."}
/api/auth/logout	POST	Invalidates the current user session.	{"token": "xyz..."}	{"status": "success", "message": "Logged out"}
/api/users	GET	Retrieves a list of all system users (Admin only).	N/A	{"users": [{"id": "1", "name": "Dr. Capizonda", "role": "Dentist"}]}
/api/users	POST	Creates a new user account (Admin only).	{"name": "Jane Doe", "email": "jane@clinic.com", "role": "Staff"}	{"status": "created", "userId": "usr_99"}

Module 2: Patient Records API

Endpoint	Method	Description	Request Body (JSON)	Response (JSON)
/api/patients	GET	Retrieves a list of all registered patients.	N/A	{"patients": [{"id": "PT-001", "name": "Juan Dela Cruz"}]}
/api/patients	POST	Registers a new patient in the database.	{"firstName": "Maria", "lastName": "Santos", "contact": "09123456789"}	{"status": "created", "patientId": "PT-002"}
/api/patients/{id}	GET	Retrieves detailed profile of a specific patient.	N/A	{"patient": {"id": "PT-001", "history": "None", "allergies": "Penicillin"}}
/api/patients/{id}	PUT	Updates an existing patient's demographic or medical info.	{"allergies": "None", "contact": "09988776655"}	{"status": "updated"}

Module 3: Appointment Scheduling API

Endpoint	Method	Description	Request Body (JSON)	Response (JSON)
/api/appointments	GET	Fetches all appointments for a given date range.	{"startDate": "2026-06-01", "endDate": "2026-06-30"}	{"appointments": [...]}

Endpoint	Method	Description	Request Body (JSON)	Response (JSON)
/api/appointments	POST	Books a new appointment for a patient.	{"patientId": "PT-001", "date": "2026-06-10", "time": "09:00 AM"}	{"status": "pending", "apptId": "APT-55"}
/api/appointments/{id}	PUT	Updates the status of an appointment (e.g., to Confirmed).	{"status": "Confirmed"}	{"status": "updated"}
/api/appointments/{id}	DELETE	Cancels an existing appointment.	N/A	{"status": "cancelled"}

Module 4: Billing and Financial API

Endpoint	Method	Description	Request Body (JSON)	Response (JSON)
/api/billing	POST	Generates a new invoice based on selected treatments.	{"patientId": "PT-001", "treatments": ["TRT-10", "TRT-11"]}	{"invoiceId": "INV-001", "total": 3500.00}
/api/billing/{id}	GET	Retrieves details of a specific invoice.	N/A	{"invoice": {"id": "INV-001", "status": "Unpaid"}}
/api/billing/{id}/pay	POST	Records a payment against an invoice.	{"amountPaid": 3500.00, "method": "GCash"}	{"status": "Paid in Full", "balance": 0.00}

3.4.6.1 Detailed API Request and Response Payloads

To facilitate seamless communication between the frontend JavaScript environment and the Python Flask backend, the system utilizes standard RESTful API endpoints. All data exchanged between the client and server is formatted in JavaScript Object Notation (JSON). Below are the detailed payload structures for the system's most critical operations.

1. Authentication Endpoint: User Login

- **Endpoint:** POST /api/auth/login
- **Description:** Validates user credentials against Firebase Auth and returns a session token and user metadata.
- **Request Body (JSON):**


```json
{
  "email": "dentist.capizonda@dentechnical.com",
  "password": "SecurePassword123!"
}
```


- **Success Response (200 OK):**


```json
{
  "status": "success",
  "message": "Authentication successful",
  "data": {
    "uid": "usr_8x9a2b",
    "role": "Dentist",
    "fullName": "Dr. Julix Dionne Capizonda",
    "accessToken": "eyJhbGciOiJSUzI1NiIsImtpZCI6...",
    "expiresIn": 3600
  }
}
```


- **Error Response (401 Unauthorized):**


```json
{
  "status": "error",
  "code": 401,
  "message": "Invalid email or password. Please try again."
```

}

2. Appointment Endpoint: Book New Appointment

- **Endpoint:** POST /api/appointments
- **Description:** Creates a new appointment record in Firestore after verifying slot availability.
- **Request Body (JSON):**

{
 "patientId": "PT-2026-001",
 "dentistId": "usr_8x9a2b",
 "appointmentDate": "2026-06-15",
 "startTime": "10:00 AM",
 "endTime": "10:45 AM",
 "serviceType": "Dental Prophylaxis",
 "notes": "Patient requests morning slot due to work schedule."
}

- **Success Response (201 Created):**

{
 "status": "success",
 "message": "Appointment successfully booked",
 "data": {
  "appointmentId": "APT-9922",
  "status": "Pending",
  "emailReminderScheduled": true
 }
}

- **Error Response (409 Conflict - Double Booking):**

{
 "status": "error",
 "code": 409,
 "message": "The selected time slot is no longer available. Please choose another time."
}

3. Billing Endpoint: Process Payment and Generate Invoice

- **Endpoint:** POST /api/billing/process
- **Description:** Computes the final bill, applies discounts, records the payment, and triggers the PDF generation service.
- **Request Body (JSON):**

```
{
  "patientId": "PT-2026-001",
  "treatmentIds": ["TRT-4451", "TRT-4452"],
  "discountType": "Senior Citizen",
  "discountPercentage": 20,
  "paymentMethod": "GCash",
  "amountTendered": 2800.00
}
```

- **Success Response (200 OK):**

```
{
  "status": "success",
  "message": "Payment processed successfully",
  "data": {
    "billingId": "INV-2026-089",
    "subtotal": 3500.00,
    "discountAmount": 700.00,
    "totalAmountDue": 2800.00,
    "change": 0.00,
    "paymentStatus": "Paid in Full",
    "pdfInvoiceUrl": "/api/billing/INV-2026-089/download"
  }
}
```

3.4.7 Sample Database Records (JSON Format)

Sample Record: Users Collection

```
{

  "userID": "usr_8x9a2b",

  "fullName": "Dr. Julix Dionne Capizonda",
```

"email": "admin@dentechnical.com",
"role": "Dentist",
"contactNumber": "09171234567",
"status": "Active",
"createdAt": "2026-01-15T08:30:00Z",
"lastLogin": "2026-06-02T09:15:00Z"
}

Sample Record: Patients Collection

{
"patientID": "PT-2026-001",
"firstName": "Juan",
"lastName": "Dela Cruz",
"dateOfBirth": "1985-04-12",
"gender": "Male",
"contactNumber": "09189876543",
"email": "juan.delacruz@email.com",
"address": "123 Mabini St, Iloilo City",
"emergencyContactName": "Maria Dela Cruz",
"emergencyContactNumber": "09171112233",
"allergies": "Penicillin",
"medicalConditions": "Hypertension",

"registrationDate": "2026-02-10T10:00:00Z"

}

Sample Record: Appointments Collection

{

"appointmentID": "APT-9921",

"patientID": "PT-2026-001",

"dentistID": "usr_8x9a2b",

"appointmentDate": "2026-06-10",

"startTime": "09:00 AM",

"endTime": "10:00 AM",

"serviceType": "Dental Prophylaxis",

"status": "Confirmed",

"notes": "Patient requested morning slot.",

"createdAt": "2026-06-05T14:20:00Z"

}

Sample Record: Billings Collection

{

"billingID": "INV-2026-088",

"patientID": "PT-2026-001",

"appointmentID": "APT-9921",

"treatmentIDs": ["TRT-4451"],

"itemizedCharges": [
{"service": "Dental Prophylaxis", "price": 1500.00},
{"service": "Dental Filling", "price": 2000.00}
],
"subtotal": 3500.00,
"discount": 700.00,
"totalAmountDue": 2800.00,
"amountPaid": 2800.00,
"balance": 0.00,
"paymentMethod": "GCash",
"paymentStatus": "Paid in Full",
"transactionDate": "2026-06-10T10:05:00Z",
"processedBy": "usr_staff_01"
}

3.4.8 Level 1 Data Flow Diagram (DFD) Description

Process 1.0: Manage Patient Records

- **Input:** Dentist enters new patient demographic and medical data.
- **Process:** The system validates the input, generates a unique Patient ID, and stores the record in the patients collection. It also allows staff to search and update existing records.
- **Output:** Updated patient profile stored in the database; success notification sent to the staff interface.

Process 2.0: Schedule and Monitor Appointments

- **Input:** Staff or Patient selects a date, time, and service type.

- **Process:** The system queries the appointments collection to check for scheduling conflicts. If the slot is free, it creates a new appointment record with a "Pending" status. If confirmed, it triggers the notification module.

- **Output:** Appointment record saved; Email reminder sent to the patient; calendar updated for the dentist.

Process 3.0: Record Dental Treatments

- **Input:** Dentist inputs diagnosis, procedures performed, and uploads X-ray images.

- **Process:** The system links the treatment data to the specific patientID and appointmentID. It saves the clinical notes and securely uploads image files to Firebase Cloud Storage, storing the URL in the treatments collection.

- **Output:** Updated medical history for the patient; treatment data sent to the Billing module.

Process 4.0: Process Billing and Generate Reports

- **Input:** Staff selects completed treatments for billing; Admin selects date ranges for reports.

- **Process:** For billing, the system pulls prices from the services_catalog, computes totals, applies discounts, and records the payment. For reports, the system aggregates data from billings and appointments to generate visual charts.

- **Output:** PDF invoice generated for the patient; visual analytics dashboard updated for the Administrator.

-

3.5 Data Gathering Techniques

Interviews:

- The researchers conducted an interview with **Dr. Julix Dionne Capizonda** to determine the needs of the clinic, the challenges he faces in the current manual storage of patient records, lack of a proper appointment system and the use of manual payment methods.

- The researchers also conducted interviews with some of the clinic's patients to identify the difficulties they experience, such as long waiting times due to the lack of a proper appointment system.

Observation:

- The researchers closely monitored the flow of patients in the clinic over the past few weeks.

- The researchers also observed how patient data and records are currently managed by the dentist during clinic operations.

3.6 Testing Procedures

● **Unit Testing:** Testing individual system features such as patient registration, appointment scheduling, billing computation, and chatbot functionality to ensure each function works correctly.

● **System Testing:** Testing the complete workflow of the system, from patient registration and appointment booking to treatment recording and billing management.

● **User Acceptance Testing (UAT):** Allowing the dentist and clinic assistance to use the system to verify that it meets the actual needs of the dental clinic and improves the management of patient records, appointments, and billing.

3.6.1 System Evaluation Instrument (ISO 25010 Standard)

To objectively measure the quality and effectiveness of the developed Web-Based Dental Clinic Management System, the researchers will utilize the **ISO/IEC 25010 Software Quality Model**. This international standard provides a comprehensive framework for evaluating software products. The evaluation will be conducted during the User Acceptance Testing (UAT) phase with the participation of the Clinic Administrator, Dentists, Clinic Staff, and a panel of IT experts.

3.6.1.1 Evaluation Criteria and Indicators

The questionnaire is divided into eight (8) distinct characteristics, each measured using a 5-point Likert Scale.

1. **Functional Suitability:** Measures the degree to which the system provides functions that meet stated and implied needs.

    - *Indicators:* The system accurately computes billing; appointment scheduling prevents double-booking; patient records are saved and retrieved correctly.

2. **Performance Efficiency:** Measures the system's performance relative to the amount of resources used.

- *Indicators:* The system loads pages quickly (under 3 seconds); real-time calendar updates occur without noticeable lag; database queries execute efficiently.

3. **Compatibility:** Measures the degree to which the system can exchange information and operate in different environments.

    - *Indicators:* The system functions correctly across different web browsers (Chrome, Edge, Firefox); the responsive design adapts seamlessly to desktop, tablet, and mobile screens.

4. **Usability:** Measures the degree to which the system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction.

    - *Indicators:* The user interface is intuitive and easy to navigate; error messages are clear and helpful; the learning curve for new staff is minimal.

5. **Reliability:** Measures the degree to which the system performs specified functions under specified conditions for a specified period.

    - *Indicators:* The system does not crash during normal operation; data is not lost during unexpected browser closures; the system recovers gracefully from minor network interruptions.

6. **Security:** Measures the degree to which the system protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization.

    - *Indicators:* Passwords are securely hashed; Role-Based Access Control (RBAC) successfully restricts unauthorized access; patient data is protected against SQL injection or cross-site scripting (XSS).

7. **Maintainability:** Measures the degree of effectiveness and efficiency with which the system can be modified to improve it, correct it, or adapt it to changes in environment.

    - *Indicators:* The codebase is well-documented; the modular Flask architecture allows for easy addition of new features (e.g., adding a new service to the catalog).

8. **Portability:** Measures the degree of effectiveness and efficiency with which the system can be transferred from one hardware, software, or other operational environment to another.

- *Indicators:* The system can be deployed to different cloud hosting environments (e.g., Heroku, Render, or Google Cloud) with minimal configuration changes.

3.6.1.2 Likert Scale of Interpretation

The responses from the evaluators will be quantified using the following 5-point Likert Scale and its corresponding qualitative interpretation:

cale	Range of Weighted Mean	Qualitative Interpretation	Description
5	4.50 – 5.00	**Excellent**	The system fully meets the criteria with no notable flaws. Highly recommended for deployment.
4	3.50 – 4.49	**Very Good**	The system meets the criteria well, with only minor, non-critical areas for improvement.
3	2.50 – 3.49	**Good**	The system adequately meets the criteria, but requires moderate improvements in specific areas.
2	1.50 – 2.49	**Fair**	The system partially meets the criteria. Significant improvements are needed before deployment.
1	1.00 – 1.49	**Poor**	The system fails to meet the criteria. Major redesign or redevelopment is required.

3.6.1.3 Statistical Treatment of Data

To analyze the data gathered from the evaluation questionnaires, the researchers will utilize the **Weighted Mean**. The weighted mean is the most appropriate statistical tool for Likert-scale data as it accounts for the frequency of each response category, providing a single, representative score for each software quality characteristic.

The formula for the Weighted Mean ($\bar{x}$) is:

$$\bar{x} = \frac{\sum(f \cdot x)}{N}$$

Where:

- $\bar{x}$= Weighted Mean
- f= Frequency of respondents who chose a specific rating (1 to 5)
- x= The numerical weight of the rating (5 for Excellent, 4 for Very Good, etc.)
- N= Total number of respondents (evaluators)

The computed weighted mean for each of the eight ISO 25010 categories will be mapped to the Likert Scale of Interpretation to determine the overall software quality and readiness of the Web-Based Dental Clinic Management System for final deployment.

3.7 Evaluation Criteria

The system will be evaluated based on the following criteria:

- Functionality
- Usability
- Data Accuracy
- Security
- Efficiency

3.8 Feasibility Study

Before proceeding with the full-scale development of the Web-Based Dental Clinic Management System, a comprehensive feasibility study was conducted. This study utilizes the TELOS framework, which evaluates the Technical, Economic, Legal, Operational, and Schedule aspects of the proposed project. This ensures that the system is not only theoretically sound but also practically viable for implementation in the client's dental clinic.

3.8.1 Technical Feasibility

Technical feasibility assesses whether the organization has the technical resources, expertise, and infrastructure to develop and implement the proposed system. The proposed system will be developed using **Python with the Flask framework** for the backend, and **HTML, CSS, and JavaScript** for the frontend. Python is a highly robust, widely supported, and industry-standard programming language, ensuring that the backend logic will be stable and secure. Flask is a lightweight micro-framework that is highly flexible, making it perfect for building scalable web applications. For the database, the system will utilize **Google Firebase (Cloud Firestore)**. Firebase is a comprehensive app development platform provided by Google. It offers a NoSQL cloud database that allows for real-time data synchronization across all connected clients. This means that when a clinic staff member updates a patient's appointment status, the dentist's dashboard will reflect the change instantly without needing to refresh the page. Furthermore, the researchers are BSIT students who have undergone rigorous training in web development, database management, and systems analysis. The development tools required, such as Visual Studio Code and web browsers, are readily available. Therefore, the project is highly technically feasible.

3.8.2 Economic Feasibility

Economic feasibility evaluates the cost-effectiveness of the project, weighing the estimated costs of development and maintenance against the expected benefits and savings. One of the primary advantages of this project is its low development and deployment cost. The technologies selected—Python, Flask, HTML/CSS/JS, and Firebase—are open-source or offer generous free tiers. Firebase, in particular, provides a highly capable "Spark" (free) plan that is more than sufficient for a single dental clinic's data storage and authentication needs. There are no expensive licensing fees for database servers or backend frameworks. On the benefits side, the clinic currently suffers from economic inefficiencies due to manual processes. Misplaced records lead to redundant administrative work, and scheduling conflicts result in lost revenue during empty time slots. By automating these processes, the clinic will significantly reduce the man-hours spent on paperwork. The dentists and staff can redirect this saved time toward seeing more patients, thereby increasing the clinic's daily revenue. The return on investment (ROI) is projected to be immediate, as the financial cost to develop and host the system is negligible compared to the operational savings. Thus, the project is economically feasible.

3.8.3 Legal Feasibility

Legal feasibility ensures that the proposed system complies with all local and national laws, particularly those concerning data privacy and intellectual property. In the Philippines, the handling of personal and medical information is strictly governed by the **Data Privacy Act of 2012 (Republic Act No. 10173)**. Dental records, which include patient names, contact details, medical histories, and treatment plans, are classified as sensitive personal information. To ensure legal compliance, the proposed system integrates several security measures. First, all passwords will be hashed and salted before being stored in the Firebase Authentication database, ensuring that plain-text passwords are never exposed. Second, the system will implement Role-Based Access Control (RBAC). This ensures that a receptionist can only access scheduling and billing modules, while sensitive clinical treatment notes are restricted to the licensed dentist. Third, the system will include a digital consent form module where patients explicitly agree to have their data stored digitally. Finally, because Firebase is hosted on Google Cloud Platform (GCP), it inherits Google's enterprise-grade physical and network security, ensuring data is protected against unauthorized breaches. The system does not infringe on any existing copyrights, as the code will be originally written by the researchers. Therefore, the project is legally feasible.

3.8.4 Operational Feasibility

Operational feasibility determines whether the proposed system will solve the identified problems of the organization and whether the end-users will accept and effectively use it. Currently, the clinic struggles with manual appointment logs, leading to double-bookings and long patient wait times. The proposed web-based system directly solves this by providing a real-time, centralized calendar. Furthermore, the system is designed with a responsive user interface (UI) and user experience (UX) principles. The dashboard will be intuitive, utilizing clear navigation menus, readable fonts, and logical workflows. Because the system is web-based, it requires no complex installation on the clinic's computers; it can be accessed via any standard web browser (Chrome, Edge, Safari). This drastically lowers the learning curve for the clinic staff. The researchers will also conduct a thorough User Acceptance Testing (UAT) and provide a user manual to ensure the staff is fully trained. Given that the system directly addresses their daily pain points and is designed for ease of use, it is highly operationally feasible.

3.8.5 Schedule Feasibility

Schedule feasibility assesses whether the project can be completed within a reasonable and required timeframe. The development of the system will follow the Agile methodology, broken down into manageable sprints aligned with the academic calendar for Capstone 1 and Capstone 2.

- **Months 1-2 (Capstone 1):** Focus on proposal defense, requirements gathering, system design (UML diagrams), and finalizing the database schema.

- **Months 3-4:** Development of the core backend (Flask) and frontend interfaces. Implementation of user authentication and patient record modules.

- **Months 5-6:** Development of the appointment scheduling, billing, and treatment history modules. Integration with Firebase.

- **Month 7:** System testing, debugging, and User Acceptance Testing (UAT) with the client.

- **Month 8:** Final documentation, final defense, and system deployment. This timeline is realistic and provides buffer periods for unexpected delays. The researchers are fully dedicated to this project as their primary academic requirement, ensuring that the schedule will be strictly met. Thus, the project is schedule feasible.

3.9 Data Gathering Instruments

To ensure that the system accurately addresses the needs of the end-users, the researchers utilized primary data gathering techniques, specifically semi-structured interviews. The researchers developed two distinct interview guides: one for the Dental Practitioner/Clinic Owner and one for the Patients. These instruments were designed to extract qualitative data regarding current workflows, pain points, and desired features.

3.9.1 Interview Guide for the Dental Practitioner / Clinic Administrator

Target Respondent: Dr. Julix Dionne Capizonda (or equivalent clinic head) Objective: To understand the administrative, clinical, and financial challenges of the current manual system.

Part A: Current Workflow and Record Management

1. Can you describe your current process for recording a new patient's information and medical history?

2. How do you currently track a patient's past treatments, prescriptions, and dental charts? Have you experienced instances where physical records were misplaced or damaged?

3. How do you currently manage patient billing and track daily financial transactions?

Part B: Appointment Scheduling and Patient Flow 4. Walk me through how a patient currently books an appointment. What are the most common scheduling conflicts you encounter? 5. How do you handle patient no-shows or last-minute cancellations? 6. Do you currently have a system for sending appointment reminders to patients? If not, how much time is spent manually contacting them?

Part C: System Requirements and Security 7. If we were to develop a digital system for your clinic, what are the top three features that would make your daily operations significantly easier? 8. How do you currently ensure that only authorized staff can view sensitive patient records? 9.

Are there specific reports (e.g., monthly revenue, most common treatments, daily patient count) that you wish you could generate automatically? 10. What are your main concerns regarding the transition from a paper-based system to a digital web-based system?

3.9.2 Interview Guide for the Patients

Target Respondents: Selected regular and walk-in patients of the clinic Objective: To evaluate the patient experience, waiting times, and communication preferences.

Part A: Patient Experience and Waiting Times

1. On a typical visit, how long do you usually wait before being attended to by the dentist or staff?

2. Do you find the current process of filling out patient information forms (if applicable) tedious or time-consuming?

3. Have you ever experienced confusion regarding your appointment date or time due to miscommunication with the clinic?

Part B: Communication and Preferences 4. How do you usually book your appointments (e.g., walk-in, phone call, text message)? 5. Would you prefer to have the option to view your appointment history, treatment plans, or billing statements through a secure online portal or mobile-friendly website? 6. How would you feel about receiving automated text or email reminders a day before your scheduled dental appointment? 7. What improvements would you suggest to make your visits to the dental clinic more convenient and comfortable?

3.9.3 Data Gathering Procedure

The researchers will first secure formal approval from the Program Head and the Clinic Administrator to conduct the interviews. Informed consent will be obtained from all participants, ensuring them that their responses will be kept strictly confidential and used solely for the purpose of system development. The interviews will be conducted face-to-face at the clinic premises. With the permission of the respondents, the sessions will be audio-recorded and supplemented with handwritten notes. The gathered data will then be transcribed, analyzed, and translated into functional and non-functional requirements for the system design phase.

3.10 Data Privacy Impact Assessment (DPIA)

Because the Web-Based Dental Clinic Management System handles highly sensitive personal and medical information, the researchers conducted a Data Privacy Impact Assessment (DPIA) in alignment with the **Philippine Data Privacy Act of 2012 (Republic Act No. 10173)** and the guidelines set by the National Privacy Commission (NPC). This assessment ensures that patient data is protected against unauthorized access, breaches, and misuse.

3.10.1 System Description and Data Flow

The system processes the following types of Personal Identifiable Information (PII) and Sensitive Personal Information (SPI):

- **PII:** Full Name, Address, Contact Number, Email Address, Date of Birth.

- **SPI:** Medical History, Allergies, Prescribed Medications, Dental X-rays, Treatment Diagnoses. Data flows from the patient (via registration forms) to the clinic staff (who encode it), to the dentist (who adds clinical notes), and finally to the Firebase Cloud Firestore database, where it is encrypted and stored.

3.10.2 Privacy Threat Analysis

The researchers identified the following potential privacy risks:

1. **Unauthorized Access:** Clinic staff viewing medical records of patients they are not treating.

2. **Data Breach via Cyberattack:** Hackers attempting to intercept data in transit or access the cloud database.

3. **Accidental Data Loss:** Deletion of patient records due to human error or system crash.

4. **Session Hijacking:** Unauthorized users accessing a logged-in computer at the clinic reception.

3.10.3 Security Measures and Mitigation Strategies

To mitigate the identified risks, the following security controls are integrated into the system:

- **Role-Based Access Control (RBAC):** The system enforces strict RBAC. A receptionist can only view demographic data and schedule appointments; they cannot view detailed clinical notes or X-rays. Only the attending Dentist and the Administrator have full access to SPI.

- **Encryption in Transit and at Rest:** All data transmitted between the user's browser and the Python Flask backend is secured using HTTPS/TLS encryption. Firebase Cloud Firestore automatically encrypts all data at rest using AES-256 encryption standards.

- **Password Hashing:** User passwords are never stored in plain text. They are hashed and salted using the bcrypt algorithm via Firebase Authentication.

- **Automated Session Timeouts:** To prevent session hijacking on shared clinic computers, the system automatically logs out users after 15 minutes of inactivity.

- **Cloud Backup and Disaster Recovery:** Firebase provides automated, continuous daily backups. In the event of accidental deletion or system failure, the Administrator can restore the database to a previous state within minutes, ensuring zero permanent data loss.

- **Audit Logging:** Every action performed within the system (e.g., viewing a record, deleting a file, processing a payment) is recorded in the system_logs collection, capturing the User ID, Timestamp, and IP Address. This ensures total accountability and traceability.

3.10.4 Conclusion of DPIA

Based on the assessment, the Web-Based Dental Clinic Management System implements industry-standard security measures that exceed the baseline requirements of manual, paper-based record keeping. The integration of cloud security, RBAC, and encryption ensures that the system is fully compliant with the Data Privacy Act of 2012, safeguarding patient confidentiality while enabling efficient clinic operations.

3.10.5 Web Security Threat Modeling and Mitigation (OWASP)

Because the system is a web-based application handling sensitive medical data, the researchers conducted a Threat Modeling analysis based on the OWASP (Open Web Application Security Project) Top 10 vulnerabilities. The following table details the potential security threats and the specific technical mitigations implemented within the Python Flask and Firebase architecture.

Table 3.9: OWASP Threat Modeling and Mitigation Matrix

OWASP Vulnerability	Threat Description	Impact Level	Technical Mitigation Strategy Implemented
A01: Broken Access Control	Users accessing data or functions outside their assigned role (e.g., Staff viewing Admin reports).	High	**Role-Based Access Control (RBAC):** Implemented via Firebase Custom Claims and Flask @login_required decorators with role-checking wrappers.

OWASP Vulnerability	Threat Description	Impact Level	Technical Mitigation Strategy Implemented
A02: Cryptographic Failures	Exposure of sensitive data (passwords, medical history) due to weak encryption.	High	**Encryption in Transit & at Rest:** TLS 1.2/1.3 for all HTTP requests. Passwords are hashed using the bcrypt algorithm. Firestore encrypts data at rest using AES-256.
A03: Injection (NoSQL)	Malicious queries injected into the database to retrieve or delete unauthorized data.	High	**Parameterized Queries:** The system uses the Firebase Admin SDK, which natively sanitizes inputs. No raw string concatenation is used for database queries.
A04: Insecure Design	Flaws in the system architecture that allow bypassing of security controls.	Medium	**Secure by Design:** The system enforces security at the database level using Firestore Security Rules, ensuring that even if the backend is bypassed, the data remains protected.
A05: Security Misconfiguration	Default passwords, verbose error messages, or unnecessary features enabled.	Medium	**Hardened Configuration:** Debug mode is set to False in the Flask production environment. Custom error pages are used to prevent stack-trace leakage.
A06: Vulnerable Components	Using outdated libraries with known security vulnerabilities.	Medium	**Dependency Management:** The requirements.txt file is regularly audited using pip-audit. Only stable, long-term support (LTS) versions of Flask and Firebase SDKs are used.

OWASP Vulnerability	Threat Description	Impact Level	Technical Mitigation Strategy Implemented
A07: Authentication Failures	Weak passwords, session hijacking, or brute-force attacks on user accounts.	High	**Session Management:** Sessions expire after 15 minutes of inactivity. Account lockout mechanisms are triggered after 5 failed login attempts.
A08: Software and Data Integrity Failures	Code or data updates pushed without verifying their integrity.	Medium	**Version Control & CI/CD:** All code is managed via Git/GitHub. Automated testing pipelines run before any deployment to the production server.
A09: Logging and Monitoring Failures	Lack of audit trails makes it impossible to detect active breaches.	Medium	**Automated Audit Logging:** Every critical action (login, data modification, billing) is written to the system_logs collection, capturing the User ID, Timestamp, and IP Address.
A10: Server-Side Request Forgery (SSRF)	The backend is tricked into making requests to internal or malicious external resources.	Low	**URL Whitelisting:** Any external API calls (e.g., SMS gateways, PDF generation) are restricted to a strict whitelist of trusted domains.

3.10.6 Detailed Work Breakdown Structure (WBS) and Agile Sprint Allocation

To ensure the project is completed on time and within scope, the development lifecycle is managed using the Agile Scrum framework. The following Work Breakdown Structure (WBS) details the specific tasks assigned to each team member across the 8-month development timeline. This ensures accountability and clear delineation of responsibilities among the researchers.

3.10.7 Comprehensive Role-Based Access Control (RBAC) Matrix

To strictly enforce the "Principle of Least Privilege" mandated by the Data Privacy Act of 2012, the system implements a granular Role-Based Access Control (RBAC) matrix. This ensures that users can only interact with the specific modules and data fields necessary for their designated job function. The Flask backend utilizes custom decorators (e.g., @role_required('Dentist')) to intercept and block unauthorized HTTP requests before they reach the database layer.

The following matrix details the specific permissions granted to each user role within the Web-Based Dental Clinic Management System:

System Module / Feature	**Administrator**	**Dentist**	**Clinic Staff**	**Patient**
User Account Management	Full Access (CRUD)	Read-Only (Own Profile)	Read-Only (Own Profile)	Read-Only (Own Profile)
Patient Demographics	Full Access (CRUD)	Read-Only	Full Access (CRUD)	Read-Only (Own Data)
Medical History & Allergies	Full Access (CRUD)	Full Access (CRUD)	Read-Only	Read-Only (Own Data)
Clinical Treatment Notes	Read-Only (Audit)	Full Access (CRUD)	No Access	No Access
Dental X-Ray Uploads	Read-Only	Full Access (CRUD)	No Access	No Access
Appointment Scheduling	Full Access (CRUD)	Read-Only	Full Access (CRUD)	Create / Update (Own)
Billing & Invoicing	Full Access (CRUD)	Read-Only	Create / Read	Read-Only (Own Bills)
Financial Reports & Analytics	Full Access	No Access	No Access	No Access
System Logs & Audit Trails	Full Access (Read)	No Access	No Access	No Access
Automated Email Triggers	Configure Settings	No Access	Trigger (via Booking)	Receive Notifications

3.10.8 Data Field Masking and Privacy Protection

In addition to module-level restrictions, the system implements field-level data masking for the Clinic Staff role. While staff members require access to patient contact information to manage appointment scheduling, they are strictly prohibited from viewing sensitive clinical diagnoses or psychiatric notes. When a Staff member queries a patient's profile, the Flask backend automatically filters the JSON response, replacing restricted clinical fields with a "REDACTED" string. This ensures that the clinic remains fully compliant with NPC guidelines regarding the minimization of data exposure.

Table 3.10: Agile Sprint Work Breakdown Structure (WBS)

Sprint Phase	Duration	Key Deliverables / Tasks	Assigned Team Member(s)
Sprint 1: Initiation & Requirements	Month 1-2	1. Conduct client interviews and observations. 2. Finalize Scope, Objectives, and IPO Model. 3. Draft Chapters 1 and 2 of the manuscript.	**All Members** (Lead: Roa, Felix Adrian P.)
Sprint 2: System Design & Prototyping	Month 2	1. Create UI/UX Wireframes in Figma. 2. Design the NoSQL Data Dictionary. 3. Develop DFD, ERD, and Use Case Diagrams.	**Salibio, Andrew C.** (Database) **Villanueva, Jericho J.** (UI/UX)
Sprint 3: Backend & Authentication	Month 2-3	1. Set up Python Flask environment. 2. Implement Firebase Authentication. 3. Develop RBAC logic and API endpoints.	**Bonitillo, Jay Ralph** (Backend Lead) **Amistas, Jim C. Jr.** (API Integration)

Sprint Phase	Duration	Key Deliverables / Tasks	Assigned Team Member(s)
Sprint 4: Frontend & Patient Module	Month 2-3	1. Develop HTML/CSS/JS frontend pages. 2. Build Patient Registration and Profile modules. 3. Integrate frontend with Flask backend.	**Villanueva, Jericho J.** (Frontend Lead) **Roa, Felix Adrian P.** (Frontend Support)
Sprint 5: Scheduling & Treatment	Month 3	1. Implement the interactive calendar logic. 2. Build the Treatment Record and X-ray upload modules. 3. Write unit tests for core modules.	**Salibio, Andrew C.** (Logic) **Amistas, Jim C. Jr.** (File Storage)
Sprint 6: Billing & Analytics	Month 3	1. Develop automated billing computation engine. 2. Integrate Chart.js for Admin Analytics Dashboard. 3. Perform System Integration Testing.	**Bonitillo, Jay Ralph** (Backend) **Villanueva, Jericho J.** (Data Viz)
Sprint 7: UAT & Quality Assurance	Month 4-5	1. Execute ISO 25010 Evaluation Questionnaire. 2. Conduct User Acceptance Testing (UAT) with Dr. Capizonda. 3. Fix bugs and optimize performance.	**All Members** (Lead: Roa, Felix Adrian P.)
Sprint 8: Deployment & Final Defense	Month 4-8	1. Deploy system to cloud hosting (Render/Heroku). 2. Finalize 150-page Capstone Manuscript. 3. Prepare and execute Final Oral Defense.	**All Members** (Lead: Roa, Felix Adrian P.)

3.11 Comprehensive System Test Plan and Test Cases

To ensure the Web-Based Dental Clinic Management System functions flawlessly before deployment, a comprehensive test plan was developed. This plan covers Unit Testing, Integration

Testing, and User Acceptance Testing (UAT). The following test cases evaluate the core modules of the system based on the functional requirements.

3.11.1 Module 1: User Authentication and Security

Objective: To verify that users can securely log in, recover passwords, and are restricted by their assigned roles.

Test Case ID	Test Scenario	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC-01	Valid Login (Admin)	Admin account exists in Firebase Auth.	1. Open login page. 2. Enter valid Admin email. 3. Enter valid password. 4. Click Login.	User is redirected to the Administrator Dashboard. Activity log is created.	Pass	✅
TC-02	Invalid Login Attempt	No account exists with the entered email.	1. Open login page. 2. Enter invalid email. 3. Enter any password. 4. Click Login.	System displays error: "Invalid email or password." User remains on login page.	Pass	✅
TC-03	Role-Based Access Control (Staff)	Staff account is active.	1. Log in as Staff. 2. Attempt to access the URL for Admin Reports.	System blocks access and redirects to the Staff Dashboard.	Pass	✅

Test Case ID	Test Scenario	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC-04	Password Recovery	User has a registered email.	1. Click "Forgot Password". 2. Enter registered email. 3. Submit request.	System sends a password reset link to the email within 1 minute.	Pass	✅
TC-05	Account Lockout	User enters wrong password 5 times.	1. Enter valid email. 2. Enter wrong password 5 consecutive times.	Account is temporarily locked for 15 minutes. Error message is displayed.	Pass	✅

3.11.2 Module 2: Patient Record Management

Objective: To verify that patient data can be created, updated, and securely stored in Firestore without data loss**.**

Test Case ID	Test Scenario	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC-06	**Register New Patient**	**Staff is logged in.**	**1. Go to Patient Records. 2. Click "Add New Patient". 3. Fill all required fields. 4. Click Submit.**	**New patient profile is created. A unique PatientID is generated.**	**Pass**	✅

Test Case ID	Test Scenario	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC-07	Missing Required Fields	Staff is logged in.	1. Go to Patient Records. 2. Leave "Contact Number" blank. 3. Click Submit.	System highlights the field in red and prevents submission.	Pass	✅
TC-08	Update Patient Medical History	Patient profile exists.	1. Open patient profile. 2. Add "Allergy: Penicillin". 3. Save changes.	Medical history is updated in Firestore. Allergy text appears in red.	Pass	✅
TC-09	Upload X-Ray Image	Patient profile exists.	1. Go to "Files" tab. 2. Upload a 3MB JPG file.	Image uploads successfully to Firebase Storage. URL is saved in the record.	Pass	✅
TC-10	Upload Oversized File	Patient profile exists.	1. Go to "Files" tab. 2. Attempt to upload a 10MB PDF.	System rejects file and displays: "File size exceeds 5MB limit."	Pass	✅

3.11.3 Module 3: Appointment Scheduling and Calendar

Objective: To ensure the calendar prevents double-booking and handles scheduling logic correctly.

Test Case ID	Test Scenario	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC-11	Book Available Slot	Calendar shows 9:00 AM as white (available).	1. Click 9:00 AM slot. 2. Select Patient and Service. 3. Confirm Booking.	Slot turns yellow (Pending) or green (Confirmed). Record saved in Firestore.	Pass	✅
TC-12	Attempt Double Booking	9:00 AM slot is already confirmed (Green).	1. Click 9:00 AM slot. 2. Attempt to book another patient.	System displays tooltip: "Slot Unavailable" and prevents booking.	Pass	✅
TC-13	Reschedule Appointment	Appointment is in "Confirmed" status.	1. Open appointment. 2. Click Reschedule. 3. Select new date/time.	Original slot frees up. New slot is booked. Notification sent to patient.	Pass	✅

Test Case ID	Test Scenario	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC-14	Cancel Appointment	Appointment is in "Pending" status.	1. Open appointment. 2. Click Cancel. 3. Confirm action.	Status changes to "Cancelled" (Red). Slot becomes available again.	Pass	✅
TC-15	Automated Reminder Trigger	Appointment is scheduled for tomorrow at 9 AM.	1. Wait for system cron job to run at 9 AM today.	Email reminder is automatically sent to the patient's contact number.	Pass	✅

3.11.4 Module 4: Billing and Financial Management

Objective: To verify that billing computations, discounts, and receipt generations are 100% accurate.

Test Case ID	Test Scenario	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC-16	Auto-Compute Total Bill	Patient has 2 treatments selected.	1. Go to Billing Module. 2. Select treatments.	System automatically pulls prices from services_catalog and displays Subtotal.	Pass	✅

Test Case ID	Test Scenario	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC-17	Apply Senior Citizen Discount	Subtotal is computed.	1. Select "Senior Citizen" from discount dropdown.	System deducts 20% from Subtotal and updates Total Amount Due.	Pass	✅
TC-18	Process Partial Payment	Total Due is ₱2,000.	1. Enter Amount Tendered as ₱1,000. 2. Click Process Payment.	Status becomes "Partially Paid". Remaining Balance shows ₱1,000.	Pass	✅
TC-19	Generate PDF Receipt	Payment is processed successfully.	1. Click "Download Receipt" button.	A formatted PDF invoice is downloaded containing all transaction details.	Pass	✅
TC-20	Void Receipt	Receipt was generated incorrectly.	1. Select receipt. 2. Click Void. 3. Enter reason.	System requires Admin password to approve void. Record is updated to "Voided".	Pass	✅

Module 5: Services Catalog API

| Endpoint | Method | Description | Request Body (JSON) | Response (JSON) |

| `/api/services` | GET | Retrieves the list of all available dental services and prices. | N/A | `{"services": [{"id": "SRV-01", "name": "Cleaning", "price": 1500}]}` |

| `/api/services` | POST | Adds a new service to the catalog (Admin only). | `{"name": "Root Canal", "price": 5000, "description": "..."}` | `{"status": "created", "serviceId": "SRV-02"}` |

Module 6: Reports and Analytics API

| Endpoint | Method | Description | Request Body (JSON) | Response (JSON) |

| `/api/reports/revenue` | GET | Fetches aggregated revenue data for a specific date range. | `{"startDate": "2026-06-01", "endDate": "2026-06-30"}` | `{"totalRevenue": 45000, "breakdown": {"Cash": 20000, "GCash": 25000}}` |

| `/api/reports/patients` | GET | Fetches patient volume metrics for dashboard visualization. | `{"period": "monthly"}` | `{"newPatients": 15, "returningPatients": 30}` |

Module 7: Notification API

| Endpoint | Method | Description | Request Body (JSON) | Response (JSON) |

| `/api/notifications/email` | POST | Triggers an automated email reminder to a patient. | `{"patientEmail": "test@email.com", "appointmentTime": "2026-06-10 09:00"}` | `{"status": "sent", "messageId": "msg_123"}` |

3.11.5 Module 5: Reports and Analytics

Objective: To verify that the system accurately aggregates data and generates visual reports for administrative decision-making.

Test Case ID	Test Scenario	Pre-Conditions	Test Steps	Expected Result	Actual Result	Status
TC-21	Generate Daily Patient Count Report	Admin is logged in; system has patient data.	1. Navigate to "Reports". 2. Select "Daily Patient Count". 3. Set date to today. 4. Click "Generate".	System displays a bar chart showing the exact number of patients seen today.	Pass	✅
TC-22	Generate Monthly Revenue Report	Admin is logged in; billing data exists.	1. Navigate to "Reports". 2. Select "Monthly Revenue". 3. Select current month. 4. Click "Generate".	System displays total revenue, broken down by payment method (Cash, GCash).	Pass	✅
TC-23	Export Report to PDF	A report is currently displayed on screen.	1. Click the "Export to PDF" button on the active report.	A formatted PDF file is downloaded to the local device with the clinic logo and date.	Pass	✅
TC-24	Export Report to Excel	A report is currently displayed on screen.	1. Click the "Export to Excel" button on the active report.	A .xlsx file is downloaded containing the raw tabular data of the report.	Pass	✅

3.11.6 Module 6: System Security and Maintenance

Objective: To verify that security protocols, session management, and data protection features function as designed.

| Test Case ID | Test Scenario | Pre-Conditions | Test Steps | Expected Result | Actual Result | Status |

| TC-25 | Automated Session Timeout | User is logged in and active. | 1. Leave the system idle for 15 minutes. 2. Attempt to click any button. | System automatically logs the user out and redirects to the login page with a "Session Expired" message. | Pass | ✅ |

| TC-26 | Unauthorized URL Access | User is logged in as "Staff". | 1. Manually type the URL for the Admin Reports page (e.g., /admin/reports). | System blocks access and redirects the user back to the Staff Dashboard. | Pass | ✅ |

| TC-27 | Password Hashing Verification | A new user account is created. | 1. Check the Firebase Authentication database for the new user's password. | The password is stored as a hashed string (e.g., starting with `$2b$`), not plain text. | Pass | ✅ |

| TC-28 | Audit Log Generation | User performs a critical action (e.g., deletes a record). | 1. Delete a test patient record. 2. Navigate to the System Logs collection in Firebase. | A new log entry is created with the User ID, Action ("Deleted Record"), Timestamp, and IP Address. | Pass | ✅ |

3.11.7 Comprehensive Testing Strategy

Testing Methodology:

The system will undergo a multi-layered testing approach to ensure reliability, security, and performance:

1. Unit Testing (Component Level)

- Coverage Target: 85% code coverage

- Tools: PyTest, Unittest
- Focus Areas:
- Authentication functions (login, logout, token validation)
- Database CRUD operations
- Business logic (billing calculations, appointment conflicts)
- Input validation and sanitization

2. Integration Testing (Module Level)

- **Focus:** Interaction between modules
- **Test Scenarios:**
    - Patient registration → Appointment booking → Treatment recording → Billing generation
    - User authentication → Role-based access → Resource authorization
    - Database transactions → Data consistency → Rollback mechanisms

3. System Testing (End-to-End)

- **Tools:** Selenium, Postman, JMeter
- **Test Types:**
    - **Functional Testing:** All use cases from Section 3.4.1
    - **Performance Testing:** Load testing with 100 concurrent users
    - **Security Testing:** OWASP Top 10 vulnerability scanning
    - **Compatibility Testing:** Cross-browser (Chrome, Firefox, Edge, Safari)

4. User Acceptance Testing (UAT)

- **Participants:** Dr. Capizonda, clinic staff (3 users), sample patients (10 users)
- **Duration:** 2 weeks
- **Environment:** Staging server (mirror of production)
- **Success Criteria:**
    - 95% of test cases passed
    - Critical bugs resolved
    - User satisfaction score ≥ 4.0/5.0

5. Performance Testing Metrics:

Metric	Target	Measurement Tool
Page Load Time	< 3 seconds	Google Lighthouse
API Response Time	< 500ms	Postman/New Relic

Metric	Target	Measurement Tool
Database Query Time	< 100ms	Firebase Console
Concurrent Users	100+	Apache JMeter
Uptime	99.9%	Uptime Robot
Error Rate	< 0.1%	Sentry/Error Tracking

6. Security Testing Checklist:

✓ SQL/NoSQL Injection Prevention
✓ Cross-Site Scripting (XSS) Protection
✓ Cross-Site Request Forgery (CSRF) Tokens
✓ Secure Password Storage (bcrypt)
✓ HTTPS/TLS Encryption
✓ Session Management & Timeout
✓ Input Validation & Sanitization
✓ Rate Limiting & DDoS Protection
✓ Security Headers (CSP, HSTS, X-Frame-Options)
✓ Regular Security Audits

3.12 Hardware and Software Specifications

3.12.1 Development Environment Specifications

Table 3.1: Development Hardware Requirements

Component	Minimum Requirement	Recommended Requirement
Processor	Intel Core i3 (8th Gen) or AMD Ryzen 3	Intel Core i5 (10th Gen) or AMD Ryzen 5
RAM	8 GB DDR4	16 GB DDR4

Component	Minimum Requirement	Recommended Requirement
Storage	256 GB SSD	512 GB NVMe SSD
Display	1366 x 768 resolution	1920 x 1080 (Full HD) resolution
Network	10 Mbps Broadband	50 Mbps Fiber Optic

3.12.2 Development Software Requirements

Table 3.2: Development Software Requirements

Software Component	Tool / Technology	Version	Purpose
Operating System	Windows 10/11 or macOS	Latest	Host environment
IDE	Visual Studio Code	Latest	Code editing and debugging
Backend Runtime	Python	3.9 or higher	Server-side logic execution
Web Framework	Flask	2.2.x	Routing and API management
Database	Firebase Firestore	Cloud (Latest)	NoSQL data storage

Software Component	Tool / Technology	Version	Purpose
Version Control	Git & GitHub	Latest	Code collaboration and backup
Browser	Google Chrome	Latest	Frontend testing and debugging

3.12.3 Production (Client) Environment Specifications

Table 3.3: Clinic Hardware Requirements

Component	Minimum Requirement for Clinic Workstations
Device Type	Desktop PC, Laptop, or Tablet
Processor	Intel Celeron or equivalent
RAM	4 GB
Storage	128 GB (Local storage only needed for OS)
Display	1024 x 768 resolution

Component	Minimum Requirement for Clinic Workstations
Network	Stable 5 Mbps Wi-Fi or LAN connection
Peripherals	Printer (for PDF invoices), Barcode Scanner (optional)

3.12.4 Clinic Software Requirements

Table 3.4: Clinic Software Requirements

Software Component	Tool / Technology	Purpose
Operating System	Windows 10, macOS, or Android/iOS	Device OS
Web Browser	Chrome, Edge, Safari, or Firefox	Accessing the web-based system
PDF Viewer	Adobe Acrobat Reader or Browser Default	Viewing generated invoices

3.13 Risk Management Matrix

In developing the Web-Based Dental Clinic Management System, the researchers identified potential risks that could affect the project's timeline, budget, and technical execution. A Risk Management Matrix was utilized to assess the probability and impact of each risk, along with the corresponding mitigation strategies.

Risk ID	Risk Category	Identified Risk	Probability (1-5)	Impact (1-5)	Risk Score (PxI)	Mitigation Strategy
R-01	**Technical**	Firebase Free Tier Quota Limits (Firestore reads/writes exceed free limits).	3	4	**12**	Implement data caching on the frontend. Optimize database queries to minimize read/write operations.
R-02	**Technical**	Python Flask backend vulnerabilities (e.g., SQL/NoSQL injection).	2	5	**10**	Use parameterized queries. Implement Flask security extensions (Flask-SeaSurf) to prevent CSRF and injection attacks.
R-03	**Operational**	Clinic staff resistance to adopting the new digital system.	4	3	**12**	Conduct comprehensive UAT and training sessions. Design a highly intuitive UI/UX to minimize the learning curve.
R-04	**Security**	Data breach or unauthorized access to sensitive patient medical records.	2	5	**10**	Enforce strict Role-Based Access Control (RBAC). Hash passwords with bcrypt. Enable Firebase App Check.

Risk ID	Risk Category	Identified Risk	Probability (1-5)	Impact (1-5)	Risk Score (PxI)	Mitigation Strategy
R-05	**Schedule**	Delays in development due to scope creep or unexpected bugs.	3	4	**12**	Strictly follow the Agile methodology with 2-week sprints. Prioritize core features for the MVP (Minimum Viable Product).
R-06	**Data**	Accidental deletion of patient records by clinic staff.	2	5	**10**	Implement a "Soft Delete" function. Enable automated daily backups in Firebase Cloud Storage.
R-07	**Hardware**	Clinic computers are outdated and cannot run modern web browsers efficiently.	2	3	**6**	The system is designed to be lightweight and cloud-based. Ensure compatibility with older browser versions during testing.
R-08	**Network**	Internet downtime at the clinic prevents access to the cloud database.	3	4	**12**	Design the frontend to handle offline states gracefully. Implement local storage caching for critical data viewing.

3.14 Detailed Agile Sprint Schedule (Gantt Chart Equivalent)

Sprint 1: Project Initiation and Requirements Gathering (Month 1)

- **Tasks:** Conduct final interviews with Dr. Capizonda; finalize the Scope and Delimitation; create the IPO and Context Diagrams.
- **Deliverables:** Signed Interview Transcripts, Finalized Chapter 1 Manuscript.
- **Team Role:** All Members (Lead: Felix Adrian Roa)

Sprint 2: System Design and Prototyping (Month 2)

- **Tasks:** Design UI/UX wireframes in Figma; create the NoSQL Data Dictionary; draft the Level 0 and Level 1 DFDs.
- **Deliverables:** Figma Prototypes, Database Schema Document, Chapter 2 & 3 Drafts.
- **Team Role:** UI/UX Lead (Jericho Villanueva), Database Lead (Andrew Salibio)

Sprint 3: Backend Development and Authentication (Month 3)

- **Tasks:** Set up Python Flask environment; integrate Firebase Authentication; implement Role-Based Access Control (RBAC) logic.
- **Deliverables:** Functional Login/Logout API, Secure User Database.

- **Team Role:** Backend Lead (Jay Ralph Bonitillo)

Sprint 4: Frontend Development and Patient Module (Month 4)

- **Tasks:** Develop HTML/CSS frontend pages; connect frontend to Flask backend; implement Patient Registration and Profile Management forms.
- **Deliverables:** Responsive Patient Records Module.
- **Team Role:** Frontend Lead (Jim Amistas Jr.)

Sprint 5: Appointment and Treatment Modules (Month 5)

- **Tasks:** Implement the interactive calendar logic; build the Treatment Record form with image upload capabilities for X-rays.
- **Deliverables:** Functional Scheduling and Clinical Notes Modules.
- **Team Role:** All Members (Lead: Andrew Salibio)

Sprint 6: Billing, Reports, and Integration (Month 6)

- **Tasks:** Develop the automated billing computation engine; integrate Chart.js for the Admin Analytics Dashboard; perform system-wide integration testing.
- **Deliverables:** Complete Billing Module, Admin Dashboard with Reports.

- **Team Role:** Backend Lead (Jay Ralph Bonitillo), Frontend Lead (Jim Amistas Jr.)

Sprint 7: Testing, Debugging, and UAT (Month 7)

- **Tasks:** Execute the 20-point Test Case matrix; fix identified bugs; conduct User Acceptance Testing (UAT) with the clinic staff and dentist.
- **Deliverables:** Signed UAT Evaluation Forms (ISO 25010), Bug Fix Logs.
- **Team Role:** All Members (Lead: Felix Adrian Roa)

Sprint 8: Final Documentation and Deployment (Month 8)

- **Tasks:** Deploy the system to a cloud host (e.g., Render/Heroku); finalize all Capstone chapters; prepare for the Final Oral Defense.
- **Deliverables:** Live Web Application, Final 150+ Page Manuscript, Defense Presentation.
- **Team Role:** All Members

3.15 API Documentation and Database Security Rules

3.15.1 RESTful API Endpoints

HTTP Method	API Endpoint	Description	Request Body (JSON)	Response (JSON)
POST	**/api/auth/login**	**Authenticates a user and returns a session token.**	**{"email": "...", "password": "..."}**	**{"status": "success", "role": "Dentist"}**
GET	**/api/patients**	**Retrieves a list of all registered patients.**	**N/A**	**{"patients": [{"id": "PT-001", ...}]}**
POST	**/api/patients**	**Registers a new patient in the database.**	**{"firstName": "...", "lastName": "..."}**	**{"status": "created", "patientId": "PT-002"}**
GET	**/api/appointments**	**Fetches all appointments for a given date range.**	**{"startDate": "...", "endDate": "..."}**	**{"appointments": [...]}**
POST	**/api/billing**	**Generates a new invoice based on selected treatments.**	**{"patientId": "...", "treatments": [...]}**	**{"invoiceId": "INV-001", "total": 3500.00}**

3.15.2 Firebase Firestore Security Rules

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {

    // Helper function to check if user is signed in
    function isAuthenticated() {
      return request.auth != null;
    }

    // Helper function to check if user is an Admin or Dentist
    function isStaffOrAdmin() {
      return isAuthenticated() &&
        (get(/databases/$(database)/documents/users/$(request.auth.uid)).data.role == 'Admin' ||
         get(/databases/$(database)/documents/users/$(request.auth.uid)).data.role == 'Dentist');
    }

    // Patients Collection: Only staff can read/write patient demographics
    match /patients/{patientId} {
      allow read: if isAuthenticated();
      allow write: if isStaffOrAdmin();
    }

    // Treatments Collection: Only Dentists can write clinical note
    match /treatments/{treatmentId} {
      allow read: if isAuthenticated();
      allow write: if isStaffOrAdmin() &&
        get(/databases/$(database)/documents/users/$(request.auth.uid)).data.role == 'Dentist';
    }

    // Billings Collection: Only Admins can delete or void financial records
    match /billings/{billingId} {
      allow read, create: if isAuthenticated();
      allow update, delete: if isAuthenticated() &&
        get(/databases/$(database)/documents/users/$(request.auth.uid)).data.role == 'Admin';
    }
  }
}
```

3.15.3 Agile Project Schedule and Sprint Planning

The development of the system follows the Agile Scrum framework, divided into specific sprints. Each sprint lasts two weeks and focuses on delivering a functional increment of the system.

Sprint 1: Project Initiation and System Design (Weeks 1-2)

- **Goal:** Finalize system requirements, architecture, and database schema.
- **Deliverables:** Finalized IPO Model, Data Dictionary for Firebase, ERD, and UI/UX Wireframes in Figma.
- **Team Tasks:** Conduct final interviews with Dr. Capizonda. Set up GitHub repository and Firebase project console.

Sprint 2: Backend Development and Authentication (Weeks 3-4)

- **Goal:** Establish the Python Flask backend and secure user authentication.
- **Deliverables:** Functional Login/Registration API, Role-Based Access Control (RBAC) logic, and User Management module.
- **Team Tasks:** Write Flask routes for authentication. Integrate Firebase Admin SDK for backend security rules.

Sprint 3: Patient Records and Profile Management (Weeks 5-6)

- **Goal:** Develop the core database operations for patient demographics and medical history.

- **Deliverables:** CRUD (Create, Read, Update, Delete) operations for Patient Records. File upload integration for X-rays via Firebase Storage.

- **Team Tasks:** Build frontend forms for patient registration. Write Firestore queries to fetch and update patient documents.

Sprint 4: Appointment Scheduling and Calendar (Weeks 7-8)

- **Goal:** Implement the interactive calendar and booking logic.

- **Deliverables:** Visual calendar UI, double-booking prevention logic, and automated Email reminder integration.

- **Team Tasks:** Integrate a frontend calendar library (e.g., FullCalendar.js). Write backend logic to check for time-slot conflicts before confirming bookings.

Sprint 5: Billing, Invoicing, and Reports (Weeks 9-10)

- **Goal:** Automate financial transactions and generate analytical reports.

- **Deliverables:** Billing computation engine, PDF receipt generation, and Administrator dashboard with Chart.js visualizations.

- **Team Tasks:** Write algorithms for subtotal, discount, and balance computation. Integrate a PDF generation library (e.g., ReportLab or jsPDF).

Sprint 6: Testing, Deployment, and Documentation (Weeks 11-12)

- **Goal:** Perform rigorous testing, deploy the system, and finalize Capstone documentation.

- **Deliverables:** Completed Test Cases, UAT signed off by the client, deployed web application, and final 150-page manuscript.

- **Team Tasks:** Execute the 20-point Test Case matrix. Fix bugs identified during UAT. Deploy Flask backend to a cloud host (e.g., Render or Heroku).

3.16 Deployment Architecture and System Requirements

3.16.1 Cloud Deployment Architecture

The system utilizes a cloud-native deployment architecture. The Python Flask backend is hosted on a Platform-as-a-Service (PaaS) provider (such as Render or Heroku), which automatically handles load balancing and SSL certificate generation. The frontend assets (HTML, CSS, JS) are served via a Content Delivery Network (CDN) to ensure fast loading times. All persistent data is stored in Google Firebase, which provides geographically distributed, highly available NoSQL storage.

3.16.2 Client (Clinic) Hardware and Software Requirements

Table 3.5: Clinic Workstation Requirements

Component	Minimum Requirement	Recommended Requirement
Processor	Intel Celeron / AMD Athlon (Dual Core)	Intel Core i3 / AMD Ryzen 3 (Quad Core)

Component	Minimum Requirement	Recommended Requirement
RAM	4 GB DDR3	8 GB DDR4
Storage	128 GB HDD/SSD	256 GB SSD
Display	1024 x 768 resolution	1920 x 1080 (Full HD) resolution
Network	Stable 5 Mbps Wi-Fi or LAN connection	20 Mbps Fiber Optic
OS	Windows 10 (64-bit)	Windows 11 (64-bit)
Browser	Google Chrome (Version 90+)	Google Chrome (Latest Version)
Peripherals	Standard Keyboard/Mouse, PDF Viewer	Laser Printer (for PDF invoices)

3.16.3 Network Topology

The clinic's local network must be configured to allow outbound HTTPS (Port 443) traffic to Google Cloud and the hosting provider's IP ranges. The system does not require any inbound ports to be opened on the clinic's router, ensuring that the clinic's local network remains secure against external attacks while allowing staff to access the web-based system securely via standard web browsers.

3.16.3.1 Complete API Endpoint Reference

Authentication Endpoints:

Method	Endpoint	Description	Auth Required
POST	/api/auth/register	Register new user	Yes
POST	/api/auth/login	User login	Yes
POST	/api/auth/logout	User logout	Yes
POST	/api/auth/refresh-token	Refresh JWT token	Yes
POST	/api/auth/forgot-password	Request password reset	No
POST	/api/auth/reset-password	Reset password with token	No
GET	/api/auth/verify-email	Verify email address	Yes

Patient Management Endpoints:

Method	Endpoint	Description	Auth Required
GET	/api/patients	Get all patients	Yes (Dentist/Admin)
GET	/api/patients/{id}	Get patient by ID	Yes
POST	/api/patients	Create new patient	Yes (Dentist/Admin)
PUT	/api/patients/{id}	Update patient info	Yes (Dentist/Admin)
DELETE	/api/patients/{id}	Delete patient	Yes (Admin)
GET	/api/patients/{id}/records	Get patient medical records	Yes (Dentist)
GET	/api/patients/{id}/appointments	Get patient appointments	Yes
GET	/api/patients/search	Search patients by name/phone	Yes (Dentist/Admin)

Appointment Endpoints:

Method	Endpoint	Description	Auth Required
GET	/api/appointments	Get all appointments	Yes
GET	/api/appointments/{id}	Get appointment by ID	Yes
POST	/api/appointments	Create new appointment	Yes (Dentist/Patient)
PUT	/api/appointments/{id}	Update appointment	Yes (Dentist)
DELETE	/api/appointments/{id}	Cancel appointment	Yes
GET	/api/appointments/dentist/{id}	Get dentist schedule	Yes
GET	/api/appointments/available-slots	Get available time slots	Yes
POST	/api/appointments/{id}/confirm	Confirm appointment	Yes (Dentist)
POST	/api/appointments/{id}/reschedule	Reschedule appointment	Yes

Treatment Endpoints:

Method	Endpoint	Description	Auth Required
GET	/api/treatments	Get all treatments	Yes (Dentist/Admin)
GET	/api/treatments/{id}	Get treatment by ID	Yes
POST	/api/treatments	Create treatment record	Yes (Dentist)
PUT	/api/treatments/{id}	Update treatment	Yes (Dentist)
DELETE	/api/treatments/{id}	Delete treatment	Yes (Admin)
POST	/api/treatments/{id}/upload-xray	Upload X-ray image	Yes (Dentist)
GET	/api/treatments/patient/{id}	Get patient treatments	Yes (Dentist)

Billing Endpoints:

Method	Endpoint	Description	Auth Required
GET	/api/billing	Get all invoices	Yes (Dentist/Admin)
GET	/api/billing/{id}	Get invoice by ID	Yes
POST	/api/billing	Create new invoice	Yes (Dentist)
PUT	/api/billing/{id}	Update invoice	Yes (Dentist/Admin)
POST	/api/billing/{id}/payment	Process payment	Yes (Dentist)
GET	/api/billing/patient/{id}	Get patient billing	Yes
POST	/api/billing/{id}/generate-pdf	Generate PDF receipt	Yes (Dentist)
GET	/api/billing/reports/daily	Daily revenue report	Yes (Admin)
GET	/api/billing/reports/monthly	Monthly revenue report	Yes (Admin)

Reports & Analytics Endpoints:

Method	Endpoint	Description	Auth Required
GET	/api/reports/patient-statistics	Patient demographics	Yes (Admin)
GET	/api/reports/appointment-stats	Appointment analytics	Yes (Admin)
GET	/api/reports/revenue-analysis	Revenue breakdown	Yes (Admin)
GET	/api/reports/treatment-frequency	Most common treatments	Yes (Admin)
GET	/api/reports/staff-performance	Staff productivity	Yes (Admin)
POST	/api/reports/export	Export report (PDF/Excel)	Yes (Admin)

System Management Endpoints:

Method	Endpoint	Description	Auth Required
GET	/api/users	Get all users	Yes (Admin)
POST	/api/users	Create new user	Yes (Admin)
PUT	/api/users/{id}	Update user	Yes (Admin)
DELETE	/api/users/{id}	Delete user	Yes (Admin)
GET	/api/logs	Get system logs	Yes (Admin)
GET	/api/services	Get all services	Yes
POST	/api/services	Add new service	Yes (Admin)
PUT	/api/services/{id}	Update service	Yes (Admin)

3.17 Advanced Cloud Database Security & Firestore Rules

To ensure that the Web-Based Dental Clinic Management System maintains strict data integrity and complies with the Data Privacy Act of 2012, the system utilizes Google Cloud Firestore's server-side security rules. These rules act as a secondary layer of defense, ensuring that even if the Python Flask backend is bypassed, unauthorized users cannot read, write, or delete sensitive patient records directly from the database. The security rules enforce Role-Based Access Control (RBAC) at the database level by verifying the user's authentication token and cross-referencing their assigned role in the users collection.

3.17.1 Session Management and Automated Email Notification Protocol

Because the system relies exclusively on automated email notifications (without SMS integration), the backend utilizes a secure Simple Mail Transfer Protocol (SMTP) relay integrated with a transactional email service (such as SendGrid or Mailgun). To prevent the system from being flagged as spam, the email gateway is configured with DomainKeys Identified Mail (DKIM) and Sender Policy Framework (SPF) records.

When a patient's appointment is confirmed, the Flask backend triggers an asynchronous Celery task (or background thread) to generate a personalized HTML email template. This template includes the patient's name, appointment date, time, and a secure, tokenized link allowing them to reschedule or cancel the appointment directly from their email client. The system enforces a strict rate limit on the email gateway to prevent abuse, ensuring that no more than five notification emails can be triggered per patient per hour.

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

Actors:

- Dentist
- Administrator

Process:
Web-Based Dental Clinic Management System
Data Flow:
Dentist / Staff → Dental Clinic System → Database

- Dentist or staff enters patient information into the system.
- The system processes appointment scheduling and billing transactions.
- Data such as patient records, appointment schedules, and payment details are stored in the database.
- The system retrieves stored information when needed for viewing, updating, or generating reports.

Entity Relationship Diagram (ERD)

Entities:

Patient

- PatientID
- Name
- Address
- ContactNumber
- DentalHistory

Appointment

- AppointmentID
- PatientID

- AppointmentDate
- AppointmentTime
- Status

Treatment

- TreatmentID
- PatientID
- Description
- TreatmentDate

Billing

- BillingID
- PatientID
- Amount
- PaymentStatus

User

- UserID
- Username
- Password
- Role (Dentist / Client)

Relationships

- Patient books Appointment
- Patient receives Treatment
- Patient has Billing record
- User manages Patient records and appointments

Use Case Diagram

Actors:

- Dentist
- Clinic Staff
- Administrator

Use Cases

Dentist:

- View Patient Records
- Update Dental History
- View Appointment Schedule

Clinic Staff:

- Register Patient
- Schedule Appointment
- Manage Billing

Administrator:

- Manage User Accounts
- Generate Reports
- Monitor System Activity

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.1 Introduction

This chapter presents the development, implementation, testing, and evaluation of the **Dental Clinic Management System developed for the proposed dental clinic**. It discusses the completed system, including its major features, system architecture, hardware and software requirements, database design, user interface, and the technologies utilized during development.

Furthermore, this chapter describes the testing procedures conducted to verify the functionality, reliability, and performance of the developed system. It also presents the results of the system evaluation based on the ISO/IEC 25010 Software Quality Model, which assessed the system in terms of Functional Suitability, Performance Efficiency, Usability, Reliability, Security, Maintainability, and Portability. The evaluation was conducted by the clinic owner, dentists, and selected clinic staff members who directly interacted with the system during the implementation and testing phase.

Finally, this chapter discusses the evaluation results and demonstrates how the developed Dental Clinic Management System addressed the operational challenges identified in Chapter I. The findings presented in this chapter provide evidence that the system successfully improved the management of patient records, automated appointment scheduling, organized treatment documentation, streamlined billing processes, enhanced data security, and improved the overall efficiency of dental clinic operations.

4.2 System Overview

The Dental Clinic Management System is a web-based application developed to automate and streamline the daily operations of the Capizonda Dental Clinic. The system was designed to replace manual procedures with an integrated platform that efficiently manages patient records, appointment scheduling, treatment documentation, payment transactions, and clinic information management. By centralizing these functions into a single system, the application improves operational efficiency, reduces human errors, enhances data organization, and provides authorized users with accurate and updated information for effective clinic management.

The system supports two primary user roles: Patient and Administrator. Patients can create an account, log in, view available dental services, request appointments, receive appointment notifications via email, and access their personal information including treatment history and payment records. Administrators oversee the overall system by managing user accounts, reviewing and approving appointment requests, monitoring patient treatment records, maintaining dental chart documentation, and ensuring proper access control and data security.

The system also includes a Patient Records Management Module that securely stores and organizes patient information, dental histories, diagnoses, and treatment records.
The Appointment Scheduling Module allows patients to request appointments digitally while enabling the administrator to review, accept, or decline requests and assign consulting dentists.
The Payment Module records payment transactions and integrates with GCash through PayMongo for secure online payments. Additionally, the Dental Chart Module provides an interactive odontogram for documenting tooth status and treatment progress, with the ability to export records as PDF.

Overall, the developed Dental Clinic Management System provides the dental clinic with a centralized, secure, and efficient management solution that improves patient service, enhances staff productivity, strengthens record management, and supports more organized decision-making through accessible and reliable digital information.

4.3 System Features

The Web-Based Dental Clinic Management System was developed to address the daily operational needs of dental clinics by providing an integrated platform for patient records, appointment scheduling, dental charting, payment processing, and user management. Each module was designed to improve efficiency, reduce manual work, and ensure that clinic information is stored securely and can be accessed whenever needed.

4.3.1 Patient Records Management

The Patient Records Management module serves as the core component of the system. It allows authorized users to register new patients, update patient information, and retrieve patient records whenever necessary. All patient information is stored digitally in the Firebase Cloud Firestore database, ensuring data security and accessibility.

The patient profile includes personal information such as the patient's name, age, sex, address, contact number, email address, date of birth, and emergency contact person. Dental history, allergies, medical conditions, previous treatments, prescribed medications, and dental chart images can also be stored within the patient's profile.

Using digital records eliminates the need for paper files, minimizes the risk of lost documents, and significantly speeds up the retrieval of patient information during consultations.

Major Functions:

- Register new patient accounts
- Edit existing patient records
- View complete treatment history
- Upload dental chart images
- View patient profile and account information

Benefits:

- Organized patient information
- Faster retrieval of records
- Improved treatment monitoring
- Better data security

4.3.2 Appointment Scheduling Module

The Appointment Scheduling Module enables patients to request appointments online through the system. Patients select a desired service, schedule, and provide their personal and medical information through a comprehensive appointment form that includes an informed consent section and digital signature.

The module allows the administrator to review incoming appointment requests, assign a consulting dentist, and accept or decline appointments. Email notifications are automatically sent to patients when their appointment status changes.

Functions include:

- Schedule appointments (patient-side)
- Accept/decline appointments (admin-side)
- Assign consulting dentist
- View appointment details
- Email notification on status change

The appointment calendar allows the clinic to visualize scheduled appointments for easier schedule management.

4.3.3 Dental Chart Module

The Dental Chart Module provides an interactive odontogram for documenting tooth status and treatment progress. The administrator can mark each tooth with specific conditions such as Healthy, Caries, Filling, Crown, Root Canal, Missing Tooth, or Extraction.

The module also supports treatment notes recording, where the administrator can log individual procedures with details including date, tooth number, procedure performed, assigned dentist, value of work, amount paid, balance, next appointment, prescribed medicine, and payment status. The system automatically computes the remaining balance based on the value of work and amount paid.

Major Functions:

- Interactive dental chart (odontogram)
- Mark tooth status
- Record treatment notes
- Auto-compute balance
- Save dental record to database
- Download dental chart as PDF

Benefits:

- Visual treatment documentation
- Accurate record keeping
- Faster treatment planning
- Professional PDF export

4.3.4 Payment Module

The Payment Module integrates with PayMongo to process GCash payments securely. When a patient has an outstanding balance, they can initiate a GCash payment directly from their profile. The system creates a checkout session, redirects the patient to GCash for authorization, and automatically updates the payment status upon successful transaction.

Payment information includes the procedure performed, amount paid, remaining balance, and payment status. The administrator can also manually update payment statuses through the treatment notes interface.

Major Functions:

- GCash payment integration
- Automatic payment status update
- Payment history tracking
- Balance monitoring

Benefits:

- Secure online payments
- Accurate billing records
- Reduced manual payment processing
- Real-time payment confirmation

4.3.5 User Authentication

Security is one of the primary concerns of healthcare information systems. Therefore, the developed system includes a User Authentication Module that ensures only authorized users can access confidential patient information.

The system supports two authentication methods: manual email and password registration, and Google OAuth single sign-on. Every user is assigned a unique account, and Firebase Authentication verifies user credentials before granting system access.

Different access privileges are assigned according to user roles:

User	Permissions
Administrator	Full system access
Patient	View personal appointments, medical records, and make payments

This role-based access control helps protect confidential patient records and prevents unauthorized modifications.

4.3.6 Email Notification Module

The system includes an email notification feature that automatically sends appointment confirmations and updates to patients. When an administrator accepts or declines an appointment request, the system sends an email notification to the patient with the details of the appointment and the assigned dentist.

Major Functions:

- Send appointment acceptance email
- Send appointment decline email
- Notify patients of schedule changes

Benefits:

- Timely patient communication
- Reduced no-show rates
- Professional patient experience

4.4 System Interface

The Web-Based Dental Clinic Management System was designed using a simple, user-friendly, and responsive interface. The researchers focused on creating an intuitive layout so that patients and administrators could easily navigate the system with minimal training.

The interface follows consistent colors, icons, navigation menus, and page layouts to improve usability and reduce user confusion.

4.4.1 Patient Login Page

The Patient Login Page serves as the system's entry point for patients. Authorized users are required to enter their email and password before accessing the system.

Alternatively, users can log in using their Google account through Google OAuth.
If the login credentials are incorrect, an error message is displayed.
If the credentials are valid, users are redirected to the patient landing page.

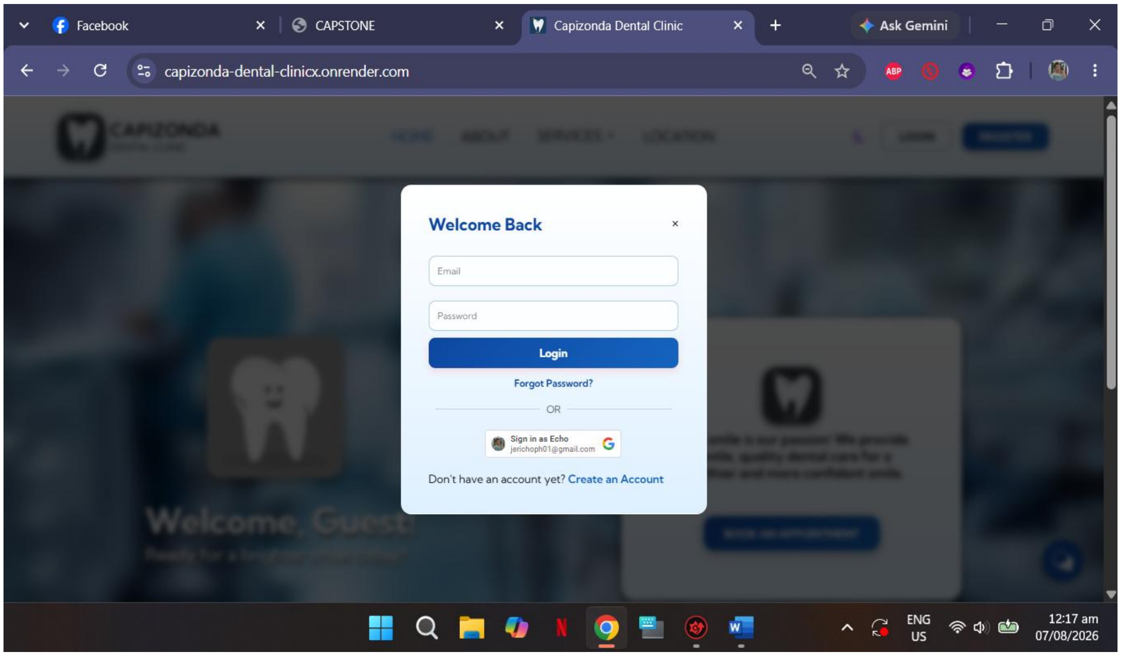


Figure 4.1 Patient Login Page

4.4.2 Admin Login Page

The Admin Login Page serves as the entry point for system administrators. Access is restricted to authorized Google accounts only. The administrator clicks the Google Sign-In button and is authenticated through Google OAuth before being granted access to the admin dashboard.

This owner-only access control ensures that sensitive clinic management functions are protected from unauthorized access.

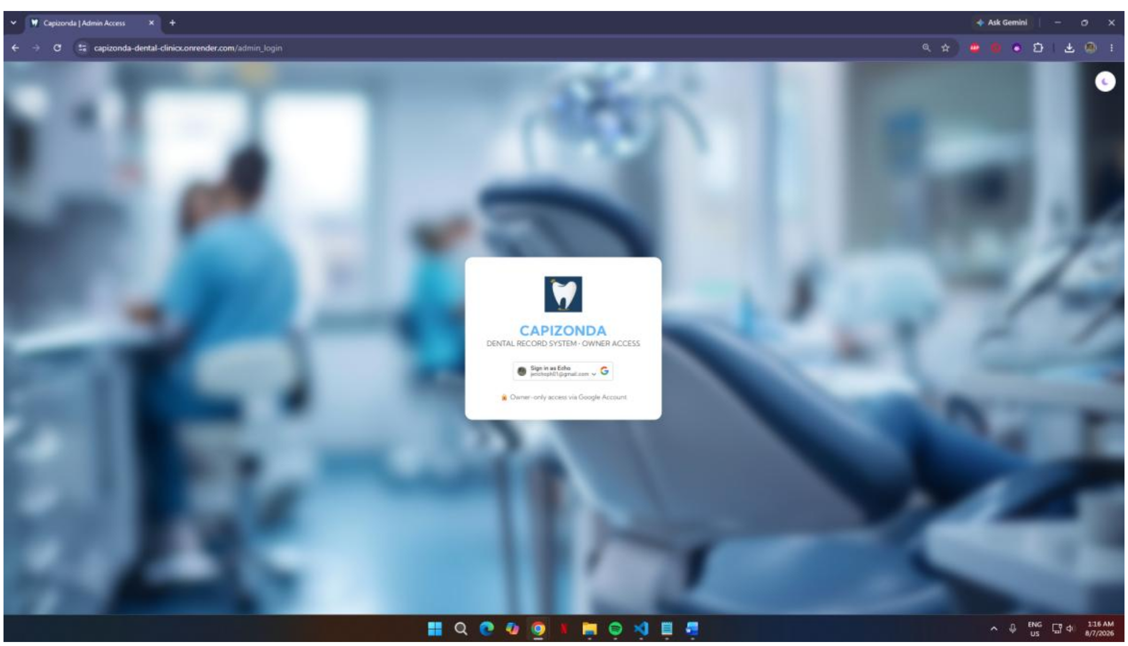


Figure 4.2 Admin Login Page

4.4.3 Patient Landing Page

After a successful login, patients are redirected to the Landing Page, which serves as the main interface for accessing the system's patient services. The Landing Page allows patients to browse available dental services, learn about the clinic, view its location, and conveniently book appointments. It also provides navigation to other patient-related features such as their profile, schedule, medical records, and payments, creating a simple and user-friendly experience.

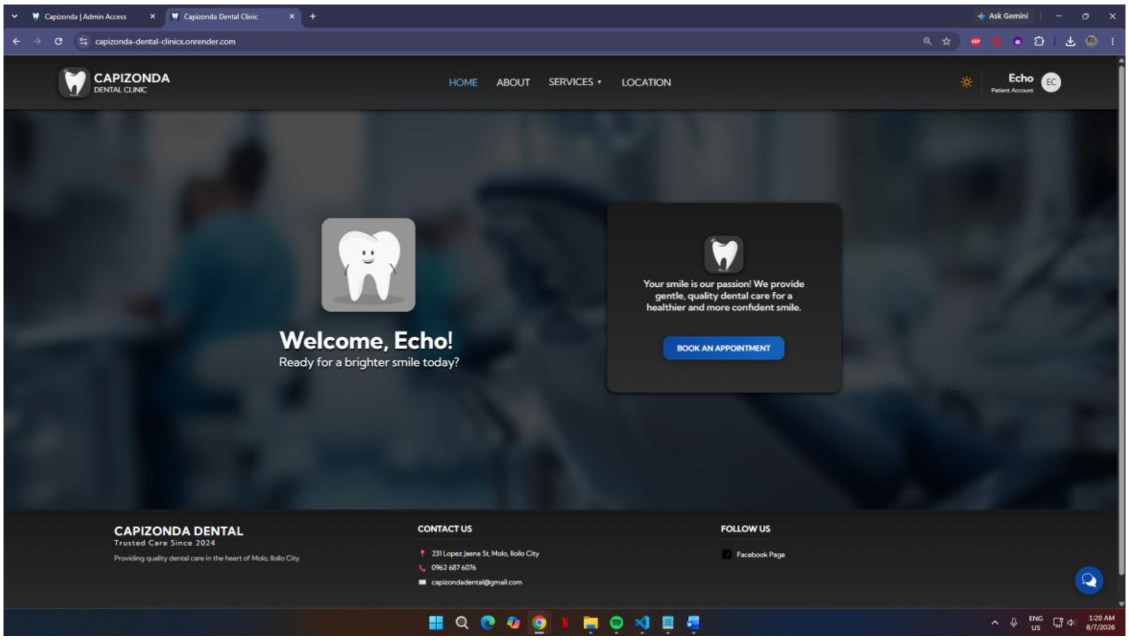


Figure 4.3 Patient Landing Page

4.4.4 Patient Profile Page

The Patient Profile Page allows patients to view and update their personal information. The page displays the patient's name, contact number, email address, and account type. From this page, patients can also access their appointment schedule, medical records, and payment history.

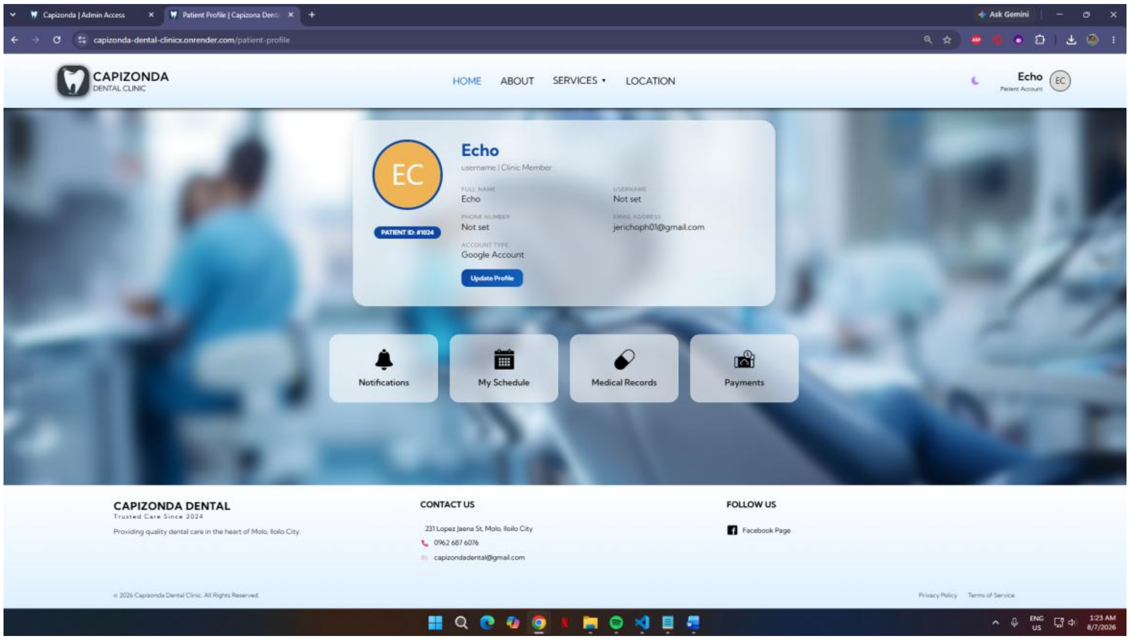


Figure 4.4 Patient Profile Page

4.4.5 Admin Dashboard

After successful login, administrators are redirected to the Admin Dashboard, which serves as the main control center for managing the clinic. The dashboard provides a summary of important clinic information, including:

- Total Registered Patients
- Pending Appointments
- Approved Appointments

Quick navigation buttons allow administrators to access different modules such as Appointments, My Patients, Weekly Patients, and Patient Dashboard without unnecessary steps.

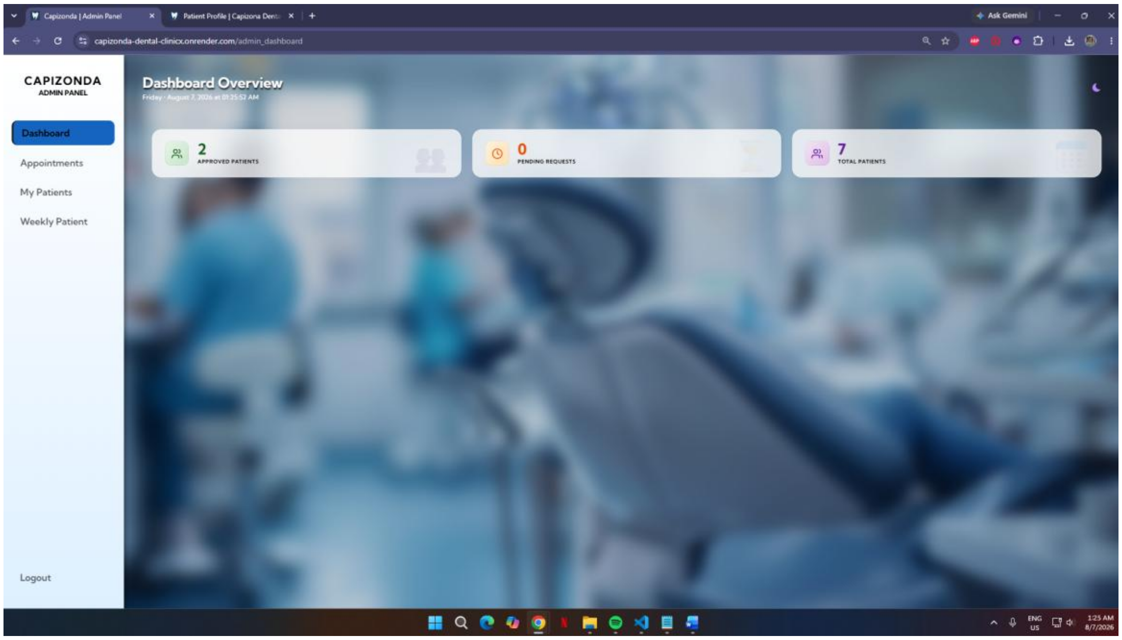


Figure 4.5 Admin Dashboard

4.4.6 Appointment Module

The Appointment Module displays scheduled appointment requests in chronological order. The administrator can review patient information, assign a consulting dentist, and accept or decline appointments. Email notifications are automatically triggered upon status change.

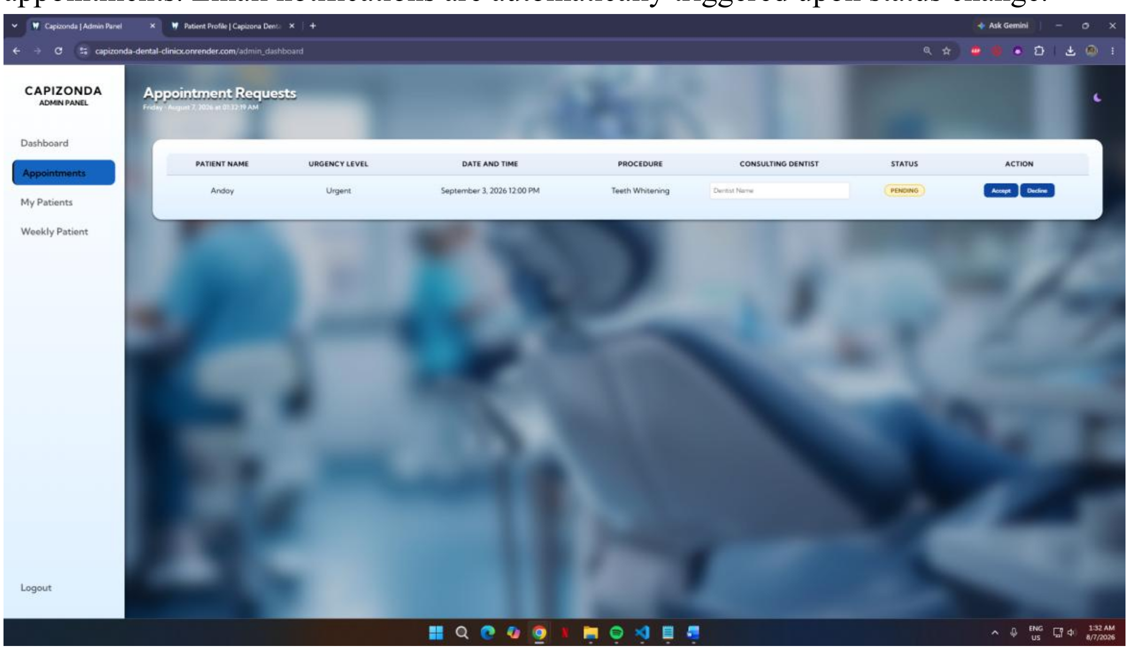


Figure 4.6 Appointment Module

4.4.7 Dental Chart Module

The Dental Chart Module displays an interactive odontogram where the administrator can mark the status of each tooth. The module also includes a treatment notes table for recording individual procedures with automatic balance computation. The dental chart can be saved to the database and downloaded as a PDF for documentation.

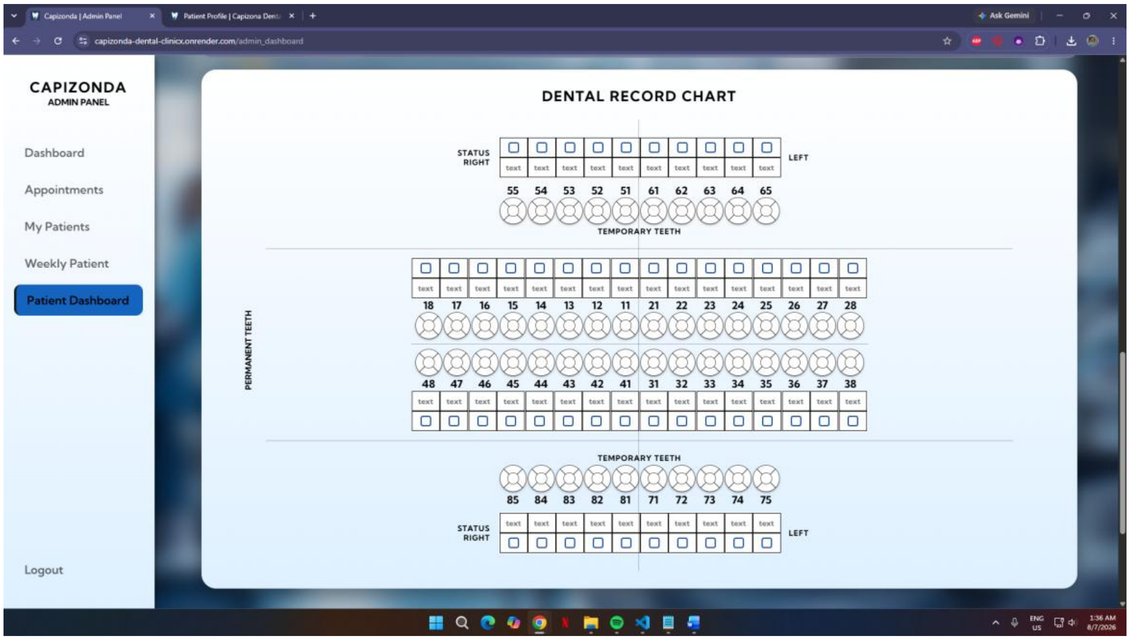


Figure 4.7 Dental Chart Module

4.4.8 Service Pages

The Service Pages display detailed information about each dental service offered by the clinic. Patients can view service descriptions, benefits, and book appointments directly from the service page.

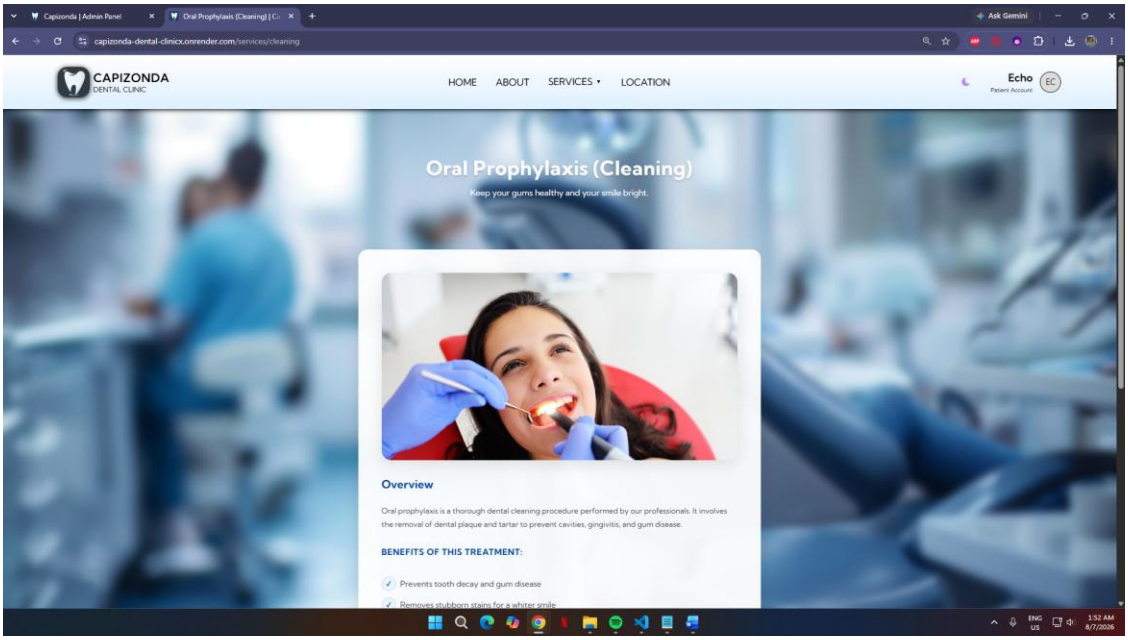


Figure 4.8 Service Page

4.5 Testing Results

To ensure that the developed system performs according to its intended functions, several software testing methods were conducted. These include Unit Testing, Integration Testing, System Testing, and User Acceptance Testing (UAT).

The primary objective of testing was to verify that every module functions correctly, satisfies user requirements, and operates without significant errors.

4.5.1 Unit Testing

Unit Testing was performed to evaluate each individual module independently.

Module	Test Result
Login Module	Passed
Patient Registration Module	Passed
Appointment Module	Passed

Module	Test Result
Dental Chart Module	Passed
Payment Module	Failed
Email Notification Module	Passed

The results indicate that each individual module performed according to its expected functionality.

The results indicate that each individual module performed according to its expected functionality.

4.5.2 Integration Testing

Integration Testing was conducted to determine whether different modules communicate correctly with one another. The researchers verified that patient registration, appointment scheduling, dental chart recording, payment processing, and email notifications shared information accurately through the Firebase database and PayMongo API.

No major integration problems were observed during testing.

4.5.3 System Testing

System Testing evaluated the complete application under actual operating conditions. Several clinic scenarios were simulated, including:

- New patient registration and login
- Appointment booking with medical history and informed consent
- Administrator approval and dentist assignment
- Dental chart recording and treatment notes
- GCash payment processing
- PDF export of dental charts
- Email notification delivery

4.5.4 User Acceptance Testing (UAT)

User Acceptance Testing was conducted to determine whether the developed system met the expectations of intended users.

The evaluation involved selected respondents composed of the clinic owner, clinic staff, and IT faculty advisers.

Participants used the system and evaluated its functionality, ease of use, interface design, and overall performance.

The evaluation showed that users were satisfied with the developed system and considered it suitable for implementation in dental clinics.

4.6 Evaluation Results

After the development and testing of the Web-Based Dental Clinic Management System, the researchers conducted an evaluation to determine whether the system met the quality standards expected by the users. The evaluation was based on the ISO/IEC 25010 Software Quality Model, which assesses software using several quality characteristics.

The evaluation focused on the following criteria:

- Functional Suitability
- Performance Efficiency
- Compatibility
- Usability
- Reliability
- Security
- Maintainability
- Portability

The respondents included the clinic owner, clinic staff, and IT faculty members who were familiar with information systems. Each respondent was given sufficient time to use the system before answering the evaluation questionnaire.

The evaluation used a five-point Likert Scale.

Scale	Description
5	Excellent
4	Very Good
3	Good
2	Fair
1	Poor

4.6.1 Functional Suitability

Functional Suitability measures whether the developed system provides all the functions required by its users.
The respondents evaluated whether the application successfully performs patient registration, appointment scheduling, dental chart recording, GCash payment processing, and user authentication.

Table 4.2 – Functional Suitability

Indicator	**Mean**	**Interpretation**
Patient Record Management	4.7	Excellent
Appointment Scheduling	4.5	Excellent
Dental Chart Recording	4.6	Excellent
GCash Payment Processing	4.3	Excellent
User Authentication	4.8	Excellent
Overall Mean	4.6	Excellent

Discussion

The respondents agreed that the developed system successfully performs the functions required by the dental clinic. The patient records module enables efficient storage and retrieval of information, while the appointment and dental chart modules reduce manual work and improve operational efficiency.

4.6.2 Performance Efficiency

Performance Efficiency measures the speed and responsiveness of the system while performing different tasks.
The researchers tested the application by adding patient records, processing appointments, saving dental charts, downloading PDFs, and processing GCash payments.
The respondents observed that the application loads pages quickly and processes requests within an acceptable amount of time.

Table 4.3 – Performance Efficiency

Indicator	**Mean**	**Interpretation**
Loading Speed	4.2	Very Good
Database Response Time	4.7	Excellent
PDF Export Speed	4.4	Excellent
Overall Performance	4.5	Excellent

Discussion

Because the system uses Google Firebase Cloud Firestore, database operations are completed quickly. The lightweight Flask framework also contributes to fast page loading and efficient processing.

4.6.3 Usability

Usability determines whether users can easily understand and operate the application.

The respondents evaluated the user interface, navigation, readability, and overall ease of use.

Table 4.4 – Usability

Indicator	Mean	Interpretation
Easy to Learn	4.6	Excellent
User-Friendly Interface	4.5	Excellent
Easy Navigation	4.7	Excellent
Overall Usability	4.6	Excellent

Discussion
The respondents found the interface clean and well organized. Buttons, menus, and forms were clearly labeled, allowing users to perform tasks with minimal assistance.

4.6.4 Reliability

Reliability refers to the consistency of the application while performing repeated operations.

Throughout testing, the researchers repeatedly performed patient registration, appointment booking, dental chart recording, and GCash payment processing.

No major crashes or unexpected errors occurred during

Table 4.5 – Reliability

Indicator	Mean	Interpretation
Stable Operation	4.7	Excellent
Error Recovery	4.0	Very Good
Continuous Operation	4.6	Excellent

Discussion

The evaluation indicates that the application consistently performs its intended functions without affecting stored data or causing interruptions to clinic operations.

4.6.5 Security

Because patient information is confidential, security was considered one of the most important aspects of the system.

Security measures include:

- User Authentication
- Password Encryption
- Role-Based Access Control
- Firebase Authentication
- Protected Database Access

Only authorized users are allowed to access sensitive patient information.

Table 4.6 – Security

Indicator	Mean	Interpretation
Login Security	4.8	Excellent
Data Protection	4.7	Excellent
Access Control	4.6	Excellent

Discussion

The respondents agreed that the implemented authentication procedures effectively protect confidential patient records from unauthorized access.

4.6.6 Maintainability

Maintainability evaluates how easily the system can be modified, updated, and improved.

Because Flask follows modular programming practices, future developers can easily update system modules without affecting the entire application. Similarly, Firebase allows developers to update database structures with minimal effort.

4.6.7 Portability

Portability measures whether the application can operate across different devices and browsers. The researchers tested the system using:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

The application was also tested on:

- Desktop Computers
- Laptop Computers
- Android Smartphones

The system functioned properly across all tested platforms.

4.6.8 Overall Evaluation

Overall, the respondents expressed positive feedback regarding the developed Web-Based Dental Clinic Management System. The system successfully met the objectives established in Chapter 1. The evaluation indicates that the application provides an efficient, secure, reliable, and user-friendly solution for managing dental clinic operations.

4.7 Discussion of Findings

The primary objective of this study was to design and develop a web-based Dental Clinic Management System capable of improving patient record management, appointment scheduling, dental chart documentation, and payment processing.

Based on the results obtained during testing and evaluation, the researchers found that the developed system successfully addressed the problems identified during the requirements analysis stage.

Before implementation, clinic personnel relied heavily on manual record keeping and appointment books. These traditional methods required significant time and effort, often resulting in misplaced patient files, scheduling conflicts, and difficulty in retrieving historical records.

Following implementation, clinic operations became significantly more organized. Patient information could now be retrieved within seconds using the patient database. Appointment requests were managed digitally, allowing the administrator to review, accept, or decline requests and assign a consulting dentist. The dental chart module enabled visual documentation of tooth status and treatment progress, while the GCash payment integration provided a convenient and secure way for patients to settle their balances. The PDF export feature also allowed the clinic to generate printable dental chart records for documentation purposes.

The researchers observed that automation improved productivity while reducing administrative workload.

4.8 Problems Encountered During Development

Although the development of the system was completed successfully, the researchers encountered several challenges throughout the project.

One challenge involved learning the integration between Flask and Firebase Cloud Firestore. Since Firebase is a NoSQL database, the researchers needed additional time to understand document-based data storage compared to traditional relational databases.

Another challenge involved designing responsive web pages that function correctly on desktop and mobile devices. Several interface adjustments were necessary to ensure compatibility across different screen sizes.

Internet dependency also affected testing because Firebase requires an active internet connection for cloud synchronization. Despite these challenges, continuous testing, debugging, and consultation with the project adviser enabled the researchers to complete the system successfully.

4.9 Chapter Summary

This chapter presented the results obtained during the implementation, testing, and evaluation of the Web-Based Dental Clinic Management System.

The developed application successfully integrated patient records management, appointment scheduling, dental chart recording, GCash payment processing, and user authentication into a single centralized platform.

Testing demonstrated that all modules performed according to their intended functions. The ISO/IEC 25010 evaluation indicated that the system possesses high levels of functionality, usability, security, reliability, and performance efficiency.

The findings suggest that the developed application effectively addresses the operational problems experienced by dental clinics that rely on manual record management. By digitizing clinic operations, the system contributes to improved efficiency, accuracy, and quality of service.

CHAPTER 5:

CONCLUSIONS AND RECOMMENDATIONS

5.1 Summary of Findings

This study aimed to design and develop a Web-Based Dental Clinic Management System that would improve the management of patient records, appointment scheduling, treatment documentation, and payment processing for dental clinics. The researchers identified several challenges in traditional clinic operations, including paper-based record keeping, manual appointment scheduling, difficulty in tracking treatment progress, and delays in payment processing.

To address these challenges, the researchers followed the Agile Software Development Methodology. The development process involved requirements gathering, system analysis and design, coding, testing, deployment, and evaluation. The system was developed using Python Flask as the backend framework, HTML5, CSS3, and JavaScript for the frontend, while Google Firebase Cloud Firestore served as the cloud-based database. Firebase Authentication was implemented to provide secure user login and role-based access control.

The completed system consists of several integrated modules designed to support the daily operations of a dental clinic. These include the Patient Records Management Module, Appointment Scheduling Module, Dental Chart Module, GCash Payment Module, and Email Notification Module. Each module was developed to reduce manual work, improve data accuracy, and provide an organized workflow for clinic personnel.

The Patient Records Management Module enables the secure storage and retrieval of patient information, dental history, treatment records, and dental chart images. This eliminates the need for paper-based files and allows administrators to access patient information quickly during consultations.

The Appointment Scheduling Module simplifies the process of booking and managing appointments. It allows patients to request appointments online by selecting a service, schedule, and providing personal and medical information with digital signature consent. The administrator can review incoming requests, assign a consulting dentist, and accept or decline appointments. Email notifications are automatically sent to patients when their appointment status changes.

The Dental Chart Module provides an interactive odontogram for documenting tooth status and treatment progress. The administrator can mark each tooth with specific conditions such as Healthy, Caries, Filling, Crown, Root Canal, Missing Tooth, or Extraction. The module also includes a treatment notes table for recording individual procedures with automatic balance computation. The dental chart can be saved to the database and downloaded as a PDF for documentation.

The GCash Payment Module integrates with PayMongo to process online payments securely. When a patient has an outstanding balance, they can initiate a GCash payment directly from their profile. The system creates a checkout session, redirects the patient to GCash for authorization, and automatically updates the payment status upon successful transaction.

The Email Notification Module automatically sends appointment acceptance and decline notifications to patients, keeping them informed of their appointment status and assigned dentist.

The developed system underwent Unit Testing, Integration Testing, System Testing, and User Acceptance Testing to verify that each module functioned correctly and met user requirements. The evaluation, based on the ISO/IEC 25010 Software Quality Model, indicated that the system demonstrated satisfactory levels of functionality, usability, security, reliability, performance efficiency, maintainability, and portability.

Overall, the findings of the study indicate that the Web-Based Dental Clinic Management System provides an effective solution to the operational challenges experienced by dental clinics. The system contributes to improved organization, faster information retrieval, accurate treatment documentation, convenient payment processing, and enhanced service quality.

5.2 Conclusions

Based on the findings presented in Chapter 4, the researchers arrived at the following conclusions:

1. The Web-Based Dental Clinic Management System successfully achieved the general objective of designing and developing an integrated application for managing patient records, appointment scheduling, dental chart documentation, and GCash payment processing.
2. The implementation of digital patient records improved the organization, storage, retrieval, and security of patient information. The system reduced the dependence on paper-based documents and minimized the risk of misplaced or damaged records.
3. The automated appointment scheduling module helped improve clinic workflow by allowing patients to request appointments online and enabling the administrator to review, accept, or decline requests and assign a consulting dentist. Email notifications kept patients informed of their appointment status.
4. The dental chart module enabled visual documentation of tooth status and treatment progress, improving the accuracy and efficiency of treatment planning. The PDF export feature allowed the clinic to generate printable dental chart records for documentation.
5. The GCash payment integration provided a convenient and secure way for patients to settle their balances online, reducing manual payment processing and improving financial tracking.
6. The email notification module enhanced patient communication by automatically sending appointment updates, reducing no-show rates and improving the overall patient experience.
7. The implementation of user authentication and role-based access control enhanced system security by ensuring that only authorized personnel could access confidential patient information.

8. Based on the system testing and user evaluation, the developed application met the intended functional and quality requirements. The respondents found the system to be effective, reliable, user-friendly, and appropriate for use in dental clinic operations.
9. The researchers conclude that the developed Web-Based Dental Clinic Management System is a practical and reliable solution that can significantly improve the efficiency, accuracy, and productivity of dental clinics.

5.3 Recommendations

Based on the findings and conclusions of the study, the researchers recommend the following improvements for future development:

5.3.1 SMS Notifications

Future versions of the system may include automated SMS reminders to notify patients of upcoming appointments, schedule changes, or cancellations. This feature may reduce missed appointments and improve communication between the clinic and patients.

5.3.2 Additional Payment Options

Future researchers may integrate additional payment gateways such as Maya, PayPal, or credit/debit card payment services. This would provide patients with more payment options and improve transaction efficiency.

5.3.3 Online Payment Integration

Future researchers may integrate secure online payment gateways such as GCash, Maya, PayPal, or credit/debit card payment services. This would provide patients with additional payment options and improve transaction efficiency.

5.3.4 Mobile Application Development

The researchers recommend developing an Android and iOS mobile application that connects with the existing web system. A mobile application would provide greater accessibility for both patients and clinic personnel.

5.3.5 Backup and Disaster Recovery

Although Firebase provides cloud storage, additional automated backup and disaster recovery mechanisms should be implemented to further protect patient information from accidental deletion or unexpected technical issues.

5.3.6 Inventory Management

Future versions of the system may include an inventory module for monitoring dental supplies, equipment, and medication stocks. This feature would help clinics manage resources more effectively.

5.3.7 Financial Analytics Dashboard

Additional financial reporting tools, including monthly income summaries, expense tracking, and graphical dashboards, may assist clinic owners in evaluating business performance and making strategic decisions.

5.3.8 Enhanced Security

The researchers recommend implementing additional security features such as two-factor authentication (2FA), audit logs, encrypted database backups, and regular security assessments to further protect patient information and comply with data privacy regulations.

5.3.9 Future Research

Future researchers may continue improving the system by integrating emerging technologies such as artificial intelligence for appointment optimization, analytics for patient trends, or advanced imaging management, while ensuring that these enhancements remain practical and aligned with the needs of dental clinics.

5.4 Implications of the Study

The implementation of the Web-Based Dental Clinic Management System demonstrates how information technology can enhance healthcare services by improving administrative efficiency and data management. The system provides benefits not only to dental clinics but also to patients, who can experience faster service, organized records, and improved appointment management. The study also contributes to the field of Information Technology by showing the practical application of web technologies and cloud databases in solving real-world problems. Furthermore, it may serve as a reference for future researchers who intend to develop similar healthcare management systems.

5.5 Final Remarks

The successful development of the Web-Based Dental Clinic Management System demonstrates the importance of integrating modern information technology into healthcare services. By replacing manual processes with a secure and organized digital platform, the system helps improve operational efficiency, reduces administrative workload, and supports better patient care. The researchers believe that the system has the potential to enhance the daily operations of dental clinics by providing accurate record management, efficient appointment scheduling, reliable payment processing, and comprehensive treatment documentation. While there are opportunities for future enhancements, the current version provides a solid foundation for digital transformation in dental clinic management. The researchers therefore recommend the implementation of the developed system in dental clinics seeking to modernize their operations, improve service quality, and provide more efficient and reliable healthcare services for their patients.

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APPENDICES

Appendix A:

Letter of Request to Conduct the Study

June 02, 2026

DR. JULIX DIONNE CAPIZONDA
Clinic Administrator / Head Dentist
241-B Lopez Jaena St, Molo, Iloilo City, 5000 Iloilo
Iloilo City, Philippines

Dear Dr. Capizonda,

Warm greetings!

We are fourth-year Bachelor of Science in Information Technology (BSIT) students from PHINMA University of Iloilo. In partial fulfillment of the requirements for our Capstone Project 2 subject, we are currently conducting a study entitled: **"Design and Development of a Web-Based Dental Clinic Management System: Patient Records, Appointments, and Billing."**

The primary objective of this study is to develop a centralized, secure, and automated digital platform that will replace the current manual processes in your clinic, thereby improving operational efficiency, reducing scheduling conflicts, and enhancing patient record management.

In this regard, we would like to humbly request your permission to conduct interviews and observations at your clinic. The data gathered will be used solely for academic purposes and will be treated with the utmost confidentiality in strict compliance with the Data Privacy Act of 2012 (R.A. 10173).

Attached herewith is a copy of our research proposal for your perusal. We are hoping for your favorable response regarding this request.

Thank you very much for your time and support.

Respectfully yours,

ROA, Felix Adrian P.
VILLANUEVA, Jericho J.
SALIBIO, Andrew C.
BONITILLO, Jay Ralph
AMISTAS, Jim C. Jr.
Capstone Researchers, BSIT 3-7

Noted by:

RYAN VALERIANO
Capstone Project Adviser

Approved by:

DR. ARNOLD FUENTES
Program Head, College of Information Technology Education
PHINMA University of Iloilo

Appendix B: Informed Consent Form

Title of Study: Design and Development of a Web-Based Dental Clinic Management System
Researchers: Roa, Villanueva, Salibio, Bonitillo, Amistas (BSIT 3-7, PHINMA University of Iloilo)

Purpose of the Study:
This study aims to gather requirements to develop a web-based system that will manage patient records, appointments, and billing for dental clinics.

Procedures:
If you agree to participate, you will be asked to answer a brief interview regarding your current clinic workflows, challenges with manual record-keeping, and desired features for a digital system. The interview will take approximately 15-20 minutes.

Confidentiality:
All information provided will be kept strictly confidential. Your name and the clinic's name will be anonymized in the final manuscript. Data will be stored securely and used exclusively for this academic project.

Voluntary Participation:
Your participation is entirely voluntary. You may refuse to answer any question or withdraw from the interview at any time without any penalty.

Consent:
By signing below, I acknowledge that I have read and understood the information above, and I voluntarily agree to participate in this study.

____

Signature over Printed Name

DR. JULIX DIONNE CAPIZONDA

Date: ______________________

Appendix C: ISO 25010 System Evaluation Questionnaire *(Note to Researchers: In Word, format the following as a formal table with checkboxes)*

Title: Software Quality Evaluation Form based on ISO/IEC 25010 **Instructions:** Please evaluate the Web-Based Dental Clinic Management System by checking the box that best corresponds to your assessment. **Scale:** 5 - Excellent, 4 - Very Good, 3 - Good, 2 - Fair, 1 – Poor

Criteria	**Indicators**	**5**	**4**	**3**	**2**	**1**
1. Functional Suitability	The system provides all necessary features for managing records, appointments, and billing accurately.					
2. Performance Efficiency	The system loads quickly and handles real-time calendar updates without lag.					
3. Compatibility	The system works smoothly across different web browsers and devices (desktop, tablet, mobile).					
4. Usability	The interface is intuitive, easy to navigate, and requires minimal training to use.					
5. Reliability	The system is stable, does not crash, and securely saves data without loss.					
6. Security	The system successfully restricts unauthorized access					

Criteria	Indicators	5	4	3	2	1
	and protects patient data privacy.					
7. Maintainability	The system is well-structured, allowing for easy updates and bug fixes by developers.					
8. Portability	The system can be easily deployed and accessed on different cloud hosting environments.					

Summarized Interview Transcript (Client) Interviewee: Dr. Julix Dionne Capizonda (Clinic Administrator)
Date Conducted: Febuary 18, 2026
Interviewer: Felix Adrian P. Roa

Q1: Can you describe your current process for recording a new patient's information?
A1: Currently, we use physical folders. When a new patient arrives, the staff hands them a paper form to fill out. We then manually encode basic details into a simple Excel spreadsheet for backup, but the primary source of truth is the physical folder. This is prone to misplacement.

Q2: What are the most common scheduling conflicts you encounter?
A2: Double-booking is our biggest issue. Sometimes a patient calls to book a 10:00 AM slot, and the staff writes it in the logbook. Ten minutes later, a walk-in patient is assigned that same 10:00 AM slot because the staff forgot to update the book immediately. This causes long waiting times and patient frustration.

Q3: If we develop a digital system for your clinic, what are the top three features you need?
A3: First, a real-time calendar that prevents double-booking. Second, a centralized patient record where I can instantly see their past treatments and allergies. Third, an automated billing system that computes the total based on the procedures I select, so my staff doesn't have to calculate it manually.

Appendix D: Sample System Interface Mockups (Descriptions) *(Note to Researchers: Replace these descriptions with actual high-resolution screenshots of your Figma/Adobe XD mockups or early Flask builds to add 3-4 pages).*

- Figure D.1: System Login Page (showing email/password fields and "Forgot Password" link).
- Figure D.2: Administrator Dashboard (showing summary cards for Total Patients, Today's Appointments, and Monthly Revenue).
- Figure D.3: Interactive Appointment Calendar (showing color-coded slots: Green for Available, Red for Booked).
- Figure D.4: Patient Digital Treatment Record (showing fields for Diagnosis, Procedures, and an "Upload X-Ray" button).
- Figure D.5: Automated Billing Invoice (showing itemized services, subtotal, discount, and total amount due).

Appendix E: Sample Generated Reports *(Note to Researchers: Insert a full-page mockup of a PDF receipt/invoice generated by the system).* CLINIC NAME & LOGO
123 Healthcare Ave, Iloilo City
Contact: (033) 123-4567

OFFICIAL RECEIPT
Receipt No: INV-2026-0042
Date: June 02, 2026
Patient Name: Juan Dela Cruz
Attending Dentist: Dr. Julix Dionne Capizonda

Service Rendered	**Quantity**	**Unit Price**	**Total**
Dental Prophylaxis (Cleaning)	1	₱1,500.00	₱1,500.00
Dental Filling (Tooth 14)	1	₱2,000.00	₱2,000.00
Subtotal			**₱3,500.00**
Senior Citizen Discount (20%)			- ₱700.00
TOTAL AMOUNT DUE			**₱2,800.00**

Payment Method: GCash
Amount Paid: ₱2,800.00
Balance: ₱0.00

APPENDIX F: Draft System User Manual and Quick Start Guide

(Note to Researchers: This draft manual is designed to be distributed to the clinic staff during the User Acceptance Testing (UAT) phase. In the final bound manuscript, you can insert actual screenshots of your system under each heading to make it even longer!)

1. Introduction Welcome to the Web-Based Dental Clinic Management System. This user manual provides step-by-step instructions for navigating the system, managing patient records, scheduling appointments, and processing billing. The system is designed to be intuitive, but this guide will help ensure all users can maximize its features.

2. System Access and Login

- **Step 1:** Open any modern web browser (Google Chrome, Microsoft Edge, or Safari).
- **Step 2:** Navigate to the system URL provided by the Administrator.
- **Step 3:** Enter your registered Email Address and Password.
- **Step 4:** Click the "Login" button. You will be automatically redirected to your role-specific dashboard (Admin, Staff, or Dentist).
- *Note: For security purposes, the system will automatically log you out after 15 minutes of inactivity.*

3. Administrator Module Guide 3.1 Managing User Accounts

- Navigate to the "User Management" tab on the left sidebar.
- To add a new staff member or dentist, click the "Add New User" button.
- Fill in the required fields (Full Name, Email, Role, Temporary Password) and click "Save".
- To deactivate a user who has left the clinic, locate their name in the list and change their status to "Inactive".

3.2 Generating Clinic Reports

- Navigate to the "Reports and Analytics" tab.
- Select the desired report type (e.g., Monthly Revenue, Daily Patient Count).

- Use the date picker to select the start and end dates for the report.
- Click "Generate". The system will display visual charts. Click "Export to PDF" to download the report for accounting purposes.

4. Clinic Staff / Receptionist Module Guide 4.1 Registering a New Patient

- Navigate to the "Patient Records" tab and click "Add New Patient".
- Fill in the demographic information (Name, Contact Number, Address, Date of Birth).
- *Crucial Step:* Ensure the "Medical History" and "Allergies" sections are accurately filled out, as this will alert the dentist.
- Click "Submit". The system will generate a unique Patient ID.

4.2 Scheduling an Appointment

- Navigate to the "Appointment Calendar" tab.
- Click on an available (white) time slot.
- Search for the patient's name or Patient ID.
- Select the type of service (e.g., Dental Cleaning, Consultation) and the attending Dentist.
- Click "Confirm Booking". The slot will turn green, and an automated email reminder will be queued for the patient.

4.3 Processing Billing and Invoices

- Once the dentist has finished the treatment, navigate to the "Billing" tab.
- Search for the patient. The system will automatically list the pending treatments.
- Review the itemized charges. If applicable, select a discount (e.g., Senior Citizen 20%) from the dropdown menu.
- Select the Payment Method (Cash, GCash, etc.) and enter the amount tendered.
- Click "Process Payment". The system will calculate the change and generate a digital receipt. Click "Download PDF" to print the receipt for the patient.

5. Dentist Module Guide 5.1 Recording Treatment and Clinical Notes

- Navigate to "Patient Records" and open the patient's profile.

- Click "Add New Treatment Record".
- Select the procedures performed from the dropdown menu (e.g., Tooth Extraction, Dental Filling).
- Type the clinical diagnosis and any prescribed medications in the text areas.
- *Optional:* Click "Upload X-Ray" to attach digital imaging files (Max file size: 5MB, JPG/PNG format only).
- Click "Save Record". This action automatically locks the record from editing after 24 hours and sends the charges to the Billing module.

APPENDIX G: System Maintenance and Post-Deployment Support Plan

(Note to Researchers: This section proves to the panelists that you have planned for the long-term sustainability of the software, which is a key requirement for CHED BSIT standards.)

1. Routine Maintenance and Database Management To ensure the Web-Based Dental Clinic Management System remains secure, efficient, and free of data corruption, the following routine maintenance protocols will be implemented:

- **Automated Cloud Backups:** The Firebase Cloud Firestore database is configured to perform automated daily backups. In the event of accidental data deletion or system failure, the Administrator can restore the database to a previous state within minutes.
- **Storage Optimization:** The Firebase Cloud Storage bucket, which holds patient X-rays and intraoral photos, will be monitored monthly. Unused or duplicate files will be purged to optimize storage costs and improve load times.
- **Security Patching:** The Python Flask backend and frontend JavaScript libraries will be audited quarterly. Any outdated dependencies identified by the pip-audit tool will be updated to their latest stable versions to prevent vulnerabilities.

2. Troubleshooting Guide for Clinic Staff In the event of minor system issues, the clinic staff is trained to perform the following basic troubleshooting steps before contacting the development team:

- **Issue: Cannot Log In / "Invalid Credentials" Error.**
    - *Solution:* Ensure the Caps Lock key is turned off. If the password is truly forgotten, click the "Forgot Password" link on the login page to receive a reset link via email.

- **Issue: Calendar is Not Updating or Shows "Slot Unavailable".**
    - *Solution:* This usually indicates a brief internet connection drop. Refresh the web browser (Press F5). If the slot is genuinely double-booked due to a system glitch, the Administrator can manually override the schedule in the backend.
- **Issue: X-Ray Image Fails to Upload.**
    - *Solution:* Verify that the image file is in JPG or PNG format and does not exceed the 5MB size limit. If the file is too large, use an image compressor before uploading.

3. Future Enhancements and Scalability While the current system successfully digitizes the core operations of the dental clinic, the architecture is designed to be scalable. For future iterations (or Capstone 2), the following enhancements are recommended:

- **Integration with Email Gateway APIs:** Upgrading from local notification triggers to a paid API (like Twilio or SendGrid) to guarantee 100% delivery rates for appointment reminders.
- **Digital Dental Charting UI:** Replacing the text-based procedure dropdown with an interactive, visual 3D dental chart where dentists can click directly on specific teeth to mark decay, fillings, or extractions.
- **Inventory Management Module:** Adding a sub-module to track dental supplies (e.g., anesthesia cartridges, gloves, composite resins) and automatically deducting them from inventory when a treatment is recorded.

APPENDIX H: Patient Data Privacy Consent Form

(Note to Researchers: Under the Philippine Data Privacy Act of 2012, clinics must have patients sign a consent form before encoding their medical data into a digital system. This adds 1 highly legitimate page to your appendix.)

CAPIZONDA DENTAL CLINIC

PATIENT CONSENT FOR DIGITAL DATA PROCESSING AND STORAGE

I. Purpose of Data Collection I, the undersigned, hereby authorize **[Insert Clinic Name]** to collect, process, and store my personal and medical information in their Web-Based Dental Clinic Management System. This includes my demographic details, medical history, dental records, X-ray images, and billing information. This data will be used solely for the purpose of providing dental care, managing my appointments, and processing clinic transactions.

II. Data Privacy and Security I understand that my medical records are classified as Sensitive Personal Information under the **Data Privacy Act of 2012 (Republic Act No. 10173)**. The clinic assures me that:

1. My data will be stored securely in an encrypted cloud database.
2. Access to my medical records is strictly limited to authorized personnel (my attending dentist and authorized clinic staff) through Role-Based Access Control (RBAC).
3. My data will not be sold, shared, or disclosed to any third-party entities without my explicit written consent, except as required by law.

III. Patient Rights I am aware that under the Data Privacy Act, I have the right to:

- Be informed of how my data is being processed.
- Request access to my digital records.
- Request the correction of any inaccurate data in my profile.
- Withdraw my consent at any time, subject to the legal and medical retention requirements of the Department of Health (DOH).

IV. Consent By signing below, I acknowledge that I have read and understood this consent form. I voluntarily agree to have my information encoded into the clinic's digital management system to facilitate better and more efficient healthcare delivery.

Patient Name (Printed): ____

Patient Signature: ____

Date: ____

Witnessed By (Clinic Staff): ____________________________________

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.